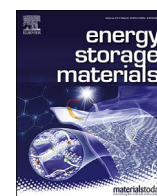




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Thermal conductivity enhancement on phase change materials for thermal energy storage: A review

Shaofei Wu^{a,b}, Ting Yan^{a,b,*}, Zihan Kuai^{a,b}, Weiguo Pan^{a,b,**}

^a College of Energy and Mechanical Engineering, Shanghai University of Electric Power, Shanghai, 200090, China

^b Institute of Distributed Energy System and Energy Storage, Shanghai, 200090, China

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ABSTRACT

Phase change energy storage technology, which can solve the contradiction between the supply and demand of thermal energy and alleviate the energy crisis, has aroused a lot of interests in recent years. Due to its high energy density, high temperature and strong stability of energy output, phase change material (PCM) has been widely used in thermal energy systems. The aim of this review is to provide an insight into the thermal conduction mechanism of phonons in PCM and the morphology, preparation method as well as thermal conductivity of composite PCMs. Phonon thermal conduction mechanism is suggested to cover three forms: phonon-phonon scattering, phonon-defect scattering and phonon-boundary scattering. Then the microcosmic factors affecting the thermal conductivity of composite PCMs are analyzed. The research progress of adding three-dimensional, two-dimensional, one-dimensional and zero-dimensional structure additives to PCM is reviewed. Besides summarizing the preparation method and microstructure, the thermal conductivity of the composite PCMs are also analyzed from the aspect of phonon thermal conductivity mechanism. Meanwhile, some novel materials, including metal organic frameworks (MOFs), titanium dioxide foam, highly graphitized network carbon, graphene foam and hexagonal boron nitride (HBN) nanoparticles, are proposed for the additives of heat storage materials. Ongoing research and development studies indicate that the challenges of the improving the thermal conductivity of PCM focus on the aspects of clarifying the phonon scattering mechanism in PCM, increasing the number of thermal conductivity chains and broadening the thermal transmission channels. Booming progress illustrates that the exploration of high performance PCM is an extremely valuable and scalable option for storing industrial waste heat and solar energy, especially for constant temperature storage and utilization. To bring the phase change heat storage solution into a broader market, more intensive studies in fields of phonon thermal conductivity mechanism, development of high performance composite PCMs and efficient and compact phase change heat storage system are still required.

1. Introduction

Latent heat storage has allured great attention because it provides the potential to achieve energy savings and effective utilization [1–3]. The latent heat storage is also known as phase change heat storage, which is accomplished by absorbing and releasing thermal energy during phase transition. Latent heat storage has the higher storage density than conventional sensible heat storage due to high enthalpy change in the phase change process. Compared to the sensible heat storage systems, latent heat storage systems require a smaller weight and volume, which brings about the relatively lower costs. In addition, latent heat storage has the

capacity to store heat of fusion nearly isothermally which corresponds to the phase transition temperature of the phase change material (PCM) [4]. Latent heat storage based on PCM can be applied in various fields, such as solar heat storage, energy-saving buildings and waste heat recycle, etc.

There are a variety of PCM to choose, but not all PCM can be used for latent heat storage. Generally, the ideal PCM available for the latent heat storage should have the following features:

- (1) Appropriate phase transition temperature to satisfy the practical application;

* Corresponding author. College of Energy and Mechanical Engineering, Shanghai University of Electric Power, Shanghai, 200090, China.

** Corresponding author. College of Energy and Mechanical Engineering, Shanghai University of Electric Power, Shanghai, 200090, China.

E-mail addresses: yt81725@126.com (T. Yan), pweiguo@163.com (W. Pan).

- (2) Large melting enthalpy to provide the high latent heat storage capacity;
- (3) High thermal conductivity to achieve the faster charging and discharging rate;
- (4) Stable thermal and chemical properties to ensure the persistent thermal storage capacity;
- (5) Non-toxic, non-corrosive, non-flammable and non-explosive to ensure the safety and harmless to surroundings;
- (6) Small enough supercooling;
- (7) Subtle volume variation during phase transition;
- (8) Abundance and easily available, cheap price to ensure the low cost.

It is hard to find the PCM can fully satisfy the above requirements. There always exist the diverse defects, such as phase separation, low heat transfer rate, supercooling, leakage in the molten state, instability of performance. PCM mainly include organic materials, inorganic materials and eutectic materials. The former is classified into paraffin, fatty acids and polybasic alcohols, while the latter is divided into hydrated salt, molten salts and metal or alloy materials. Various PCM, which can be used for the latent heat storage, are shown in Fig. 1. Almost all pure PCM have a common shortcoming of low thermal conductivity except for metallic PCM [5]. Hence, thermal conductivity enhancement is one of the main issues for the PCM in the application field of the latent heat storage. Many efforts have been made to improve the thermal-physical properties of PCM. Besides, a great deal of theoretical and experimental studies were carried out to investigate the heat transfer mechanism of various latent heat storage systems [6]. At present, the main methods to improve the thermal conductivity of PCM is to add high thermal conductivity matrix and chemically treat the surface of additive. These include modification of surfaces and grafted functional groups, and addition of porous three-dimensional (3D), two-dimensional (2D), one-dimensional (1D) and zero-dimensional (0D) structure additives. Modification and grafted functional groups can improve the compatibility of materials and reduce interfacial thermal resistance, but the success rate of modification is lower and the operation is more complicated. The addition of thermal conduction matrix can form thermal conduction chain, reduce phonon scattering and accelerate heat transfer. However, high content will significantly reduce the heat storage capacity of PCM. Therefore, appropriate addition amount and experimental conditions should be considered when selecting methods to improve thermal conductivity of PCM.

In this review, the microscopic mechanism of enhancing thermal conductivity on the composite PCMs were systematically analyzed. Further, we introduce preparation, morphological characterization and microscopic explanation of the composition PCM added a variety of new

substrates. We also emphasize that to enhance the thermal conductivity of the composite PCMs and to solve the research problems and emphases for the upcoming wide application are urgently needed. The contents of this review article are shown in Fig. 2. In conclusion, we need to have a thorough understanding of the system and apply technical knowledge to decrease the phonon scattering caused by one or other factors, and employ appropriate techniques to enhance the thermal conductivity of the system.

2. Mechanisms of thermal conductivity

Although many studies have been carried out in this field, the research on phonon thermal conductivity in PCM is relatively few from the perspective of micro mechanism and material morphology. How to improve the thermal conductivity of PCM is still an important research topic [7,8]. Many factors affect the thermal conductivity of composite PCMs, especially for phonon scattering as one important factor limiting the thermal conductivity of composite PCMs. The study of phonons is the most fundamental description of heat conduction at the microscopic level. In addition, the phonon is the driving force responsible for transferring heat from one end to the other. In microscale heat transfer, the key research to improve the thermal conductivity of composite materials lies in the effective phonon transmission around the axis point. Of course, pretending the phonon as a heat carrier can be propagated randomly without any hindrance, it is the best way to transmit phonon, and it is very likely to reach with desired thermal conductivity in a specific composite material. In composites, the three main factors controlling phonon transmission are phonon velocity, specific heat capacity and mean free path. The thermal transfer of composite PCMs and interfacial phonons is not limited to these factors, because these factors are also directly or indirectly affected by various factors, such as temperature, atomic weight, density and binding energy.

The process of heat transfer is shown in Fig. 3 including: (a) heat is first transmitted to atoms on the surface of the sample; (b) surface atoms gain vibrational energy; (c) the heat energy of the surface atoms is then transferred to the neighboring atoms at the same speed; (d) thermal energy is then diffused to the sample in the form of vibration waves; (e) after diffused by heat energy in a common vibration mode or phonon throughout the crystal; (f) when heat reaches the back of the sample, it is then partially transferred to the surrounding environment by conduction or radiation.

2.1. Lattice vibration and phonon transmission

Heat transport can be achieved through electrons and lattice vibrations in the solids. For the metallic PCM, heat transport is dominated by

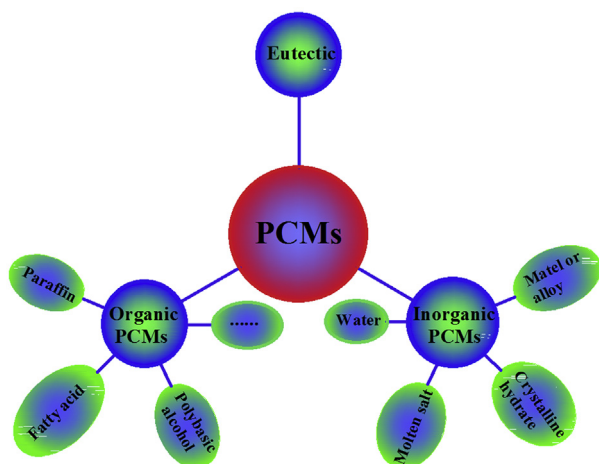


Fig. 1. Classification of phase change materials.

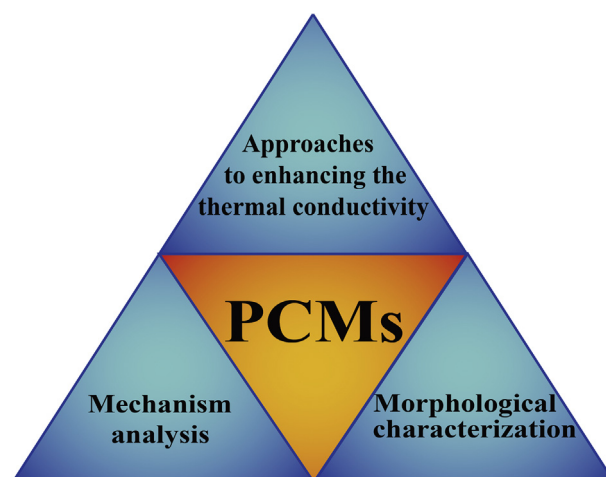


Fig. 2. Structure of this review article.

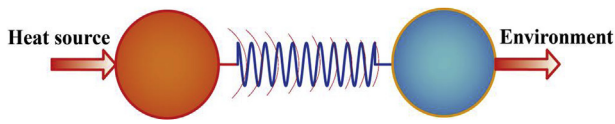


Fig. 3. Heat transfer diagram in crystal heat conduction.

electrons. While most inorganic nonmetallic PCM have electrons bound in their ionic crystal lattice, but free electrons are few and cannot be heat conduction carriers [9]. For the nonmetallic PCM, heat conduction mainly relies on the vibration of the crystal lattice [10]. Lattice vibration is the main mechanism of heat conduction for these PCM. As a result, the phonons are the dominant heat carriers in these PCM. Obviously, heat transport by lattice vibration is addressed in this work [10–12].

The particles in the crystal vibrates near the equilibrium position even at a low temperature. Hence, the lattice of the PCM can be regarded as a coupled mass and spring system, as shown in Fig. 4. The vibration will deviate from the equilibrium position when the crystal is subjected to heat. Afterwards the high energy phonons are generated and diffuses, which forms the temperature gradient. As a result, the heat will flow from the hot end to the cold end in the presence of the temperature gradient. Fig. 5 shows the temperature gradient between the hot end and cold end due to the diffusion of the phonons. Thus, the heat conduction is a nonequilibrium phenomenon [13].

As above mentioned, phonons play a pivotal role in the heat transfer of non-metallic materials. Commonly, the stronger is phonon diffusion in a material, the larger is its ability to heat conduction and the higher thermal conductivity. Conversely, the phonon diffusion is obstructed in non-metallic materials, it will result in the low thermal conductivity. Indeed, different phonon scatterings inside the non-metallic materials is the fundamental reason for the deterioration of phonon diffusion. Phonon scattering can shift or even reverse the direction of energy transport, and thus is the dominating reason for the poor heat conduction in the non-metallic materials. In composite materials including the PCM, the phonon scattering is primarily caused by the mismatch between the filler-matrix and the PCM. Specifically, phonon scattering mainly contributes to the interfacial thermal resistances and thus degrades the thermal current. There are three types of phonon scattering: phonon-phonon scattering, phonon-defect scattering and phonon-interface scattering [14], as shown in Fig. 6.

2.1.1. Phonon-phonon interactions

Phonons arise from lattice vibration and are quanta of lattice vibration. Phonons have the significantly particle-like vibrational modes of a crystal and therefore can be treated as quasi-particles. Different waves

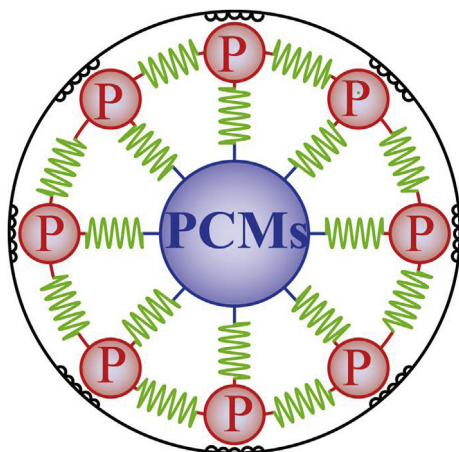


Fig. 4. Spring vibration heat transfer system composed of crystal atoms and abstract phonons.

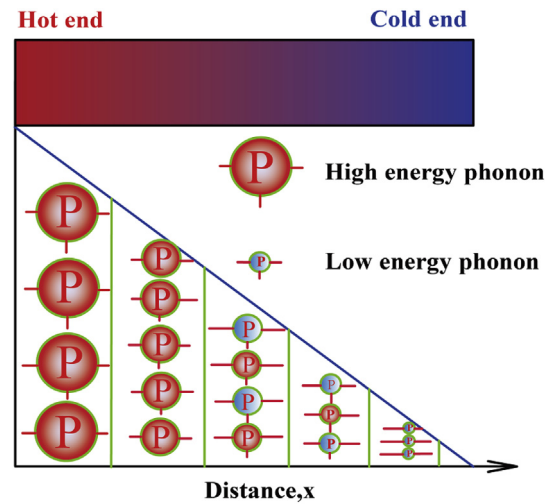


Fig. 5. Temperature curve with length and phonon energy diagram.

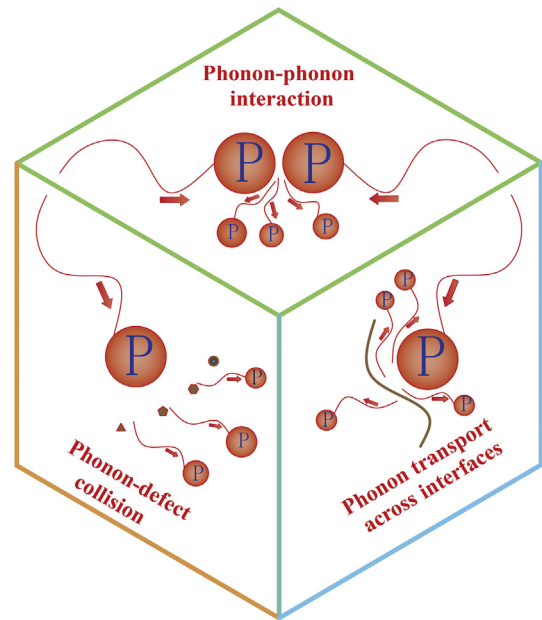


Fig. 6. Schematic diagram of three phonon scattering modes.

propagating all the time makes a number of phonons taken part in the interactions. However, the phonon numbers are not conserved due to the creation or annihilation of phonon. It is a remarkable fact that three-phonon scattering is the most important interactions between the waves, which determines the thermal conductivity. Furthermore, three-phonon scattering has two fundamentally different kinds of diagrams: either two phonons interact and combine into the third phonon, or energy is divided from one mode to two other modes. Phonon-phonon collision can be usually classified into two types: normal process (N process) and Umklapp process (U process) [15,16]. Three-phonon processes obey the following equations [17]:

Case (a):

$$\hbar\omega_1 + \hbar\omega_2 = \hbar\omega_3 \quad (1a)$$

$$\hbar k_1 + \hbar k_2 = \hbar k_3 \quad (1b)$$

Case (b):

$$\hbar\omega_1 = \hbar\omega_2 + \hbar\omega_3 \quad (2a)$$

$$\hbar k_1 = \hbar k_2 + \hbar k_3 \quad (2b)$$

Case (c):

$$\hbar \omega_1 + \hbar \omega_2 = \hbar \omega_3 \quad (3a)$$

$$\hbar k_1 + \hbar k_2 = \hbar k_3 + G \quad (3b)$$

Since $\hbar \omega$ is energy of phonon with the angular frequency ω , the first relation of these cases represents the energy conservation. Where k is the wave vector, and G is the reciprocal lattice vector. A phonon doesn't have an ordinary mechanical momentum, but $\hbar k$ has some of its attributes and can be perceived as the phonon quasi-momentum. Therefore, equations (1b) and (2b) reflect the momentum conservation. The N processes obey both energy and momentum conservation whereas the U processes only conforms to the energy conservation. As depicted in Fig. 7, Case (a) and Case (b) are the N processes and Case (c) is the U process.

If there is no interaction between phonons, the thermal energy is propagated through the perfect crystal without the defects and the thermal conductivity will be infinite. However, the heat transfer is a random process and this random nature of the conduction process brings the temperature gradient [18]. The thermal energy does not simply and directly proceed in a straight path from one end to the other, but diffuses through the crystal and undergoes the frequent collisions. Among then the average distance traveled by a phonon between collisions is called the mean free path. At sufficiently low temperature, lattice vibrations will be frozen at the ground state and it is hard to be excited for the phonon. Only low frequency phonon (or long wavelength acoustic modes) and small phonon number (or wave number modes) are excited. As the probability of the collisions between phonons is very small and therefore causes the long mean free path [19]. Moreover, phonon-boundary scattering is the dominant mechanism that affects the improvement of thermal conductivity. For the most part, the temperature dependence of thermal conductivity λ is dependent on the specific heat capacity and thus is proportional to T^3 where T is the absolute temperature [20]. The number of excited phonons increases with the rise of the temperature, and thus the probability of the collisions between phonons becomes large. Meanwhile, mean free path becomes shorter and the thermal conductivity is increased. At high temperature, the number of phonons is proportional to T . The effect of anharmonic coupling between different phonons limits the mean free path. While the mean free path is proportional $1/T$ at high temperatures. As a result, thermal conductivity is proportional to $1/T$.

2.1.2. Phonon-defect scattering

Lattice defects could give rise to the scattering of phonons, such as: phonons are reflected, diffracted or refracted, all of which have a negative impact on the phonon transport. Lattice defects mainly include point defects, dislocations, vacancy defects, impurities, etc. Phonon collisions with the defects or a crystal boundary don't change the energy of individual phonons. But the presence of the defects makes the mean free path

shorten and therefore leads to a lower the intrinsic thermal conductivity of the crystal. Even if there are only small amounts of impurities in the crystal structure, however, they can have a significant impact on the thermal conductivity. Apparently, the impurities in the crystal inevitably produce the additional thermal resistance, and thus lead to the decrease of the thermal conductivity.

2.1.3. Phonon-boundary scattering

When phonon travels across the boundaries of crystal of finite size, it gets scattered and this affects the effective mean free path of phonon. Impact of phonon-boundary scattering is most significant at low temperature, where the mean free path of phonon is the longest. At room temperature and above, it can also contribute to scattering especially in materials like very thin silicon layers [21]. In nanostructures, when the mean free path of phonon approaches to the characteristic scale of materials, thermal transport deviates from the Fourier law [22,23]. Phonon transport greatly varies when characteristic length of materials is smaller than the mean free path of the phonon. In simpler terms, heat transfer is also a function of the characteristic length of materials especially in nano-structures or low dimension materials [23–25]. When phonon approaches boundary, it can lead to two types of boundary scattering. One is specular phonon-boundary scattering where the reflection of phonon from the boundary is specular in nature [26]. The other is diffuse and phonon is scattered at various angles. Both of these scattering modes pose serious bottlenecks in heat transport across interfaces. In composite PCMs, phonon scattering is mainly attributed to the interfacial thermal resistances which arise from acoustic mismatch at filler-matrix interface [27].

Three kinds of phonon scattering mechanisms will inevitably generate thermal resistance in the heat transfer process of composite PCMs, but the existence of scattering phenomenon leads to a significant reduction in thermal conductivity, such as phonon-phonon scattering, phonon-defect scattering and phonon-boundary scattering, etc.

2.2. Factors influencing effective thermal conductivity of PCM

The heat conduction inside materials is caused by the collision of various microscopic particles. As a result, the variation of crystal structure or any microstructure of the material will change the thermal conductivity of materials. The complexity of the heat conduction mechanism through composites also arises from the many factors. For the crystalline materials, the lattice thermal conductivity can also be theoretically estimated through Debye formulation Eq. (5) [12,28–30]:

$$\lambda = \frac{1}{3} \int C_v(\omega) \times v(\omega) \times l(\omega) d\omega \quad (5)$$

Where C_v is the heat capacity of the phonons per unit volume, v is the phonon group velocity, l is the phonon mean free path.

According to the calculation expression of thermal conductivity, the direct factors affecting the composites thermal conductivity are the

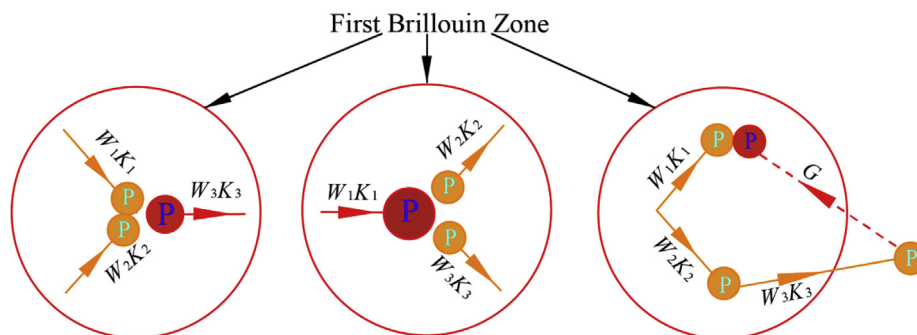


Fig. 7. Three-phonon N process (a) and U process (b).

specific heat capacity (C_v), phonon group velocity (v) and phonon mean free path l . The individual parameter of analyzing the effect of phonon thermal conductivity by phonon scattering mechanism from the microscopic perspective is shown in Fig. 8.

2.2.1. Heat capacity C_v

For the nonmetallic PCM, heat capacity C_v is temperature-dependent. At an extremely low temperature, the specific heat capacity of the phonon is proportional to T^3 [29]. At relatively high temperature, the heat capacity becomes basically constant, which approximates to $3R$ (R is the gas constant).

2.2.2. Phonon group velocity v

Phonon group velocity is a vital factor in the microscopic theory influencing the thermal conductivity of composites. Phonon group velocity, which is the transmission velocity of a wave packet, can be obtained from the dispersion relation. In addition, the phonon scattering is intimately related to the frequency. The higher is the frequency of phonon, the stronger is the phonon scattering. The continuous low-frequency modes associated with the acoustic phonons. The group velocity the acoustic phonons is large. However, the phonon group velocity is so small that it can be negligible for the short wavelength optical modes near the edge of the Brillouin zone. The optical phonons contribute to the specific heat, but only a little bit to the heat flux. Hence, the long wavelength acoustic phonon branch accounts for the thermal transport [31,32]. At the long wavelength limit or continuum limit, the group velocity reaches the acoustic velocity C_s in a crystal, which is associated with the elastic stiffness E , and mass density ρ in the long wave-length region [14,33].

$$\lim_{q_{ph} \rightarrow 0} v = \lim_{q_{ph} \rightarrow 0} \frac{d\omega}{dq_{ph}} = C_s = \sqrt{\frac{E}{\rho}} \quad (6)$$

2.2.3. Phonon mean free path L

There are three main factors: inelastic phonon–phonon scattering resulted from the anharmonicity effect, phonon–defect scattering and phonon–boundary scattering, all of which can impact on the mean free path. In the preparation of composites, the large phonon mean free path is expected and the effective phonon mean free path L_{eff} can be described by the following Matthiessen's rule [29,30]:

$$\frac{1}{L_{eff}} = \frac{1}{L_{ph}} + \frac{1}{L_d} + \frac{1}{L_b} \quad (7)$$

Where L_{ph} , L_d , and L_b are the mean free path arisen from the phonon–phonon, phonon–defect, and phonon–boundary scattering, respectively. Three phonons mean free paths deriving from the different scattering patterns can be shown in Fig. 9. All phonon scattering processes can make the mean free path reduce. Only one of the factors among which brings about the shortest mean free path, will have a prominent influence. Here, we will analyze the dominant factors.

2.2.3.1. Phonon–phonon scattering mean free path, L_{ph} . As above mentioned, the lattice vibration can be analogous to the spring mass system. Two lattice waves haven't interactions and a single wave doesn't decay for the harmonic theory. Moreover, the elastic coefficients don't vary. The heat transport could proceed with no obstruction and therefore cause an infinitely large thermal conductivity. In a real crystal, the elastic

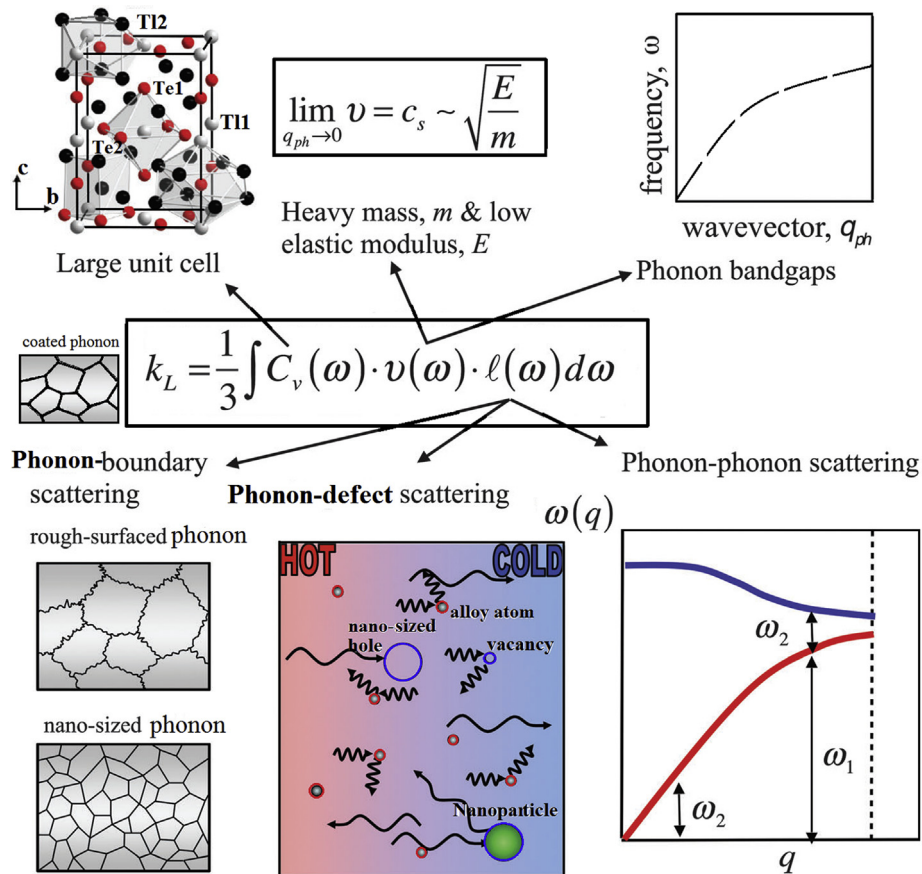


Fig. 8. Strategies of phonon engineering categorized into methods affecting each component of phonon thermal conductivity.

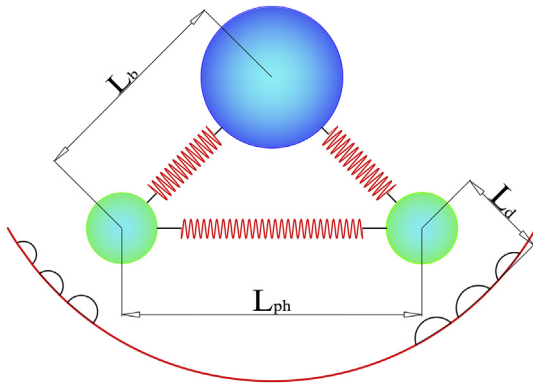


Fig. 9. Phonon mean free pathways for three different phonon scattering modes.

coefficients may be changed because of the distortions of the lattice arisen from the atomic displacements. When the phonon propagates, the change of the elastic coefficients will influence the heat transfer. If the temperature is low enough for the number of phonons to be so small that even N processes can be neglected [34]. However, the phonon excitation and three-phonon scattering rate augments with the increase of the temperature. The N processes become the dominant. At the higher temperature, all phonon vibrational modes are thermally excited. The main mechanism of thermal resistivity is phonon–phonon umklapp scattering with the attendant large momentum change in the collision. The U processes are more prominent and account for the decrease of the thermal conductivity at a relatively high temperature [35].

2.2.3.2. Defect scattering mean free path, L_d . Phonons can be regarded as pseudo-particles. In this sense, kinetic theory can be used to explain phonon impurity scattering. According to kinetic theory, the mean free path due to an impurity is

$$L_{im} = \frac{1}{\sigma_{sc}(\chi) \times \eta} \quad (8)$$

Here, η denotes the impurity concentration. σ_{sc} is the scattering cross-section, which is a function of the size parameter χ . The size parameter χ is defined as the wave-vector q of an incoming phonon multiplied by the radius of the scatter r , i.e., $\chi = qr$. The important parameter in impurity scattering is the scattering cross-section. When the size parameter approaches zero, the scattering cross-section forms according to the Rayleigh law. In the Rayleigh scattering regime, the scattering cross-section varies as $\sigma \sim d^6 \omega^4$, where d is the size of the scattering particle, and ω is the phonon frequency. For atomic substitutions in alloys, $d \sim 1 \text{ \AA}$. The above relation suggests that short wavelength or Brillouin zone edge phonons are scattered much more effectively than mid- and long-wavelength phonons. At the other limit, where the size parameter approaches infinity, the scattering cross-section is independent of frequency.

In porous matrix materials, impurities are usually defects, vacancies, dislocations, etc., which are on the order of the atomistic scale. Therefore, in porous matrix materials, the scattering cross-section follows the Rayleigh law, i.e. $L_d \sim \omega^{-4}$. Although short-wavelength phonons can be scattered by porous matrix atoms or empty space, long-to mid-wavelength phonons are not affected by the scattering of porous matrix and still have strong thermal conductivity. Therefore, by decreasing the width of grain boundary and increasing the grain size to reduce the volume percentage of grain boundary in the crystal, and transforming the amorphous state of grain boundary into grain state, the scattering of short-wavelength phonons can be significantly reduced, and the total thermal conductivity of crystal composites can be increased [36].

2.2.3.3. Boundary scattering mean free path (L_b). Phonon often suffers strong scattering at the boundary or interface of the crystal when it

propagates. For the PCM working at room temperature or above, boundary scattering is often important in practice. In the composites PCMs, the interface between the filler-matrix and the PCM is the main barrier for the thermal transport. Phonon transport characteristics differ greatly at interfaces in bulk than that of nano-scale region. As particle size decreases to nanometer scale, interface can lead to significant phonon scattering. The heat flow within the composite materials is impeded by the presence of the interfacial resistance, which decreases the benefit of adding high thermal conductive filler [37,38]. As a result, the thermal conductivity of the composite materials is lower than that of the high thermal conductive filler. In composite PCMs, the phonon transport can be hindered by the phonon scattering due to the interfacial thermal resistance. There occurs a temperature discontinuity at the interface between two different materials. This temperature discontinuity is resulted from the thermal boundary resistance (Kapitza resistance). The thermal boundary resistance in the interfaces between two different materials is becoming increasingly important especially on the nanometer length scales [39]. Moreover, either the different lattice constants or even different crystal structure between two different crystals will give rise to the lattice mismatch.

3. Approaches to enhancing the thermal conductivity of PCMs and their mechanisms

Thermal conductivity of PCM can be predicted and improved by analyzing the microcosmic heat transfer mechanism. Morphological features of materials play an important part in heat conduction mechanism. Fig. 10 involves the main content of this chapter. Firstly, the methods of the chemical modification to improve the thermal conductivity are summarized. Then the preparation methods, morphological characterization and micro explanation of the composite PCMs are emphatically discussed, which adds 3D, 2D, 1D and 0D structure additive to improve the thermal conductivity. Finally, we provide some useful conclusion and references to explore the high performance composite PCMs.

3.1. Chemical modification of matrices

3.1.1. Doping

The influence of high heat conducting materials, which doped matrix on the properties depends not only on the doping amount of the matrix and its intrinsic properties, but also on the existing form of the doped matrix in high heat conducting materials. When no chemical reaction between the doped matrix and the medium of the high heat conducting particle, it is mainly in the form of mischcrystal, uniformly dispersed in the matrix or filled in the pores of the basis material. However, the doped matrix reacts with the medium of the heat conducting particle, it may exist in the form of carbide or oxide. No matter how the doping matrix exists, it is affected the microstructure of the high heat conducting material and the results are change the physical and chemical properties of the composite PCMs. In a word, the effect of macroscopic doping on the micro morphology of composites is mainly reflected by changing the pore quantity and pore size of matrix. The micro-doping results in great changes in the surface phase composition and structure of high thermal conductivity matrix. In addition, the thermal conductivity of composites is improved by increasing the number of thermal conduction channels and reducing phonon scattering. In order to improve the thermal conductivity and cyclic stability of PCM, porous materials with doped elements can be added to improve the microstructure and surface compatibility of PCM.

Researchers have done a lot of work about doped with special elements to an additive substrate. Atinafu et al. [40] prepared the form-stable binary eutectic PCM by encapsulating myristic acid (MA)/stearic acids (SA) mixture into porous carbon matrixes. The nitrogen (N) heteroatom is introduced the porous carbons and forms mesoporous N-doped carbons (NPC) in situ synthesis to improve the

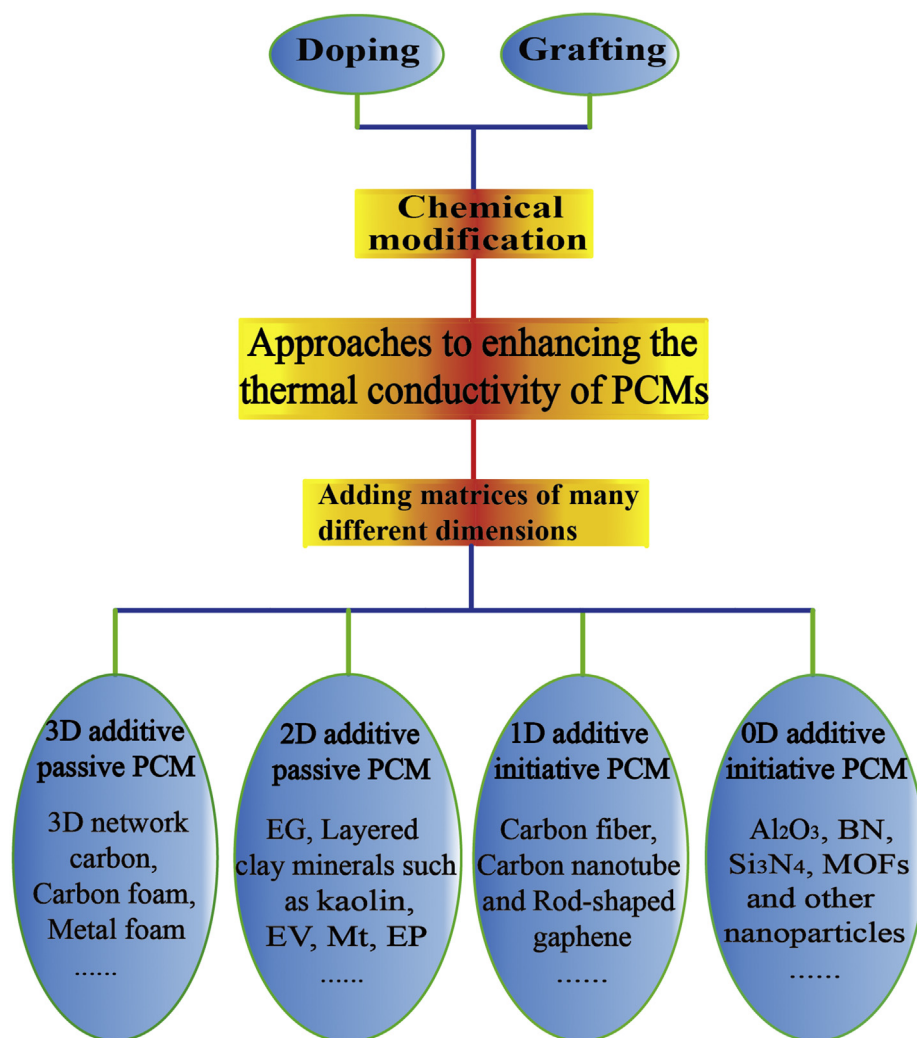


Fig. 10. Approaches of thermal conductivity enhancement.

thermal conductivity, thermal energy storage and conversion performance. The form-stable loading content of PCM in the NPC/MA-SA composites without leakage reaches 88 wt%. Fig. 11 is thermal conductivity and heat flow of porous carbon matrix/MA-SA composite PCMs and the experimental results show that the NPC/MA-SA composite PCMs exhibit excellent thermal conductivity and heat storage capacity. The enthalpy of the NPC/MA-SA composites are 164.33 kJ/kg with the phase change temperature of 49.45 °C. The latent heat is up to 45.69% higher than that of composites by post synthesis route. Moreover, the thermal conductivity of NPC/MA-SA composite PCMs increases by 117.65% than pristine PCM and is about 43.06% larger than that of composites by post synthesis route. The introduction of N heteroatom plays a key role in changing the textural properties of carbon materials. The nitrogen content, N–C arrangement of amino functional groups, graphite properties and interconnected porous carbon is conducive to form the heat transfer network.

Mehrali et al. [41] prepared palmitic acid (PA)/nitrogen-doped graphene (NDG) composite PCMs. NDG is obtained by chemical doping with nitrogen of graphene oxide (GO) using the hydrothermal process. PA is dissolved in the toluene solution with the different mass percentages (1–5 wt%). In order to make the nano particles uniformly disperse, the high power ultrasonication was utilized to break down the aggregation of the NDG. Fig. 12 (a) exhibit the field emission scanning electron microscopy (FESEM) images of the NDG. From the diagram, it can be seen that the NDG keeps the 2D single-layer graphene structures, which has high

specific surface area and volume ratio. The NDG nanosheets are randomly stacked and the porous structure resembles crumpled silk veil waves. Fig. 12 (b) and (c) show the transmission electron microscopy (TEM) images of the NDG and PA/NDG composite PCMs. We can see that the NDG is enclosed by membranes made from the PA. It suggests that the PA adsorbed by the network structure of the NDG. The heat storage density of this novel form-stable PA/NDG composite PCMs are 199.48 kJ/kg. Furthermore, the addition of the NDG makes the thermal conductivity and specific heat capacity of the PCM remarkably augment. The thermal conductivity of the PA/NDG composite PCMs are 1.73 W/(m·K) and is up to about 518% higher than that of the pure PA (0.28 W/(m·K)). The reason is that the heat-conducting carbon network formed by the NDG layer enhances the effect of heat transfer. With the increase of NDG content, the speed of thermal conductivity increased to a maximum followed by a slow decrease to a constant value and thermal conductivity reach the maximum value. In addition, the form-stable composite PCMs have an excellent thermal reliability.

Chen et al. [42] studied the impacts of interfacial interactions between different small molecular organic PCM and N-doped hierarchical carbon on the thermal energy storage properties. The in-situ N-doped hierarchical 3D nanoscale porous carbon was derived from one-step high temperature pyrolysis of polyaniline hydrogel. Fig. 13 is shown the preparation procedures of composite PCMs. The results suggest that one-step pyrolysis process is beneficial to impregnate effectively organic small molecule PCM to promote the phonon transport and the thermal

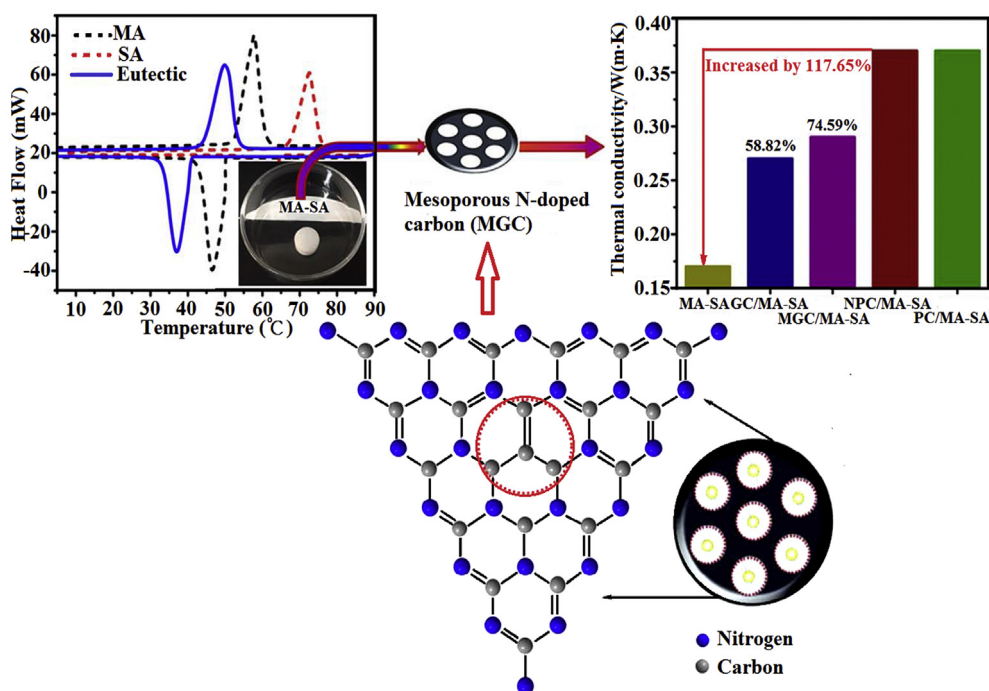


Fig. 11. Thermal conductivity and heat flow of porous carbon matrix/MA-SA composite PCMs.

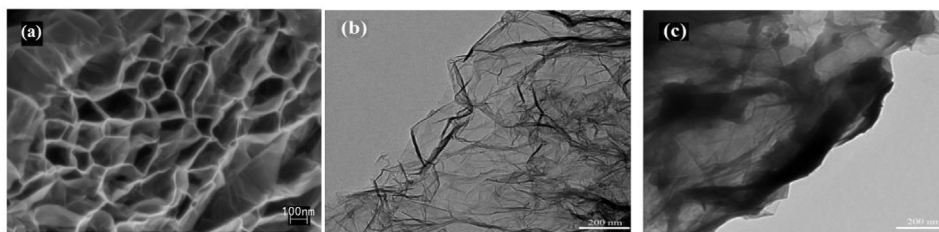


Fig. 12. (a) Field emission SEM images of NDG, TEM images of (b) NDG and (c) PA/NDG composite PCM.

transfer is enhanced with the rise of the pyrolysis temperature. The improvement of the thermal conductivity ascribes to the synergistic effect of highly interconnected graphitized network carbon and rich graphitic nitrogen, which can also accelerate phonon group velocity and increase the mean free path to promote faster heat flow. Besides, the induced hydrogen bond intensity between N-doped hierarchical carbon and PCM, and space restriction effect of the hierarchical pore on the PCM determine the nanoconfinement behaviors of PCM [43]. It can increase the densification degree of PCM and carbon skeleton, thereby reducing interfacial thermal resistance and significantly improving thermal conductivity.

3.1.2. Graft functional groups

Grafting or post synthesis is a method of introducing organic groups into the surface of mesoporous channels through chemical reactions between organic precursors and functional groups on the superficies of mesoporous materials. Organic functional groups are introduced into the channels of mesoporous by grafting and further reactions are made to produce new active centers. As the active centers generated are close to each other, many available organic functional groups have no significant effect on the mesoporous structure of the matrix material. However, most of the unevenly distributed organic functional groups are distributed on the orifice and the external surface, which is lead to the blocking of mesoporous channels and thus reduce the pore volume. Therefore, the characteristics of introducing organic functional groups into composite materials: (1) as the proportion of organic groups increases, the pore size

of the matrix materials gradually decreases; (2) the mesoporous structure of the matrix material is not changed; (3) the hydrophilic/hydrophobic and thermal conductivity of composite materials can be regulated by changing the type of organic groups.

Based on the above characteristics, the supporting material has been modified by grafting to improve its comprehensive properties, such as Wang et al. [44] fabricated polyethylene glycol (PEG)/carbon nanotube (CNT)@Cr-MIL-101-NH₂ form-stable composite PCMs via the impregnation method. PEG/CNT@Cr-MIL-101-NH₂ composite PCMs synthesis diagram, morphological characterization and thermal conductivity curve are shown in Fig. 14. The CNT@Cr-MIL-101-NH₂ is obtained via the situ method to synthesis Cr-MIL-101-NH₂ and the surface of the CNTs is modified by the Cr-MIL-101-NH₂ nano-particles. The PCM is absorbed by the MOFs nanoparticles due to the surface tension force and capillary action. As a result, the MOFs closely connected with CNT to form a porous structure of MOFs can eliminate the leakage of liquid phase PEG. Therefore, it forms 3D network structure and the structure creates the heat transfer network [45]. Moreover, the interfacial thermal resistance between CNT@Cr-MIL-101-NH₂ and PCM can be reduced and mean free paths of phonons increases. Further, the thermal conductivity of the PEG/CNT@Cr-MIL-101-NH₂ composite PCMs increases by 100.9% than without the introducing CNTs. Li et al. [46] utilized paraffin and grafted CNTs to prepare the composite PCMs. Fig. 15 shows the formation process and morphological images of pristine CNTs and CNTs/paraffin composite PCMs. The pristine CNTs (PCNTs) are more tendency to agglomerate owing to van der Waals' interaction and the use of chemical

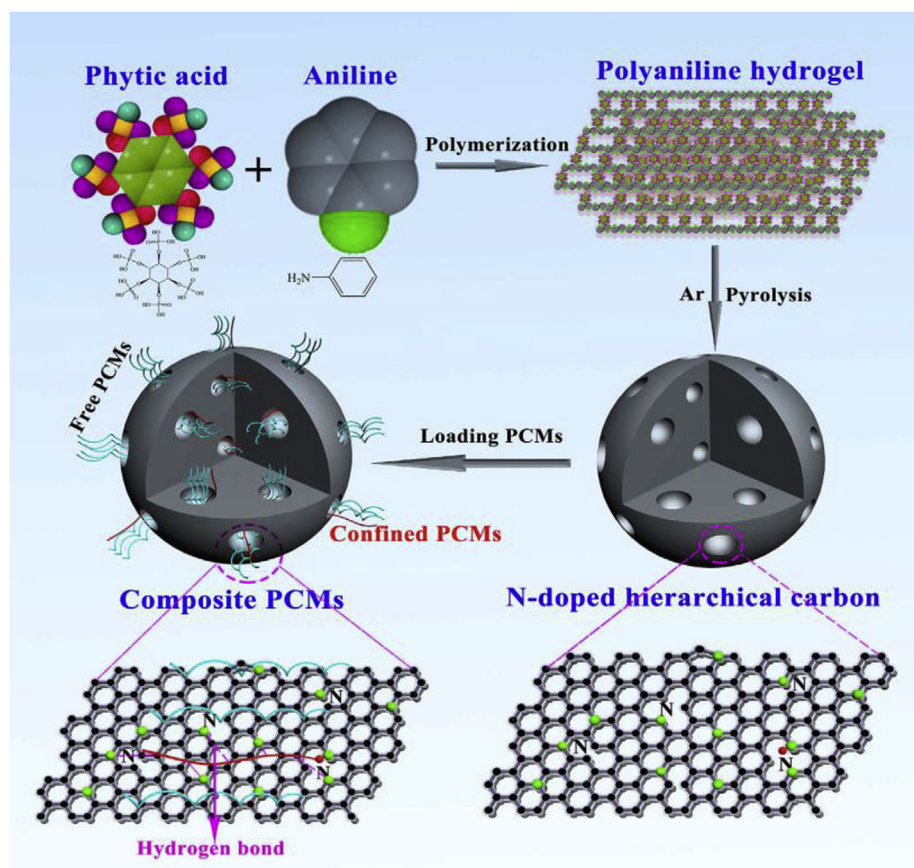


Fig. 13. Preparation schematic illustration of the N-doped hierarchical porous carbon based composite PCMs.

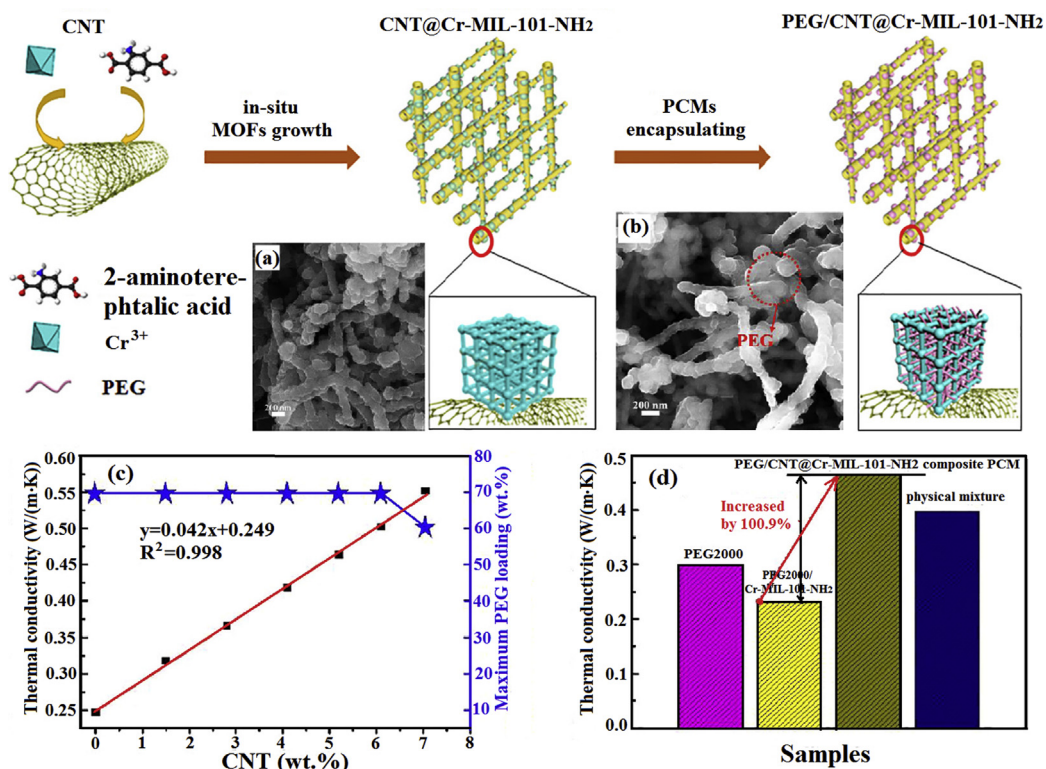


Fig. 14. PEG/CNT@Cr-MIL-101-NH₂ composite PCM synthesis diagram, microstructure and thermal conductivity curve.

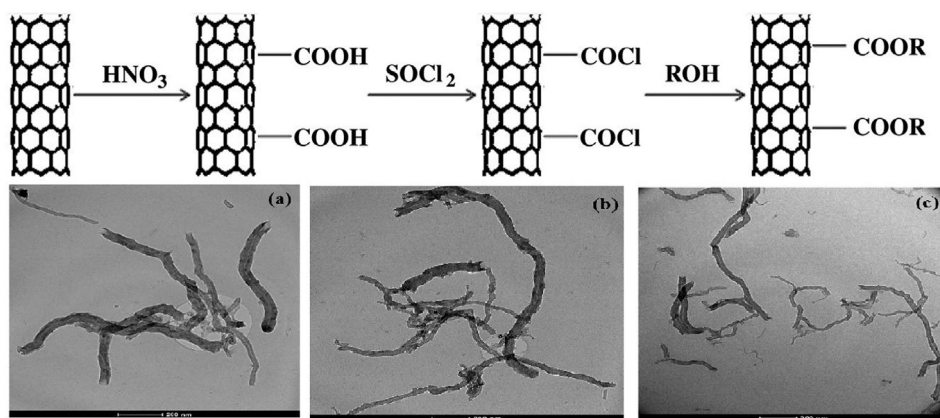


Fig. 15. TEM images of the formation process and products of grafted CNTs: (a) CNT-C8, (b) CNT-C14 and (c) CNT-C18.

modification improves the dispersibility of CNTs [47]. Further, the thermal conductivity of grafted CNTs/paraffin composite PCMs is higher than that of PCNTs/paraffin and the thermal conductivity of composite PCMs augment with the increase of CNTs content. When the CNTs content is 4 wt.%, the thermal conductivity of CNTs/paraffin, CNTs-C₈/paraffin, CNTs-C₁₄/paraffin and CNTs-C₁₈/paraffin composite PCMs are 0.6454 W/(m·K), 0.7683 W/(m·K), 0.7783 W/(m·K), 0.7903 W/(m·K), respectively. Because the well-grafted CNTs, which has larger carbon numbers and longer carbon chains, can blend successfully into the paraffin. Hence, grafted CNTs with longer carbon chains have the smaller the interface thermal resistance and larger thermal conductivity with the same CNTs content. After grafting treatment, the interfacial thermal resistance between the grafted CNTs and paraffin is reduced. According to mechanisms of thermal conductivity, the phonon-boundary scattering is significantly weakened and the phonon average free path is remarkably increased. As a result, the thermal conductivity is improved.

Table 1 summarizes the effects of doping and graft functional groups methods on enhancing the thermal conductivity of composite PCMs. It's indicated that chemical treatment is helpful to improve the thermal conductivity of composite PCMs. The main reason is that the chemical treatment can form a 3D thermal conductivity network and reduce the surface thermal resistance, which is conducive to promoting the rapid transfer of heat along the thermal conductivity chain.

3.2. Fabrication and characterization of composite PCMs

Mixing a material having high thermal conductivity with a PCM can be classified as two modes. One mode is to prepare a form-stable composite PCM, in which the PCM is absorbed into the pores within the supporting material to completely avoid the liquid leakage after the PCM took place the phase change from solid to liquid. This mode is called passive composite PCMs. It mainly forms a heat-conducting chain in the

middle of the composites through 3D structures (such as highly graphitized network carbon, carbon foam and metal foam, etc.) or 2D structural additives (expanded graphite and clay minerals, etc.) to reduce phonon scattering to improve thermal conductivity; The other one is associated with adding a small amount of a material with high thermal conductivity into the PCM to just improve its thermal conductivity, the obtained mixture can be considered as a initiative composite PCMs since the liquid leakage of the PCM cannot be avoided. This mode mainly forms a heat conduction channel after the PCM is added to the 1D structure (carbon fiber, carbon nanotube and rod-shaped geaphene, etc.) or 0D structure additive (nanoparticle). Therefore, it can increase the phonon mean free path to improve thermal conductivity.

3.2.1. Passive composite PCMs using three-dimensional structure additives

The 3D structure additives have many morphological structures, such as graphite foam (GF), highly graphitized network carbon and metal foam etc. Due to the 3D structure additives can easy form the inter-connected pore network and it can provides conductive pathways for enhancing the heat transfer in PCM matrix. Besides, the 3D structure additives, which also can avoid the leakage of the composites in the melting state and reduce the supercooling degree, are the ideal choice for the preparation of the high performance PCM [48].

3.2.1.1. Addition of graphite foam in passive PCM. Graphite foam (GF) is a high-performance porous 3D structure additive with high porosity, low density, effective thermal conductivity up to 200 W/(m·K) [49], high temperature resistance, corrosion resistance and other characteristics, which can also be mixed with most PCM compatible [50]. At the same time, compared with metal foam, graphite foam has a higher thermal conductivity, so it has been widely concerned [51,52].

Karthik et al. [53,54] used Resorcinol-Formaldehyde (RF) resin as the precursor and polyurethane foam as the carbon source by sacrificing the

Table 1

Summary of chemical modification for thermal conductivity enhancement in composite PCMs.

Method of chemical modification	PCMs	Thermal conductivity of PCMs k_p W/(m·K)	chemical processing	Composite PCMs	Thermal conductivity of Composite PCM k_c W/(m·K)	Improvement (%)	References
Doping	MA-SA	0.17	Nitrogen doped porous carbon	NPC/MA-SA	0.37	217.65%	[40]
	PA	0.28	Nitrogen doped graphene	PA/NDG	1.73	617%	[41]
Graft functional groups	Paraffin	0.28	Mesoporous carbons	Paraffin/C-900-85	0.35	125%	[43]
	PEG	0.298	CNTs grafted MOFs	(PEG)/CNT@ MOFs	0.464	200.9%	[44]
	SA	0.18	MIL-101-NH ₂	SA/MOFs	0.51	283.3%	[45]
	Paraffin	0.2312	Grafted CNTs-C18	Paraffin/CNTs	0.7903	341.68%	[46]
	n-octadecane	0.19	Octadecyl amine- grafted graphene oxide (GO-ODA)	n-octadecane- melamine formaldehyde (MF) resin/GO-ODA	0.26	136.84%	[47]

macro-porous template to prepare highly interconnected 3D graphite foam composite matrix. The graphite foam matrix was impregnated with erythritol to prepare a composite PCMs. Fig. 16 shows the preparation of GF/erythritol composite PCMs and SEM morphological characteristics diagram. It is found that the thermal conductivity of erythritol/GF composite PCMs (3.77 W/(m·K)) is five times higher than that of pure erythritol (0.72 W/(m·K)). Morphologies of the samples are shown in Fig. 16. It is observed that due to the capillary force and surface tension of the porous GF, the pores of the GF are filled with erythritol. The developed high thermal conductivity porous matrix of GF easily forms dense thermal conduction channels in the composite PCMs. Further, accelerate the phonon group transmission velocity and improve the thermal conductivity of composite PCMs.

Zhang et al. [55] grew stable 3D diamond foam (DF) network structure on diamond-like foam by hot filament assisted chemical vapor deposition (CVD) technology and then added it to paraffin by vacuum impregnation method to prepare high quality DF/paraffin composite PCMs. Fig. 17 shows the morphological characteristics of DF schematic diagram of a DF consisting of a 3D interconnected diamond network and a continuous DF with high thermal conductivity. It can be observed that the 3D macro-porous skeleton of high thermal conductivity DF is uniformly compact and interconnected. In addition, the foam matrix is completely covered by continuous diamond film, ensuring efficient and continuous heat transfer. When the volume fraction of DF is very low (1.3 vol%), the thermal conductivity of DF/paraffin (6.7 W/(m·K)) is 162% higher than that of carbon fiber (CF)/paraffin (2.56 W/(m·K)) and 2580% higher than that of pure paraffin (0.25 W/(m·K)). When the thermal conductivity of diamond particles (1800 W/(m·K)) is large, even if the volume fraction of diamond particles (20 vol%) is increased, the thermal conductivity of diamond particles/paraffin is only 0.45 W/(m·K), far lower than that of CF/paraffin, which is sufficiently to demonstrate that the 3D continuous structure can promote heat transfer. It is observed that abundant rhombohedral interconnection network can provide a faster thermal conduction channel, which can significantly reduce phonon-phonon and phonon-boundary scattering. The rhomboid structure can be used to reduce the interfacial phonon scattering and improve the effective thermal conductivity. In addition to the preparation of graphite foam composite PCMs, Mesalhy et al. [56] also conducted experiments and simulations on the thermal storage performance of the materials with different porosity in the system. In this way, the best porosity of graphite foam can be provided reference for practical application.

3.2.1.2. Addition of highly graphitized network carbon in passive PCM. Highly graphitized 3D network carbon is a promising and candidate matrices for possessing high thermal conductivity, which can

provide superior interconnected paths for phonon propagation. Therefore, adding highly graphitized 3D network carbon to the PCM can effectively improve the thermal conductivity of the composite PCMs.

Chen et al. [57] closely connected the carbon quantum dots (CQDs) of acetone and divinylbenzene (DVB) by one-step calcination method, and under the synergism of appropriate pyrolysis temperature and cross-linking reaction, a controllable 3D graphitized mesoporous and macro-porous heat conducting network carbon 3D celosia-like were constructed. PEG/3D porous carbon composite PCMs were obtained by using melt impregnation method to infiltrate PEG into 3D graphitized network carbon pores. The preparation process and micromorphology are shown in Fig. 18. The porous carbon material can significantly reduce the binding force of PEG penetration into microporous and mesoporous scaffolds. Compared with pure PEG (0.28 W/m·K), the thermal conductivity of composite PCMs (0.94 W/m·K) increased by 236% and the phase change enthalpy is almost close to the theoretical value. In addition, the composite PCMs with the addition of heat-conducting network carbon has excellent thermal stability. Among them, the improvement of the thermal conductivity of PEG/3D porous carbon composite PCMs is mainly due to the increase of thermal conductivity channel in the 3D carbon skeleton and the increase of the phonon mean free path, which is conducive to the enhancement of its thermal conductivity.

Li et al. [58] synthesized CNT-porous carbon (PC) 3D porous core support structure by high temperature carbonization method using CNT and ZIF-8 MOFs. This material has high porosity and large specific surface area. SA/CNTs/PC composite PCMs were prepared by melting impregnation method under the action of capillary force and fixed in the pores of PC wall. The preparation process and micro-morphology are shown in Fig. 19. The close combination of SA and CNTs can significantly reduce the interface thermal resistance. It can provide a continuous channel for phonon transfer and realize the rapid heat transfer in the composites. In addition, SA/CNT-PC form stable composite PCMs have good thermal conductivity (1.023 W/m·K). Compared with SA/PC composite PCMs (0.310 W/m·K), the thermal conductivity of SA/CNT-PC composite PCMs increased by 222.6%. It can be concluded that 3D PC is an additive to improve the thermal conductivity of PCM.

Kholmanov et al. [59] continued to study the microstructure and thermal conductivity of highly carbonized porous network structures. UGF-CNTs composites were synthesized by using continuous ultrathin graphite foams (UGFs) and CNT as raw materials through nickel-based template-assisted CVD method. Erythritol/UGF-CNTs composite PCMs were prepared by adsorption of erythritol to the 3D network structure of porous UGF-CNTs by melting impregnation method. The preparation process and the morphology of the porous material are shown in Fig. 20. They found that the thermal conductivity of UGF-CNTs/erythritol composites was 1.8 times higher than that of UGF/erythritol composites.

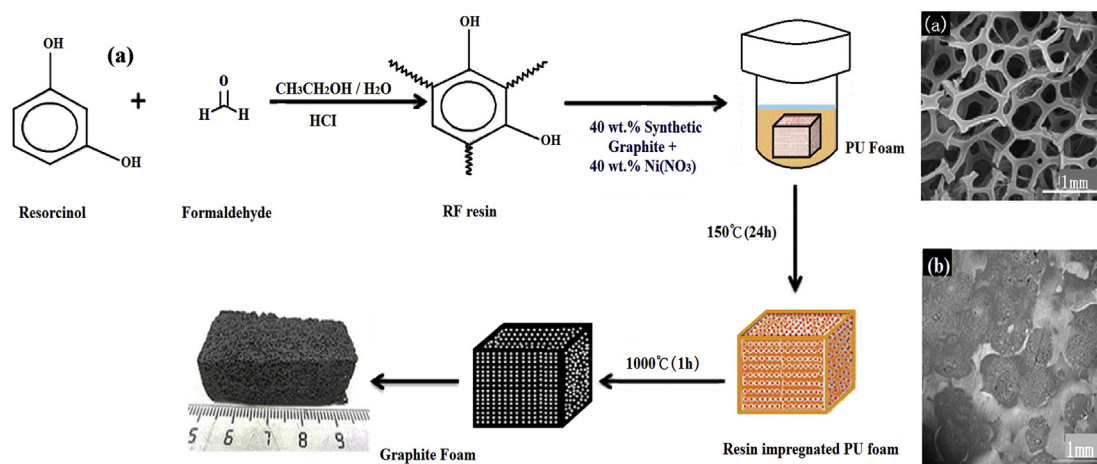


Fig. 16. Preparation of graphite foam/erythritol composite PCMs and SEM microstructure diagram.

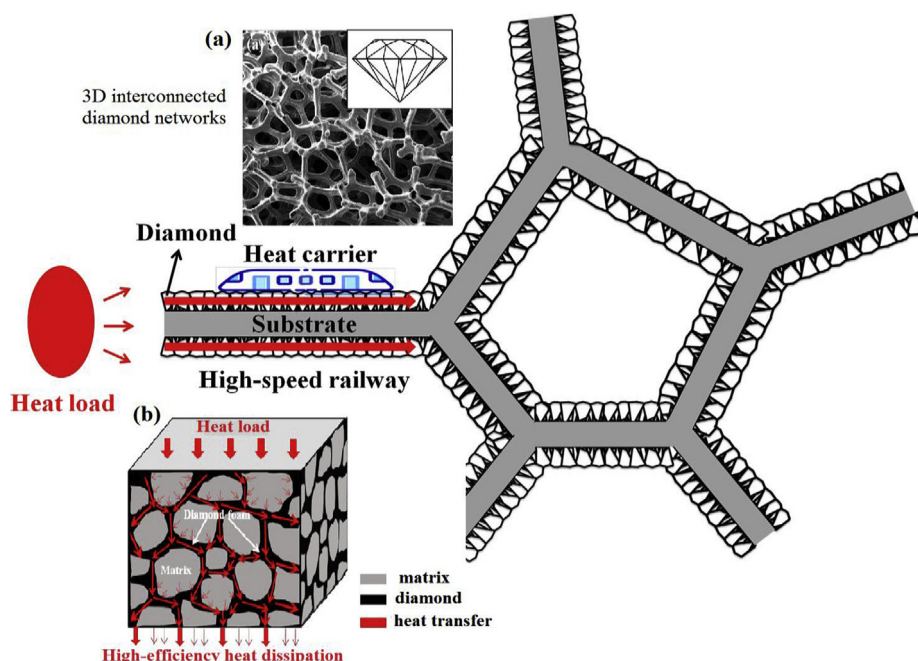


Fig. 17. SEM diagram shows the diamond foam schematic diagram of a diamond foam consisting of a 3D interconnected diamond network and a continuous diamond foam with high thermal conductivity.

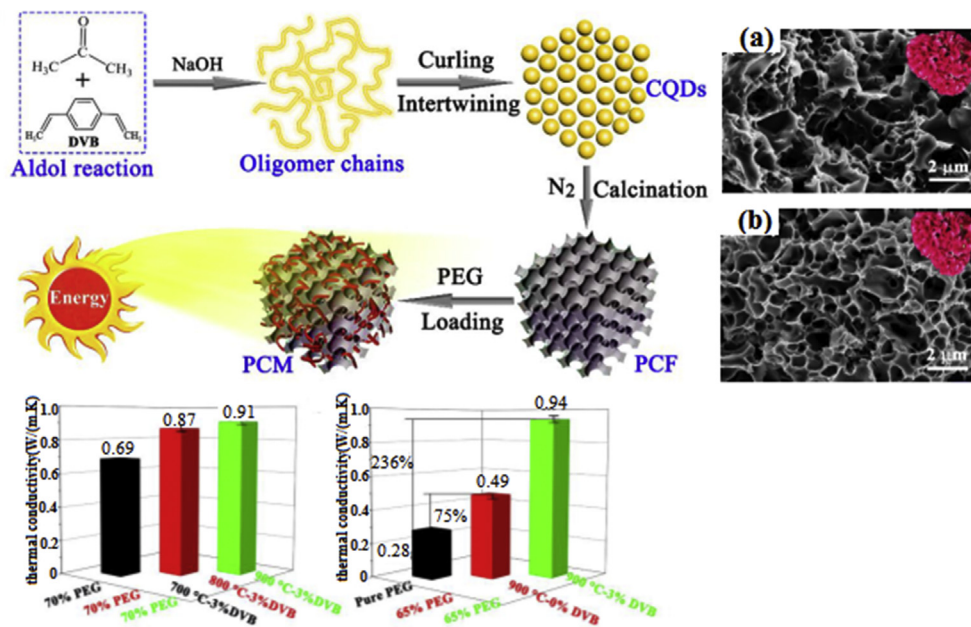


Fig. 18. Schematic illustration and thermal conductivity of PEG8000/CQDs-derived 3D porous carbon composite PCMs.

However, the effective thermal conductivity of the UGF-CNTs porous structure composite synthesized by the nickel-based template-assisted CVD method increased from 2.9 to 0.2 W/(m·K) to 4.1 and 0.3 W/(m·K), compared with the simple mixture with the same UGF and CNT addition amounts of 1.8 and 0.8 wt%, respectively. It indicates that the 3D porous network additive formed by the bond between UGF and CNT can effectively improve the thermal conductivity of PCM. The reason may be that the thermal conduction speed is significantly improved by expanding the thermal conduction channel.

Yang et al. [60] synthesized porous cellulose/GNP aerogel with high thermal conductivity by using graphene nanoplatelets (GNPs) and microcrystalline cellulose through typical Sol-gel method, and encapsulated PEG into 3D network structure of porous cellulose/GNP aerogel by

vacuum-assisted impregnation method, which can effectively prevent liquid PEG leakage. When GNPs content is 5.3 wt%, the thermal conductivity of the composites is 1.35 W/(m·K), which is 463% higher than the composites without GNPs (0.24 W/(m·K)). The addition of high thermal conductivity matrix is the main reason for the improvement of thermal conductivity of composite PCMs. However, after the rapid heat treatment, the graphite oxide still has oxygen components and structural defects, which are not conducive to phonon transmission. But the high thermal conductivity of PEG/GNPs composites should be attributed to the high intrinsic thermal conductivity of defect-free GNPs. Compared with the original graphite sheet, the annealed GNPs are more effective in improving the thermal conductivity of the composites, because they are more likely to form a conductive network conducive to thermal

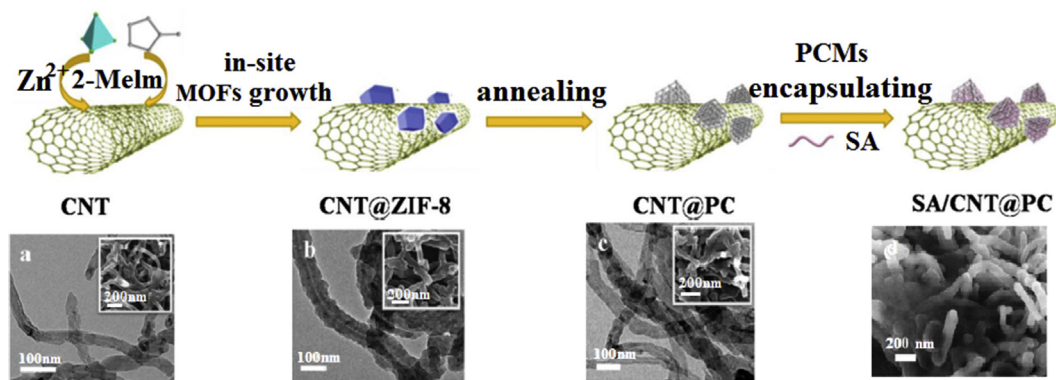


Fig. 19. Schematic illustration for the synthesis of SA/CNT@PC composite PCMs and the TEM and SEM images of (a) CNTs; (b) CNT@ZIF-8; (c) CNT/PC; (d) SA/CNT@PC.

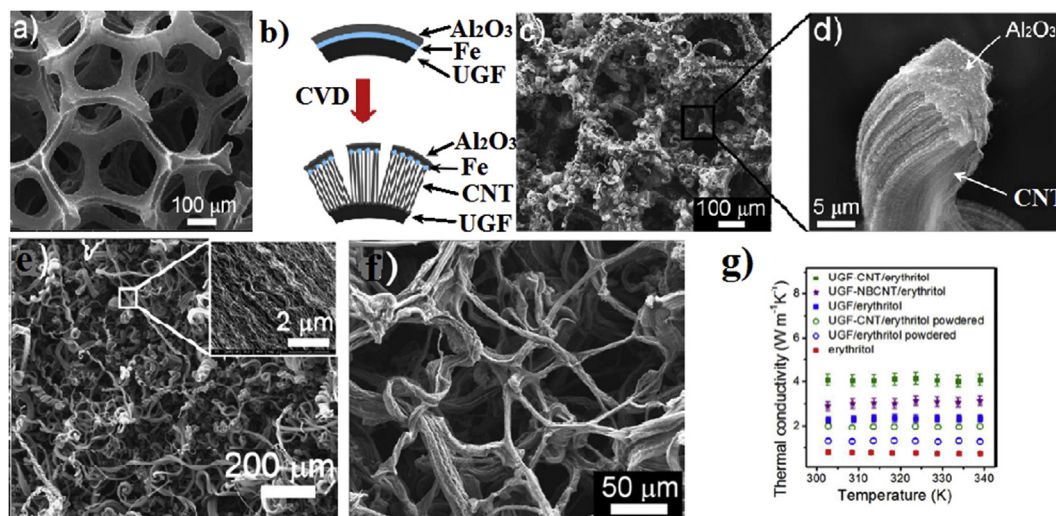


Fig. 20. SEM image of neat UGF, UGF-CNT hybrid structure, CNT tip covered by an Al_2O_3 layer, UGF-CNT hybrid foam, CNT networks formed inside the UGF pore and schematic of CNT growth on the UGF, and the thermal conductivity data of composite materials.

conductivity, thereby reducing phonon scattering.

3.2.1.3. Addition of metal foam in passive PCM. Due to its large specific surface area per unit volume, high thermal conductivity inherent in the solid skeleton, and strong mixed disturbance characteristics of heat flow in the curved channel, the metal foam ligament destroys the heat flow boundary layer, thereby increasing the axial and radial heat transfer of the heat transfer skeleton and further improving the heat transfer rate. Adding PCM to the continuous skeleton structure and high porosity void inside the metal foam can significantly improve the thermal conductivity of the PCM under the condition of slightly reducing the heat storage capacity of the composite PCMs. At present, the most widely studied metal foams include copper foam [61,62], aluminum foam [63], nickel foam [64–67] and titanium dioxide (TiO_2) foam [68]. Fig. 21 shows the structure of copper foam (a), aluminum foam (b) nickel foam (c) [62] and titanium dioxide foam. The effective thermal conductivity of composite PCMs can be improved by adsorbing the composites into the pores of metal foam by vacuum impregnation method.

3.2.1.3.1. Passive PCM embedded with copper foam. It is well known that copper foam has porous structure, which possess larger surface area to volume ratio and they have a strong mixing ability. Moreover, the thermal conductivity of copper skeleton is high ($401 W/(m \cdot K)$) [69] and has enough more the number of thermal conductivity chains, which is conducive to the rapid transmission of heat flow. Hence, lots of metal foam are as well commonly used as matrices for thermal conductivity enhancement of thermal energy storage system. Researchers often add

copper foam with different porosity and pore size in PCMs to improve the heat transfer rate of inorganic and organic PCMs, and continue to explore excellent properties of metal foam based composite PCMs.

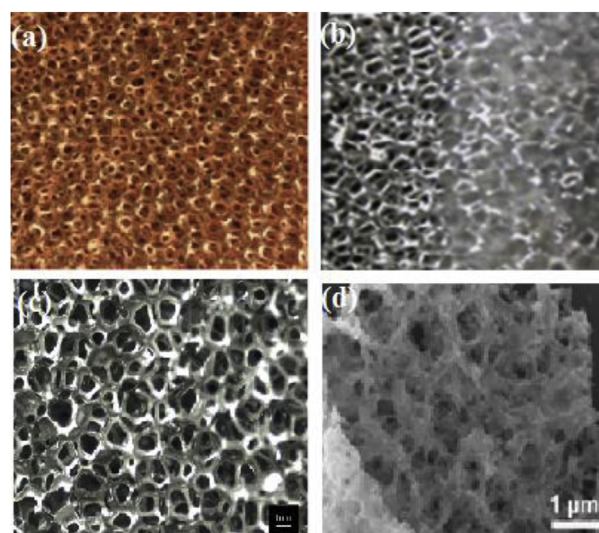


Fig. 21. Copper foam (a), aluminum foam (b), nickel foam (c) and titanium dioxide foam (d).

Compared with pure copper, copper foam with a cellular structure has better load, adsorption capacity and thermal conductivity. Using inorganic PCM and copper foam to synthesize composites can significantly improve its heat transfer performance. Such as Li et al. [70] first added 2.0 wt% disodium hydrogen phosphate hydrate (DHPD) as nucleating agent and 0.5 wt% carboxymethyl cellulose (CMC) as thickening agent to the sodium acetate trihydrate (SAT). It is found that the thermal conductivity of SAT/copper foam composite PCMs could be significantly increased with the decrease of the porosity of copper foam. This indicates that the copper foam additive with low porosity is more beneficial to improve the thermal conductivity of composite PCMs. The pore size of copper foam ranged from 5 to 25 PPI, and the porosity ranged from 92.4% to 97.3%. The thermal conductivity of copper foam/SAT composite PCMs ranged from 6.8 to 3.3 W/(m·K). When the porosity of the copper foam matrix is 92.4%, the thermal conductivity (6.8 W/(m·K)) is approximately 11 times that of the pure SAT thermal conductivity (0.59 W/(m·K)). It is possible that the copper foam formed many thermal conduction channels in the composite PCMs, which accelerated the heat transfer rate. However, there are also study on the preparation of composites with organic PCM and copper foam. For instance, Chen et al. [71] proved that copper foam has ability to enhance the heat transfer rate of paraffin through experimental research. They are adsorbed paraffin into the network structure of powders of styrene-*b*-(ethylene-isobutylene)-*b*-styrene (SEBS) triblock copolymer by direct impregnation method. On the basis, it mixed with high-density polyethylene (HDPE) material uniformly and copper foam to prepared SEBS/paraffin/HDPE/copper foam form-stable composite PCMs. The thermal conductivity of SEBS/paraffin/HDPE (1:6:0.5) is 0.272 W/(m·K) while the thermal conductivity of SEBS/paraffin/HDPE/copper foam form-stable composite PCMs are increased to 2.142 W/(m·K). The thermal conductivity of the form-stable composite PCMs calculated by the empirical formula is 2.612 W/(m·K). On the one hand, due to the low thermal conductivity of SEBS/paraffin/HDPE (1:6:0.5), the thin layer adsorbed by the composite onto the surface of the copper foam will increase the interface thermal resistance and reduce the

overall thermal conductivity. On the other hand, the accuracy of the empirical formula is mainly verified by the experiment of heat transfer between water or air and metal foam. The natural convection of the composite PCMs are relatively weak, which is promote the heat transfer inside the composite PCMs. In a word, copper foam can obviously promote the thermal conductivity of PCM and it is an excellent thermal conductivity additive matrix.

3.2.1.3.2. Passive PCM embedded with aluminum foam. Aluminum foam can be divided into two types: open-hole aluminum foam and closed-hole aluminum foam [72–74]. Closed-hole aluminum foam is commonly used as insulation material, but open-hole aluminum foam is often used for enhanced heat transfer. As a kind of ultra-light porous metal material, open-hole metal foam is often used as a material to enhance heat transfer due to its small density, large surface area, good flow performance and high thermal conductivity of metal skeleton (167 W/(m·K)) [75]. Filling metal foam as a skeleton into PCM can improve the heat transfer performance of the phase change heat transfer device [76,77]. In addition, the thermal conductivity of composites is also closely related to the impregnation rate, pore size and porosity etc.

Jiang et al. [63] used aluminum foam to enhance the heat transfer performance of PCM. They took slice paraffin and SA as the PCM, which adsorbed the PCM into the bulk porous aluminum foam under different pressures by directly impregnate method, to prepare two kinds of parameter composite PCMs. Furthermore, they studied the microstructure of the two composite PCMs and thermal performance. Fig. 22 shows the optical micrograph and SEM micrograph of SA/aluminum foam (a) and paraffin/aluminum foam (b) composite PCMs under normal pressure (c, d). The black color is the PCM area and the white color is the aluminum foam area in the figure. There is no crack in the interface, which is because of the high permeability of low temperature organic PCM, van der Waals force of the metal foam, and the strong cohesion of organic and metal foam, so that the composites have good impregnation effect. It can be seen from the SEM micrograph that PCM has a good impregnation effect in aluminum foam. For PCM/aluminum foam

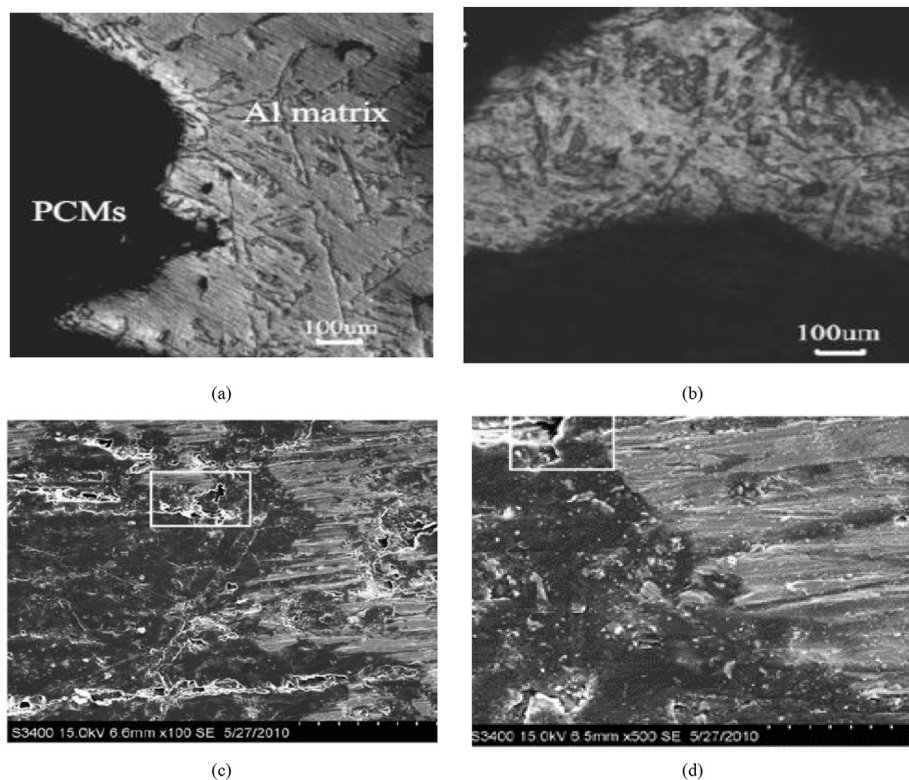


Fig. 22. Optical micrographs of SA/Al foam (a, c) and paraffin/Al foam (b, d) composite PCMs prepared by embedding method and constant pressure impregnation method.

composites, the impregnation rate of PCM in the pores of aluminum foam is more than 80%, and the number of remaining pores is small. The lower residual porosity in the paraffin/aluminum foam composites indicate that the impregnation effect of paraffin is better than that of SA. The impregnation of PCM improved the thermal performance and made its ideal energy saving materials.

In order to research the effects of different pore sizes and porosity on the thermal conductivity of composites, Hu et al. [78] continued to study the thermal conductivity enhancement of metal foam/paraffin by numerical simulation and theoretical prediction. They filled paraffin into the pores of honeycomb cubic aluminum foam with symmetrical structure, and used pore-scale numerical simulation to study the thermal properties of porosity and pore size on composite PCMs. The simulation results showed that the melting time of composite PCMs are 75.9% shorter than that of pure paraffin. When the pore size is 10 PPI, the effective thermal conductivity of PCM with a porosity of 95% and 70% is 3.3 W/(m·K) and 24.6 W/(m·K), respectively, which is 17 times and 123 times higher than that of pure paraffin (0.287 W/(m·K)). The possible reason is that the smaller porosity can coarsening the heat conducting aluminum skeleton and widen the heat flow channel, thus it can enhance thermal conductivity. Meanwhile, the simulation results are compared with the results of the thermal conductivity prediction model established by Xu et al. [79] and Bhattacharya et al. [80], as shown in Fig. 23. With the increase of porosity, the effective thermal conductivity gradually decreases, while the effective thermal conductivity of composite PCMs with different pore sizes of the same porosity (such as 5 PPI, 10 PPI) is roughly the same, indicating that the effective thermal conductivity is basically not affected by the pore size. In short, the thermal conductivity of composite PCMs is more affected by porosity than pore sizes. Therefore, the pivotal step to enhance the thermal conductivity is to select the appropriate porosity of metal foam.

Further, for studying the effect of porosity and phase change behavior, which are involve in improving thermal conductivity of metal foam/paraffin composite PCMs. Atal et al. [81] utilized two different porosity aluminum foam in a shell and tube heat storage unit to study the thermal conductivity of pure paraffin (0.25 W/(m·K)), paraffin/aluminum foam composite PCMs and pure aluminum foam (167 W/(m·K)) configurations. Because of the high thermal conductivity of aluminum foam, the addition of high thermal conductivity aluminum foam to paraffin can significantly improve the heat transfer performance of paraffin. When aluminum foam with equivalent porosity of 95% and 77% is added to the paraffin, the effective thermal conductivity of the composite PCMs are 4.192 W/(m·K) and 17.29 W/(m·K), respectively. It can be clearly observed that paraffin/aluminum foam composite PCMs with small porosity have relatively high thermal conductivity, and the heat storage and release time of shell and tube heat storage device is shorter. It

is assumed that the heat transfer rate can be significantly increased by applying aluminum foam to the inner wall of the shell and tube of the heat storage device, because the thermal resistance between the tube wall and the aluminum foam is greatly reduced. From the above, it is also a good way to investigate the thermal conductivity of metal foam through heat storage experiment, which can more directly explore the effect of porosity on heat exchanger heat transfer performance.

3.2.1.3.3. Passive PCM embedded with nickel foam. Nickel foam is a kind of metal foam with homogeneous 3D network structure, its porosity is between 88% and 99%, and its specific surface area can reach 1000–9000 cm²/cm³, which is conducive to improve its stability and heat flux density [82]. In order to give full play to the high thermal conductivity of open-hole nickel foam network and overcome the instability of the function caused by the binder, functional materials can be grown directly on the nickel foam. The preparation of nickel foam based composite PCMs using nickel foam as the framework can reduce the contact thermal resistance between the PCM and nickel foam skeleton and increase the heat transfer capacity between the interfaces [65]. Furthermore, the optimal additives will be selected to compare their thermal conductivity by adding different single-component foams or bimetallic foams into PCM.

According to the specific needs of practical applications, suitable pore size of nickel foam can be selected. Oya et al. [67] prepared composite PCMs by vacuum impregnation method with erythritol and porous nickel with pore sizes of 100, 300 and 500 μ m. Fig. 24 shows the morphological characteristics of nickel foam with pore sizes of 100, 300 and 500 μ m and thermal conductivity curve. It is observed that with the decrease of pore size, the damage degree of porous structure to the skeleton increases, the skeleton becomes thinner and the connectivity decreases. Furthermore, they have tested the thermal conductivity of composite PCMs. The thermal conductivity of nickel foam/erythritol composite PCMs with the nickel foam pore size of 100, 300 and 500 μ m are 6.62, 9.36, 11.6 W/(m·K), respectively. The results show that, the thermal conductivity of composite PCMs rises with the increase of the aperture. The biggest thermal conductivity of composite PCMs are 11.6 W/(m·K), which is approximately 15 times larger than that of pure erythritol (0.733 W/(m·K)). The microstructure can be concluded that the effective thermal conductivity of the nickel foam/erythritol composite PCMs with small pore size is reduced due to the narrow heat conduction channel caused by the thin framework connecting rod structure.

For the above research, the composite PCMs with nickel foam matrices to enhance thermal conductivity is effective. Liang et al. [83] continues to synthesize polydimethylsiloxane graphene-nickel foam (PDMS-G-NF) by coating graphene nanosheets onto nickel foam followed via surface modification with PDMS, is used as porous supporting material to prepare form-stabilized composite PCMs. SEM photos is

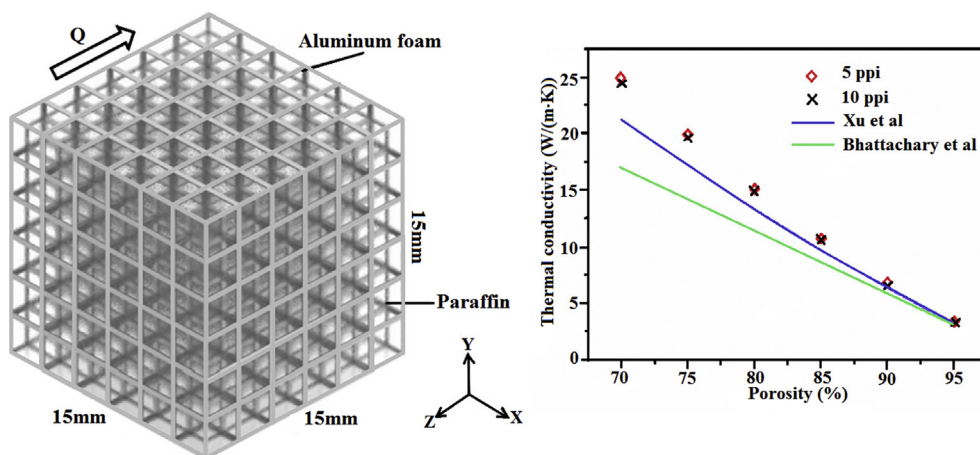


Fig. 23. Composite PCMs numerical model and the diagram of effective thermal conductivity changing with porosity.

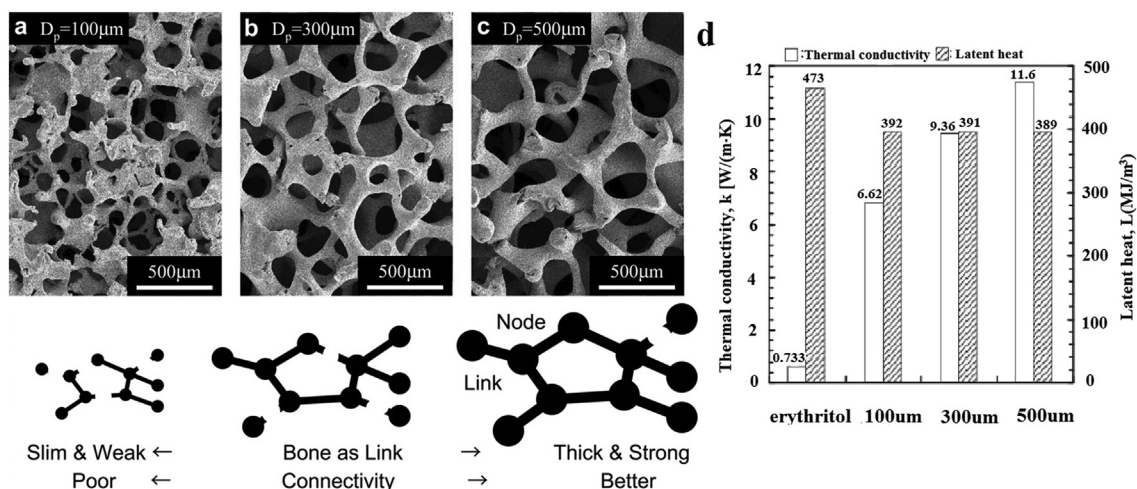


Fig. 24. SEM images of porous nickel with 100, 300, 500 μm in pore size, together with schematic bone structure and the effective thermal conductivity and volume heat storage density of each composite PCMs.

performed to evaluate the morphology of composite materials. It can be seen that the treated nickel foam exhibited nano/micro-size along with rough surface composed of stacked layers of graphene sheets. The higher magnification image reveals more clearly the graphene nanosheets with wrinkled surface topology and folding edges on the skeleton of nickel foam, whose intrinsic hydrophobicity is favorable to the absorption of PCM. Fig. 25 shows the morphological characteristics of the G-NF and the pores of PA/PDMS-G-NF are homogeneously filled by PA, which illustrates that n-carboxylic acids have been impregnated into PDMS-G-NF due to the effect of superoleophilicity and the surface absorption of graphene. Since the pores in the PDMS-G-NF supporting materials are filled by the PA, the thermal conductivity of the composites ($2.262 \text{ W}/(\text{m}\cdot\text{K})$) at PA loading fraction of 59.02 wt% increased about 14 times as compared with that of the PA ($0.162 \text{ W}/(\text{m}\cdot\text{K})$). Due to the porous property of PDMS-G-NF, heat can be transferred rapidly in the

pore skeleton, which can further increase the phonon mean free path, weaken the heat flow resistance and significantly enhance its thermal conductivity.

The comparison of the promoting effect of copper foam and nickel foam on the thermal conductivity of paraffin has practical application value. Xiao et al. [64] also used vacuum impregnation method to prepare different metals foam to investigate their thermal conductivity. They have prepared paraffin/nickel foam and paraffin/copper foam composite PCMs. Further, the impregnation rate and thermal conductivity of paraffin impregnated in vacuum are compared. Preparation and morphological characteristics of paraffin/metal foams composite PCMs are shown in Fig. 26. The surface porosity obtained by image analysis is between 90%–94%, while the volume porosity predicted by mass fraction is about 97%. Compared with pure paraffin wax, the thermal conductivity of the composite PCMs are significantly improved, and the

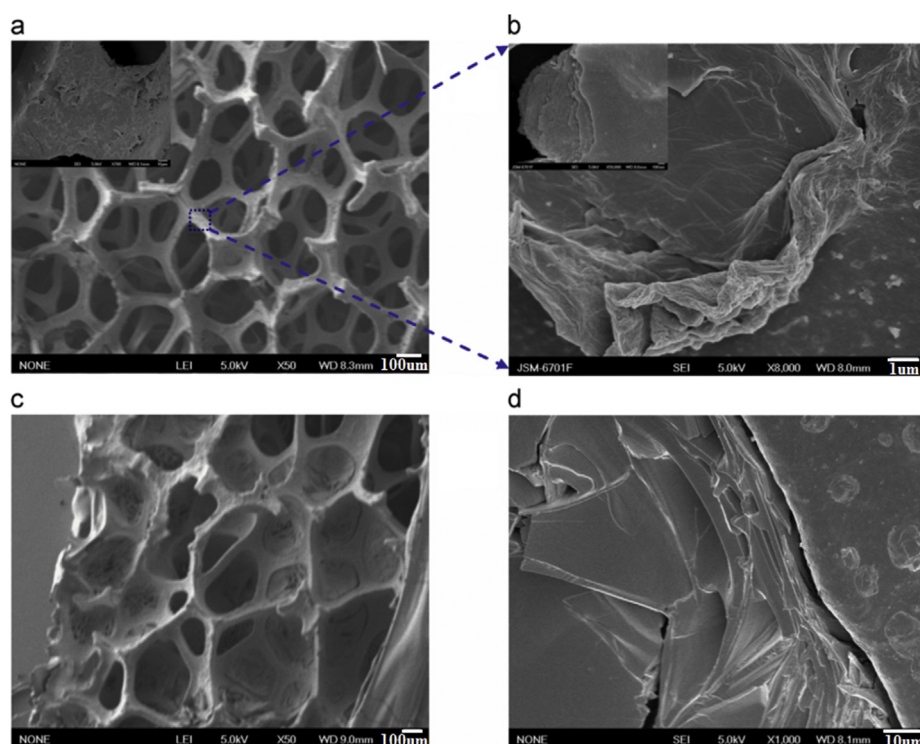


Fig. 25. SEM images of (a) G-NF, (b) graphene films on the surface of G-NF, (c) PA-PDMS-G-NF and (d) structure of pores of PA-PDMS-G-NF.

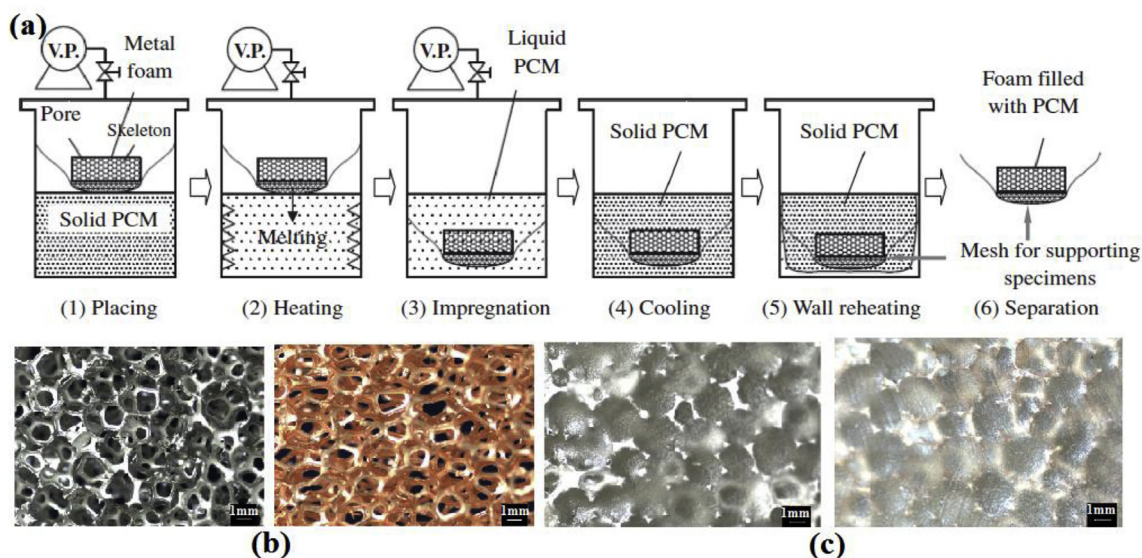


Fig. 26. Preparation of paraffin/Al foam and paraffin/Cu foam composite PCMs (a) as well as images of metal foam (b) and paraffin/metal foam composite PCMs (c).

thermal conductivity of paraffin/nickel foam composite PCMs are almost three times that of pure paraffin. On the other hand, the comparison of the promoting effect of copper foam and nickel foam on the heat conduction of paraffin shows more obviously improved effect of copper foam and more conducive to the thermal conductivity additive.

Huang et al. [84] also compared and analyzed the thermal properties of the copper foam and nickel foam-based composites. They selected myristyl alcohol (MA) as the PCM and metal foam as the supporting framework to prepare MA/metal foam composites by vacuum melting infiltration method. Fig. 27 shows the microstructure of MA/metal foam composite PCMs before and after the preparation of nickel foam and copper foam with pore sizes of 40, 70 and 90 PPI. As the large number of metal foam skeletons support MA, it can be completely wrapped by the metal foam and the surface of the composite PCMs are more smoothly. Furthermore, the thermal conductivity of the nickel foam and copper foam composite PCMs with three pore sizes are tested. The results show that the thermal conductivity of the MA/nickel foam composite PCMs (40 PPI) are 0.4768 W/(m·K), which is 1.80 times higher than that of pure MA (0.17 W/(m·K)). Meanwhile, the thermal conductivity of MA/copper foam composite (40 PPI) is 1.452 W/(m·K), which increased by 7.51 times. Because of the larger the hole, the lower the thermal conductivity. Further, the larger the pore size of metal foam results the larger the interface area, the higher the thermal resistance and the lower the thermal conductivity. The thermal conductivity of MA/copper foam composite PCMs with the same pore size is higher than MA/nickel foam composite PCMs. In addition, with the increase of the density of MA/metal foam composite PCMs, the thermal conductivity significantly increased, and the heat transfer rate of the heat storage unit per unit volume accelerated. Moreover, within a reasonable range, increase the mass fraction of metal foam and improve the thermal conductivity of composite PCMs.

3.2.1.3.4. Passive PCM embedded with titanium dioxide foam. In addition to the three typical metal foams mentioned above, TiO₂ foam has attracted attention as a supporting matrix for PCMs because of their advantages of environmentally friendly, low cost, fireproof material with high thermal conductivity. For a PCM confined in a porous structure, the interfacial interactions between the PCM and the porous skeleton are the decisive factors in latent heat storage performance. Thereby, it is inevitable to explore the preparation of TiO₂ foam and composite PCMs, and to test thermal and mechanical properties of pore skeleton.

It is well known that metal foams have high thermal conductivity, which is suitable to enhance heat transfer rate of composite PCMs. Hence,

how to prepare porous TiO₂ foam stably and efficiently is extremely important. Yu et al. [85] used biopolymer alginate as the sacrifice template for TiO₂ nucleation template and promoted the formation of porous TiO₂ nanostructure, to synthesize layered porous TiO₂ foam by soft template method. Its microscopic morphology can be obtained that TiO₂ material has a spongy hierarchical mesoporous structure with a wide distribution of pore diameter. Open skeleton, which be generated TiO₂ nanoparticles with uniform morphology and size, are closely connected and agglomerated to form macro-porous foamed materials. Because this method has the advantages of uniform pores and stable skeleton, it is used to synthesize porous matrix in composite PCMs. Li et al. [86] synthesized layered porous TiO₂ foam by soft template method and prepared *n*-octadecane/TiO₂ foam composite PCMs by introducing *n*-octadecane (OC) under vacuum conditions. Fig. 28 shows the preparation process (a), microstructure (b), porous structure (c) and thermal conductivities (d) of porous TiO₂ foam and composite PCMs. The carrier is TiO₂ crystal and the product has layered porous structure. Due to the effect of capillary force and interfacial tension, the nanopores are wrapped with abundant OC, which is closely bound to the surface of layered porous TiO₂ foam. Furthermore, when OC content is 50 wt%, the thermal conductivity of the composite is up to 0.38 W/(m·K) with the addition of porous TiO₂ foam with high thermal conductivity. The thermal conductivity of the composite is 138% higher than that of pure OC (≈ 0.16 W/(m·K)). The results show that the thermal conductivity of OC can be effectively improved by adding porous TiO₂ foam as substrate. Deng et al. [87] continued to prepare a flower-like TiO₂ nano-structure to support PEG. Schematic diagram of the synthesis process and micro-structure of FLN-TiO₂ (a) and preparation process of PEG/FLN-TiO₂ composite PCMs (b) are shown in Fig. 29. In this encapsulating PCM, flower-like TiO₂ is not only adding matrices to improvement thermal conductivity but also supporting materials to prevent PCM leakage. Further, the phase change temperatures of the composite PCMs shifted slightly as a result of the physical force between the PEG and the TiO₂ nanostructure. Li et al. [88] synthesized porous TiO₂ foam (PTF) by particle-stabilized emulsion method to prepare paraffin/PTF composite PCMs by vacuum impregnated method. SEM images show that the prepared PTF have 3D interpenetrating structure and uniform pore. But solid paraffin has obvious influence on pore structure and morphology of PTF. Fig. 30 shows the schematic diagram of the preparation process, micro-structure (a) and thermal conductivity (b) of porous TiO₂ foams and composite PCMs by synthesized and prepared method. With the increase of paraffin content, the porosity of samples shrinks, and the

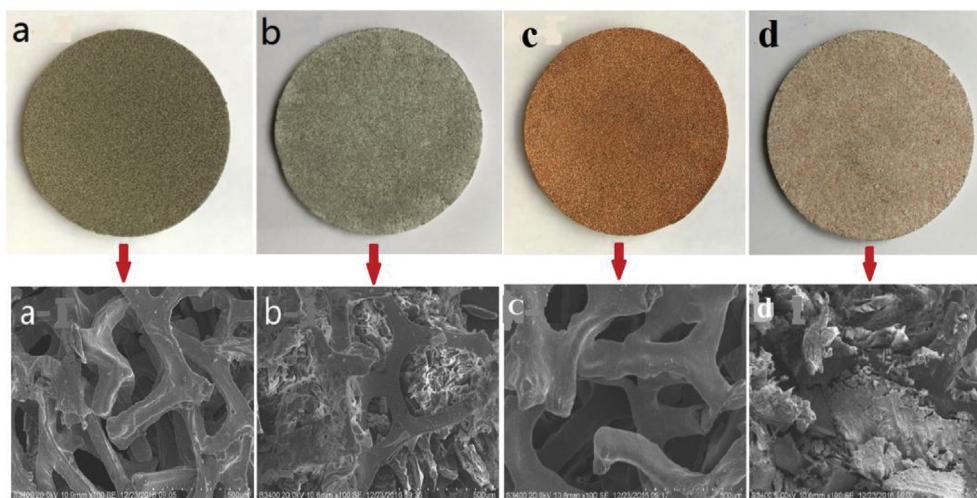


Fig. 27. SEM photographs and real products of nickel foam (a, b) and copper foam (c, d) before and after preparation with pore size 40 PPI.

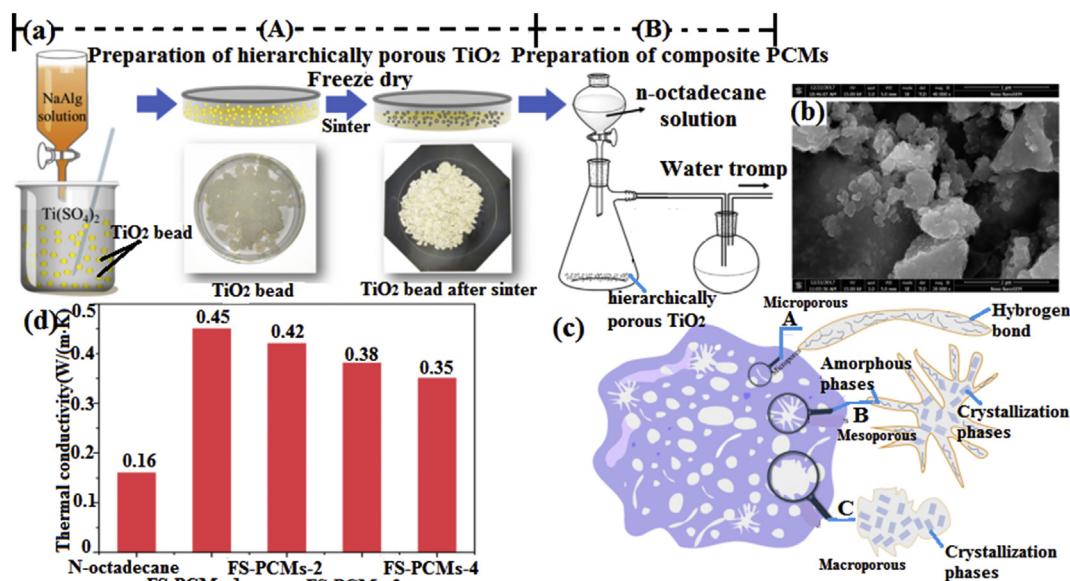


Fig. 28. Preparation process (a), microstructure (b), porous structure (c) and thermal conductivities (d) of porous TiO₂ foam and composite PCMs.

corresponding embedding rate decreases. In order to further improve the thermal conductivity and adsorption capacity of PTF, sucrose is added into the emulsion and carbonized in situ to form carbon/PTF (PTFC). The paraffin is successfully impregnated into the porous framework to obtain a stable paraffin/PTFC (PTFCP) composite PCMs. As a result, the thermal conductivity of PTFCP composite PCM is 0.302 W/(m·K), which improved about 0.092 W/(m·K) compared with the pure paraffin (0.21 W/(m·K)) because of the high thermal conductivity of TiO₂. Further, the thermal conductivity of PTFCP composite PCM is 1.059 W/(m·K), 5.04 times of the pure paraffin. The results show that the high thermal conductivity carbon layer on the surface of TiO₂ particles can improve the thermal conductivity of composite PCMs to some extent and provide another solution for enhancing the thermal conductivity of PCM.

At the same time, Sol-gel method is another feasible method to prepare porous TiO₂ foam, which possess the advantages of molecular uniformity, low reaction temperature and fast synthesis. Kimling et al. [89] induced synthesis of nanometer porous TiO₂ microspheres using alginate as sacrificial template by Sol-gel method. The microtopography shows that the microsphere surface is highly porous and consists of interconnected fibers. With the increase of Ca²⁺ concentration and alginate concentration, a denser porous network structure can be observed,

resulting in the formation of TiO₂ foam. This method provides a new idea for the preparation of foam matrix of PCM. Li et al. [90] used amorphous TiO₂ and SA to synthesize SA/TiO₂ encapsulate PCM by Sol-gel method and continued to study the influence of the pH values during the synthesis process. At the same time, they also adopted CNTs into the composite PCMs to enhance the thermal conductivity. The results show that microcapsules are more easily formed under acidic than alkaline conditions. After adding CNTs, which are uniformly attached to the surface of microcapsules. Due to the good thermal conductivity of CNTs and abundant thermal channel, the heat transfer rate and thermal conductivity of composites can be improved. In a word, based on TiO₂ foam as matrices has high thermal conductivity, stable chemical and stable structure, it has important application value in preparing high performance composite PCMs.

All in all, compared with pure metal materials and carbon black, metal foam and graphite foam are an excellent additive in improving the thermal conductivity of PCM due to its porous structure, specific surface area and low density. Hence, it has a better prospect used as matrices to improve heat transfer rate of composite PCMs, which has a crucial function for the practical applications in a greater demand to the fast heat storage and release of system. Various 3D structure additives for thermal

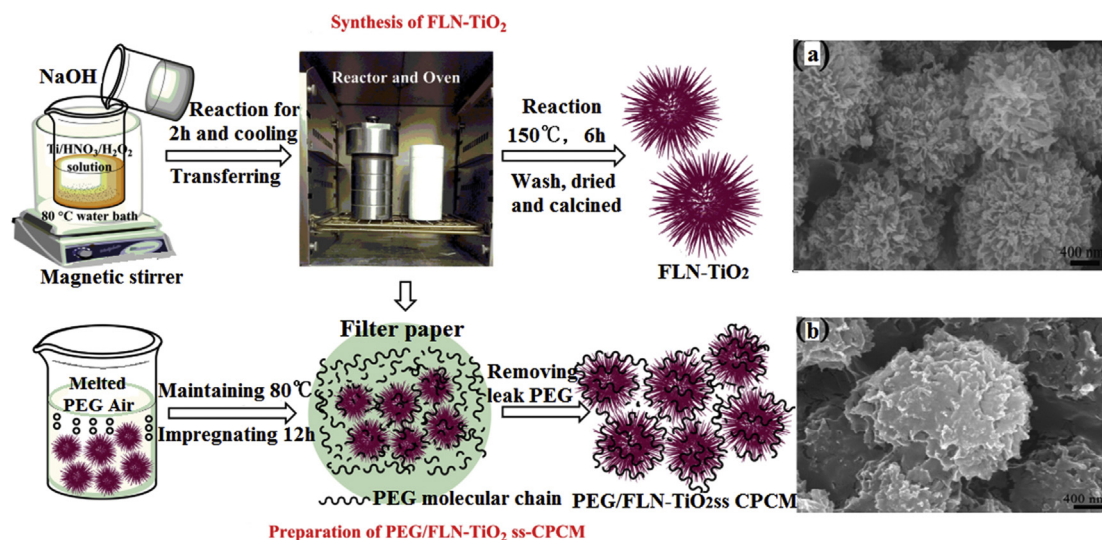


Fig. 29. Schematic diagram of the synthesis process and microstructure of FLN-TiO₂ (a) and preparation process of PEG/FLN-TiO₂ composite PCM (b).

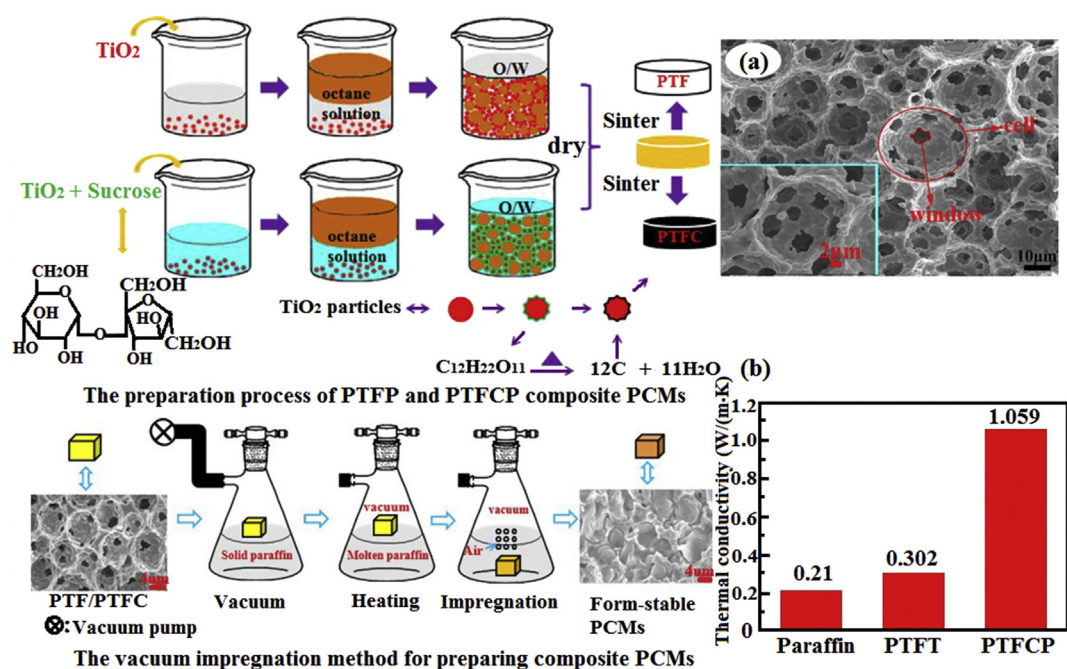


Fig. 30. Schematic diagram of the preparation process, microstructure (a) and thermal conductivity (b) of porous TiO₂ foams and composite PCMs by particle-stabilized emulsion method and the vacuum impregnation method.

properties of PCM/3D structure additives composites are listed in Table 2.

3.2.2. Passive composite PCMs using two-dimensional structure additives

Since the expanded graphite (EG) has a similar 2D lamellar structure as the clay mineral, it has good compatibility with the PCM and prevents its leakage property. EG has abundant network pore and large surface area, where is endowed with relatively more active sites, surface activities and lots of heat conduction chains, which can improve the thermal conductivity of the composite PCMs [91]. In addition, most clay minerals also have pore network structure, large surface area and relatively high thermal conductivity. Therefore, the passive PCM can be quickly adsorbed into the pore structure of clay minerals to form-stable composite PCMs. However, due to the small pore openings and narrow interval between layers of clay minerals, the PCM added are limited under certain circumstances. The interaction between PCM and clay minerals mainly

depends on hydrogen bond, capillary force, van der Waals force and surface tension. These interactions can limit the PCM in the molten state to the clay mineral structure and prevent their leakage. SiO₂ is used as the supporting material to add clay minerals to the PCM, which could effectively prevent PCM leakage due to its three-layer reticulated crystal structure [92–97]. Expanded perlite, sepiolite, attapulgite and expanded vermiculite have large specific surface areas [98–105]. The interlayer spacing of kaolin is small, making it difficult for the PCM to compound with kaolin [106]. Diatomite and bentonite have great porosity and pore space, and it can also effectively prevent the leakage of PCM [98,99,107, 108]. As clay minerals have good dispersion, they are not only widely used in form-stable PCM, but also used as common additives in micro-encapsulated composites to enhance their thermal conductivity. Plentiful matrices of 2D structure additives for thermal conductivity enhancement of metal foam/PCM composites are listed in Table 3.

Table 2

Comparison of thermal conductivity of passive composite PCMs formed by three-dimensional structure additives and preparation methods.

PCM type	Thermal conductivity of PCM k_p W/(m·K)	Three-dimensional structure additives	Preparation methods		Thermal conductivity of composite PCM k_c W/(m·K)	Fraction of matrices	Improvement (%)	Ref.
Paraffin wax and erythritol	0.2, 0.81	Ultrathin-graphite foams (UGFs)	Chemical vapor deposition (CVD)		2.30, 2.81	UGFs = 1.23 vol%	1150%, 346.9%	[49]
Paraffin	Solid: 0.21 Liquid: 0.12	Graphite foam matrix saturated	Molten embed		–	–	–	[51]
Paraffin	0.354	Carbon foam	Vacuum assistance impregnation method		1.198	Carbon foam = 13.93 wt%	338.4%	[52]
Erythritol	0.72	Graphite foam	Incipient wetness impregnation method		3.77	Graphite foam = 25 wt%	523.6%	[53]
Paraffin wax	Solid: 0.24, Liquid:0.11	Graphite foam	Impregnation method without any vacuum		Solid: 2.6, Liquid:1.8	Graphite foam = 22 wt%	Solid:1083%, Liquid:1636%	[54]
Paraffin	0.25	Three-dimensional (3D) diamond foam (DF)	Vacuum impregnation method		6.70	3D DF = 1.3 vol%	2680%	[55]
Poly-ethylene glycol (PEG)	0.28	3D network carbon	Physical mixing and impregnation method		0.94	Divinyl benzene = 3 wt%	336%	[57]
SA	0.341	CNTs/PC	Wet impregnation method		1.023	CNTs/PC = 25 wt%	300%	[58]
Erythritol	Solid: 0.78 Liquid: 0.326	UGFs	Ni template-assisted CVD method		4.09 ± 0.3	UGF = 1.8 wt% CNT = 0.8 wt%	Solid: 486%– 563%	[59]
PEG	0.24	Cellulose/GNP aerogel	Vacuum-assisted impregnation method		1.35	GNPs = 5.3 wt%	563%	[60]
PCM type	Thermal conductivity of PCM k_p W/(m·K)	Three-dimensional structure additives	Preparation methods	Porosity	Pore size	Thermal conductivity of composite PCM k_c (W/(m · K)) or shortened time (s)	Improvement (%)	Ref.
Paraffin	0.3	Copper foam	Molten embedding method	0.95	5 PPI	68s	120.5%	[62]
SA and sliced paraffin	0.24	Bulk porous Al foam	Vacuum impregnation method	65–70%	<1 mm	–	–	[63]
Paraffin	0.305	Nickel foam and copper foam	Vacuum impregnation method	Nickel foam: 97.17% copper foam: 96.58%	Nickel foam: 5 PPI copper foam: 10 PPI	Paraffin/nickel foam: 1.2; Paraffin/copper foam: 4.9	393.4%; 1606.6%	[64]
Paraffin	0.354	Copper foam and nickel foam	Vacuum impregnation method	Copper foam: 88.89% nickel foam: 90.61%	Copperfoam:25 PPI nickel foam: 25 PPI	Paraffin/copper foam: 16.01 Paraffin/nickel foam: 2.33	4522.6%; 658.2%	[65]
Erythritol	0.733	Spherical graphite (SG) and nickel particles	conventional dispersion technique	–	–	Erythritol/SG: 4.72	643.9%	[66]
Erythritol	0.733.	Porous nickel	Vacuum impregnation method	0.85	500 nm	Erythritol/porous nickel:11.6	1582.5%	[67]
Paraffin	0.302	Porous TiO ₂ foam (PTF)	No surfactant impregnation method	0.88	30 nm	Paraffin/PTF-carbon: 1.059	350.7%	[68]
Paraffin	Solid:0.3, Liquid:0.1	Copper foam	Vacuum impregnation method	0.97	25 PPI	–	–	[69]
Sodium acetate trihydrate (SAT)	0.59	Copper foam	Vacuum impregnation method	0.924	15 PPI	Modified SAT/copper foam: 6.8	1154.5%	[70]
SEBS/paraffin/HDPE	0.272	Copper foam	Direct impregnation method	0.98	20 PPI	SEBS-paraffin-HDPE/copper foam: 2.142	787.5%	[71]
n-eicosane	Solid:0.423, Liquid:0.146	Aluminium foam	Numerical simulation	0.94	8 PPI	Aluminium foam = 6 wt%	400%–500%	[74]
Paraffin	0.2	Copper/aluminium foam	Numerical simulation	0.95	–	Paraffin/copper-aluminium foams: 7.4	3700%	[75]
Paraffin wax	0.2	Aluminum foam	Pore-scale numerical method	0.95	10 PPI	Paraffin/aluminum foam: 24.6	12300%	[77]
Paraffin wax	0.25	Aluminum foam	Vacuum impregnation method	0.95 0.77	–	Paraffin/aluminum foam: 17.29 and 4.192	6916%; 1676.8%	[81]
Paraffin	0.19	Graphene coated nickel (GcN) foam	Chemical vapor deposition (CVD)	0.9	12.7 PPI	Paraffin/graphene-nickel foam: 4.6	2300%	[82]

(continued on next page)

Table 2 (continued)

PCM type	Thermal conductivity of PCM k_p W/(m·K)	Three-dimensional structure additives	Preparation methods		Thermal conductivity of composite PCM k_c W/(m·K)	Fraction of matrices	Improvement (%)	Ref.
Paraffin	0.162	Graphene–nickel foam	Melting blend method	–	–	Paraffin/PDMS-G-NF: 2.262	1396.3%	[83]
Myristyl alcohol (MA)	0.1701	Nickel foam	Vacuum melting infiltration	–	40 PPI	MA/nickel foam: 0.4768	280%; 851%	[84]
n-octadecane	0.16	TiO ₂ foam	Vacuum infiltration method	–	–	MA/copper foam: 1.4520		
Paraffin	0.21	Porous TiO ₂ foam	Vacuum impregnation method.	0.87	–	n-octadecane/TiO ₂ foam: 0.38	238%	[86]
						Paraffin/carbon/PTF: 1.059	504%	[88]

3.2.2.1. Addition of expanded graphite in passive PCM. Expanded graphite is one of the most popular additives due to its inherent advantages of high thermal conductivity, chemical stability, low density, large surface area, high surface activity, good adsorption performance and biocompatibility, EG has been widely studied and applied in the application of composite PCMs [109,110].

Many scholars have studied the methods of adding EG into the PCM to improve the thermal conductivity of the composite PCMs. Wu et al. [111] synthesized different parameters SA/EG form-stable composite PCMs, in which liquid SA was adsorbed into EG additives with different contents and the mixture was compressed into blocks of shapes stabilized with different densities by melting adsorption and pressing molding. Fig. 31 shows the microstructure of pure EG and SA/EG composite PCM. The figure shows that EG matrix has a worm-like microporous network structure, which has high surface activity and large specific surface area, thus is easy to absorb molten SA. In addition, the compressed composite PCM samples show regular graphite sheets. Since the thermal conductivity of EG is much higher than that of SA, it presents obvious anisotropy after compression. The maximum difference between axial thermal conductivity (6.41 W/(m·K)) and radial thermal conductivity (23.27 W/(m·K)) is nearly 4 times. When EG content is less than 25 wt%, the thermal conductivity of EG/SA composite PCM increases with the increase of EG content. While EG content exceeds the optimal value of 25 wt%, the thermal conductivity of EG/SA composite PCM slowly decreases. Furthermore, the thermal conductivity of EG/SA composite PCM changes similarly with the filling density. With low EG content, the formation of thermal conductivity chain can be promoted with the increase of EG content, while the regularity of graphite sheet layer can be reduced with high EG content. For low-density samples, the air thermal resistance can be reduced as the density increases, while the high-density samples can increase the interfacial thermal resistance and reduce the thermal conductivity of form stable composite PCMs.

Amin et al. [112] investigated the thermal properties of beeswax/graphene composite PCMs. Morphology and preparation of beeswax/graphene nano-PCM is shown in Fig. 32. The graphene nanosheets disperse well in beeswax and have no agglomeration phenomenon. The melting point decreases but the solidification point increases with the increase of the mass fraction of the graphene nanosheets. Compared with the pure beeswax, the melting enthalpy increases by 22.32% due to the Brownian motion and particle clustering mechanisms. The thermal conductivity of the composite PCMs increases when the mass fraction of the graphene nanosheets varies from 0.05% to 0.3%. Exactly, the enhancement of thermal conductivity has a linear relationship with the increase of graphene content within a certain range. With the concentration of graphene continues to increase, the thermal conductivity enhancement is nonlinear because of the agglomeration of the nanoparticles. This indicates that in the low concentration stage, heat conduction channels are formed in the interior of composites, while the concentration of graphene increased, the agglomeration of graphene nanocrystalline sheets make the heat conduction channels disordered

and their heat transfer is consumed. However, the thermal conductivity decreases gradually with the increase of temperature for the same graphene mass fraction. The thermal conductivity of the composite PCMs reach to 2.89 W/(m·K) when the temperatures is 40 °C and the graphene concentration is 0.3%. The main reason is that the heat conduction process is weakened, while the convective heat transfer process, which is dominated by beeswax liquid, is enhanced and thus its heat conduction performance is reduced to some extent.

Tian et al. [113] used paraffin as the PCM, ethylene-vinyl acetate (EVA) as supporting material, EG and carbon fiber (CF) of different dimensions as high thermal conductivity matrix to synthesize form-stable composite PCMs. Both the dimension and the structured of EG and CF are different, which leads to the different contact ways between EG and CF. The experimental results indicate there exists synergistic enhancement effect of CF and EG on the thermal conductivity because of the complementary structure and morphology between CF and EG. Fig. 33 shows the morphological characterization, and α , β , γ represent point contact, line contact and face contact, respectively. The thermal conduction mechanism is different in the heating and cooling processes and gives rise to the difference of synergistic thermal conduction between the cooling process and the heating process. The phonons play a crucial part in the heat transfer for the solid phase composite PCMs. The 1D rod-like structure of CF makes the paraffin and EVA form the orienting arrangement, which is in favour of the phonon transport. The convection becomes the principal heat transfer way with the generation of the phase change in the heating process. It is opposite to the cooling process. The optimal content of CF is between 10 wt% and 20 wt%, otherwise some CF will agglomerate corresponding to the case of ϵ in Fig. 33. Meanwhile, the crystalline structure of EVA would be destroyed and caused the leakage of PCM. Moreover, more interfaces will be generated and makes thermal resistance augment, which will obstruct the heat transfer. Hence, the thermal conductivity rises with the increase of CF content when the content of CF is under the optimal range. Conversely, the thermal conductivity will decline rapidly as soon as the CF content exceeds the optimal range. In addition, the horizontal thermal conductivity of composite PCMs are higher than the longitudinal thermal conductivity. The high temperature not only excites molecular chains of EVA and paraffin but also facilitates the phonon propagation. As a result, the thermal conductivity of the form-stable composite PCMs augments with the rise of the temperature.

Zhang et al. [114] investigated the thermophysical properties of high-temperature composite PCMs utilizing ternary carbonates Li₂CO₃–Na₂CO₃–K₂CO₃ with three different carbon materials: EG, graphene nanosheets (GNSs) and multi-walled carbon nanotubes (MWCNTs). Fig. 34 shows the structure of carbonates with carbon materials. EG has the micron-level size and loose porous worm-like structure, which is special 3D layered structures and tends to form the continuous heat conduction path. The carbonate salts are not only prone to crystallize on the surface of the EG and but also implant into the pores of the EG. For the GNSs, the 2D plane structure makes the thermal

Table 3
Thermal properties of two-dimensional structure additive/PCM passive composite PCMs.

PCM type	Thermal conductivity of PCM k_p W/(m·K)	Two-dimensional structure additives	Preparation methods	Fraction of matrices	Thermal conductivity of composite PCM k_c W/(m·K)	Improvement (%)	References
Stearic acid	–	Montmorillonite and SiO ₂	self-assembly method	61.5 wt%	–	–	[93]
n-tetradecane (Tet)	0.3721	Polystyrene-silica (PS-SiO ₂)	In-situ polymerization method	1.0 wt%	Tet@PS-SiO ₂ : 0.4341	116.7%	[95]
n-octadecane	0.1505	Inorganic silica	sol–gel method	50 wt%	n-octadecane/silica: 0.6213	412.8%	[96]
Paraffin	0.2	Al ₂ O ₃ and SiO ₂	Direct-synthesis method	4 wt%; 2 wt%	Paraffin/Al ₂ O ₃ : 0.919; Paraffin/SiO ₂ : 0.724	459.5%; 362%	[97]
Galactitol hexa myristate (GHM) and galactitol hexa laurate (GHL)	0.11, 0.10	Diatomite, perlite, vermiculite and EG	Vacuum impregnation method	55, 67, 52 wt% and 5 wt%	GHM/vermiculite-EG: 0.24; GHL/diatomite-EG: 0.24	218.2%; 240%	[98]
Xylitol penta palmitate (XPP) and xylitol penta stearate (XPS)	0.10, 0.11	Gypsum, cement, perlite, diatomite, vermiculite and EG	Vacuum adsorption method	22, 17, 67, 48 42 wt% and 10 wt%	XPP/gypsum: 0.28; XPS/cement-EG: 0.20	280%; 181.8%	[99]
Paraffin	0.2	Expanded perlite	Direct mixing method	20 wt%	Paraffin/expanded perlite: 0.52	260%	[100]
Lauric acid/stearic acid (LA-SA)	–	Modified sepiolite	Melting adsorption method	62.5 wt%	–	–	[101]
Stearic–capric acid eutectic (S–C)	–	Activated-attapulgit (a-ATP)	Vacuum fusion absorption method	50 wt%	–	–	[102]
Capric–palmitic acid	–	Attapulgit	Melting blending and vacuum absorption method	65 wt%	–	–	[103]
SA	0.21	Modified expanded vermiculite (MVE)	Direct vacuum impregnation method	47.1 wt%	SA/MEV: 0.52	247.6%	[104]
Polyethylene glycol (PEG)	0.06	Expanded Vermiculite (EV) silver nanowire (Ag)	Physical blending and impregnation method	21.9 wt%, 19.3 wt%	PEG/EV-Ag: 0.68	1230%	[105]
Paraffin	0.2	Kaolin	Vacuum impregnation method	18 wt%	Paraffin/kaolin: 0.77	385%	[106]
Paraffin	–	Diatomite	Vacuum impregnation method	20 wt%	–	–	[107]
SA	0.26	Graphite and bentonite	Vacuum impregnation method	31 wt%	SA/G-B: 0.77	296.2%	[108]
SA-BA eutectic mixture	0.3396	EG	Melting blend - solidification form stable	5.381	EG = 16 wt%	1584.51%	[109]
Paraffin	0.369	EG	Self-assembly method	10.37	EG = 30 wt%	2810%	[110]
SA	0.18	EG	Melting blend - solidification form stable	23.27	EG = 25 wt%	13000%	[111]
Beeswax	0.25	Graphene	Ultrasonic method	2.89	Graphene = 0.3 wt%	1156%	[112]
EVA and paraffin	–	EG and CF	Vacuum impregnation	2.10	EG = 70%, CF = 30%	–	[113]
Li ₂ CO ₃ –Na ₂ CO ₃ –K ₂ CO ₃	1.674	EG, GNSs and MWCNTs	Solution- evaporation technique	2.663, 2.225, 2.089	EG = 1.0 wt% GNSs = 1.0 wt% MWCNTs = 1.0 wt%	159.1%, 132.9%, 124.8%	[114]
Octadecanol (OC)	0.4336	Expanded perlite (EP) and EG	Vacuum impregnation method	15 wt%	OC/EP-EG: 5.8550	1350.3%	[115]
Paraffin	0.33	Kaolin	Vacuum impregnation method	40 wt%	Paraffin/kaolin: 0.413	125.2%	[116]
Lauric-myristic-stearic acid (LA-MA-SA)	0.262	Al ₂ O ₃ -loaded expanded vermiculite (aEV/AO)	Vacuum impregnation method	29.3 wt%	LA-MA-SA/aEV-AO: 0.671	256.1%	[117]
PEG	0.3612	Diatomite/AgNP	Vacuum impregnation method	31%, 7.2 wt%	PEG/diatomite-AgNP: 0.82	227%	[118]
SA	0.267	Flake graphite-carbon nanofiber-modified bentonite (FG–CNF–Bm)	Vacuum impregnation method	41.18 wt%	SA/FG–CNF–Bm: 2.803	1049.8%	[119]
Capric acid (CA)	0.18	Bentonite clay, EG	Vacuum impregnation method	5 wt%	Bentonite/CA/EG: 0.77	427.8%	[120]

conductivity along this plane enhance. When carbonate salts permeate the crystal structure of GNSs, it can be attached to MWCNTs via the different styles and forms composite PCMs. However, the thermal conductivity perpendicular to the GNSs plane is a little smaller than EG. Among three carbon materials, MWCNTs provide the minimum levels of improvement on the thermal conductivity of ternary carbonates. MWCNTs ascribes to the 1D structure and the heat conduction path is

based on the axial direction. The experimental results indicate that three carbon materials express the significant improvement in the thermal conductivity. EG is the best fruitful and the thermal conductivity increases by 59.1%. The average thermal conductivity of carbonate salts with EG, GNSs and MWCNTs is 2.663, 2.225 and 2.089 W/(m·K), respectively. This phenomenon indicates that the phonon group velocity of heat flow is the fastest in the 3D EG carbon skeleton, which results in

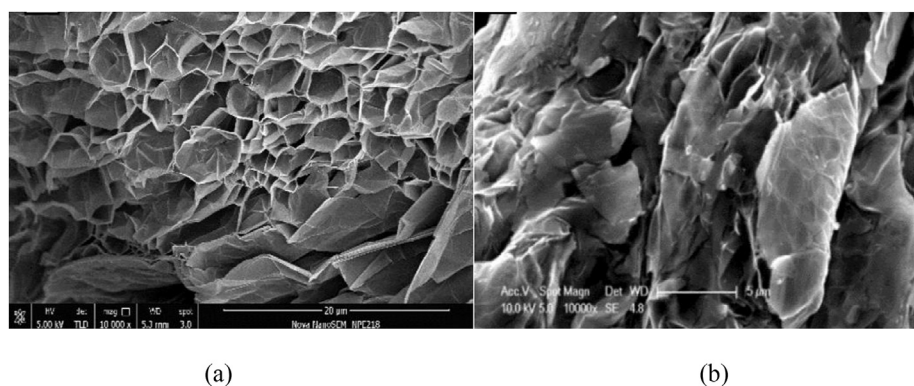


Fig. 31. SEM pictures of the form-stable EG/SA CPCMs: (a) EG matrix; (b) EG/SA sample.

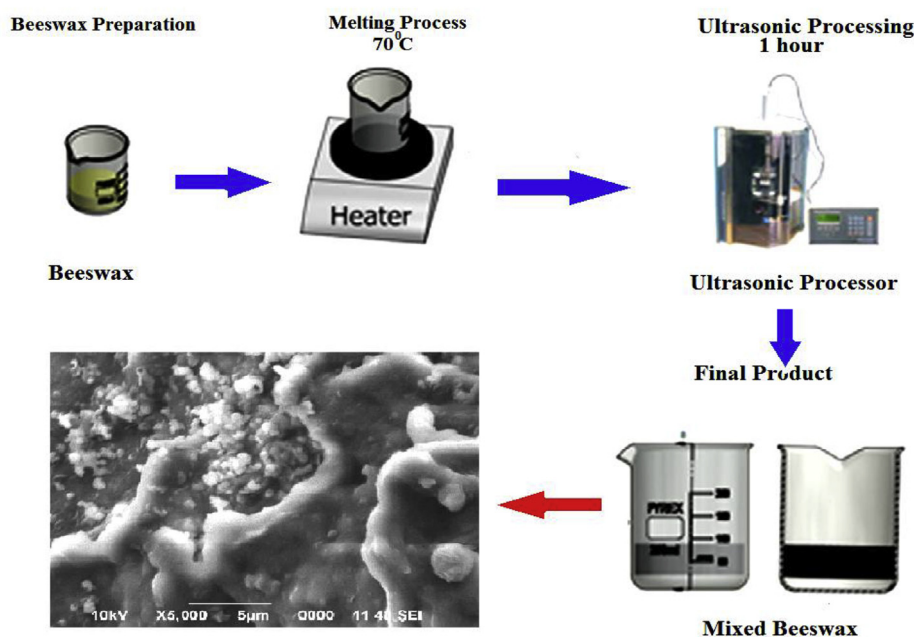


Fig. 32. Preparation of beeswax/graphene nano-PCM and SEM image.

the highest thermal conductivity.

3.2.2.2. Addition of clay minerals in passive PCM. Because of its porous properties, clay minerals can be used as the supporting matrix of PCM to prevent leakage, which has been studied by many researchers. Peng et al. [93] selected montmorillonite (Mt) as wall material and SA as core material, and obtained Mt/SA microcapsule composite PCMs coated with SiO₂ by tetraethyl orthosilicate (TEOS) hydrolysis and condensation polymerization using self-assembly method. Fig. 35 (a₁, a₂) shows the morphological characteristics of Mt/SA microcapsule composite PCMs. The polymerized particles of Mt/SA microcapsule adhere to each other without free Mt. It can be particularly visible that Mt forms effective self-assembled nested microspheres on the surface of SA. But it seems to only destroy some Mt/SA microcapsules with SA attached to the surface and the nucleocapsid structure is slightly deformed. Moreover, the Mt is hydrophobic due to the insertion of cetyltrimethylammonium bromide (CATB). The specific synthesis process of Mt/SA microcapsules is shown in Fig. 35 (a). Firstly, Mt and SA droplets are self-assembled through hydrophobic and electrostatic interaction. Secondly, TEOS are added to the surface of the SA droplet coated with Mt for hydrolysis and condensation. In the end, it can construct the silica gel matrix network, which can improve the structural stability of Mt/SA microcapsule. Fig. 35 (b) is a schematic diagram of the heat conduction pathway of Mt/SA

microcapsules. Mt and SA droplets are self-assembled by surfactant to form microcapsules, and the SiO₂ silica gel matrix network is constructed to form the heat conduction network. The reason is that hydrogen bands may be generated between SA, Mt and SiO₂ matrix. The hydrogen band in Mt/SA microcapsules can improve the thermal conductivity of microcapsules. The surface convection heat transfer can be promoted through the silica gel shell, and then the heat is transferred from Mt to SA.

Fu et al. [95] continued to study the effect of microcapsule slurry flow on thermal conductivity. They employed n-tetradecane (Tet) as core and polystyrene (PS) and SiO₂ as composite shell, a novel Tet@PS-SiO₂ nano-encapsulated PCM slurry for cold energy storage to obtain by dispersing nano-encapsulated PCM into base fluid by Sol-gel method. Fig. 36 shows the synthesis schematic of Tet@PS-SiO₂, TEM images of Tet@PS and Tet@PS-SiO₂. It indicated that the thermal conductivity of the slurry can be enhanced by the grafting SiO₂ of on the PS shell. It's also worth mentioning that the thermal conductivity of slurry can be strengthened owing to micro-convection effects in the case of moving. The reason for the enhanced thermal conductivity may be that the SiO₂ shell has a high thermal conductivity and its convective heat transfer increases during the flow process.

Zhang et al. [96] also taken OC as the core material and SiO₂ as the wall material to prepare OC/SiO₂ microcapsule PCM by using Sol-gel method. The encapsulation of OC is shown in Fig. 37. Specifically,

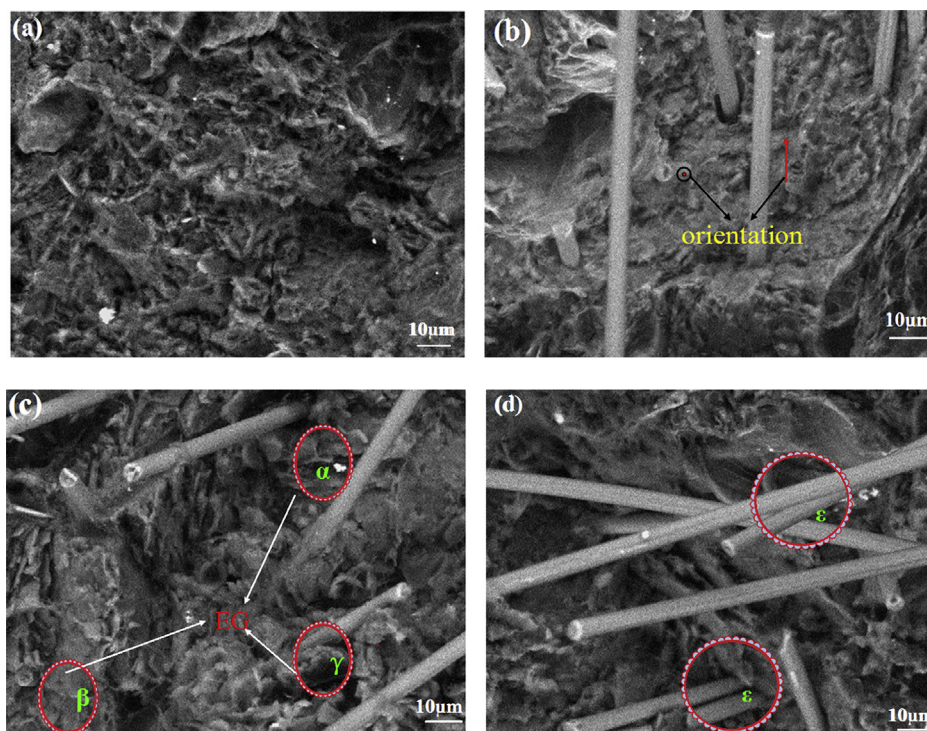


Fig. 33. SEM images of the FSPCMs: (a) FSPCM-CF0, (b, c) FSPCM-CF20 and (d) FSPCM-CF40.

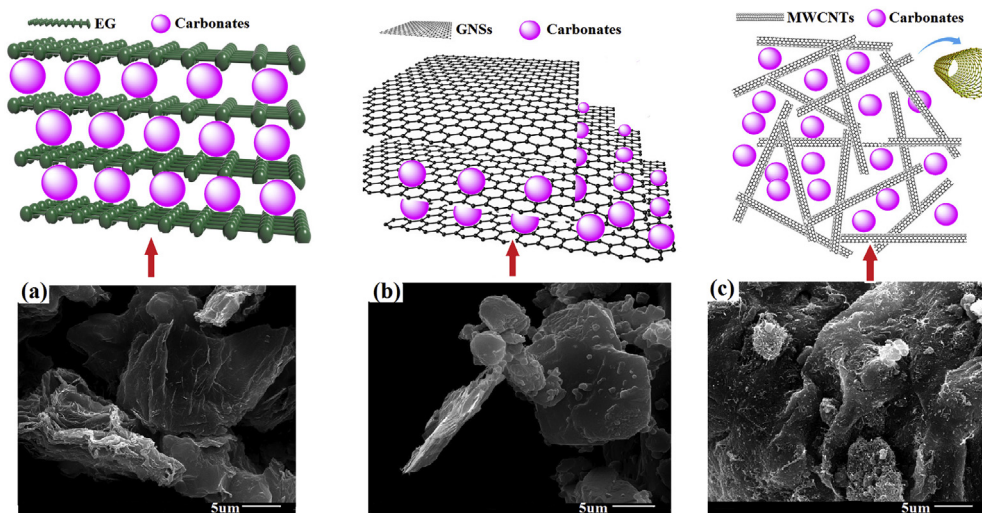


Fig. 34. SEM diagram and structure diagram of carbonate composite with three additives: (a) EG, (b) GNSs, (c) MWCNT.

silica sol homogeneous solution encapsulating OC is obtained through hydrolysis and condensation reaction, silica sol-gel process finished with ethanol washing get OC/SiO₂ microcapsules. It is observed that the silicon microcapsules are spherical in shape and have clear core-shell structure by showing SEM images in Fig. 37. In addition, when microcapsules are synthesized at pH = 2.45, the polycondensation rate matched the self-assembly of the oligomer on the surface of the OC micelles and the polycondensation between the silanol groups, which enabled the spherical microcapsules to maintain a complete and compact smooth shell. Moreover, the thermal conductivity of solid pure OC is 0.1505 W/(m·K), and that of pure silica gel shell is 1.296 W/(m·K). Obviously, the thermal conductivity of silicon microcapsules increased with the addition of SiO₂. When the mass ratio of OC/TEOS is 1:1, the thermal conductivity of the microcapsule increased to 0.6213 W/(m·K). However, when the mass ratio of OC/TEOS is increased to 7:3, the

thermal conductivity of the microcapsules only increased to 0.3751 W/(m·K), indicating that the porous and thin packaging shell is conducive to heat transfer. We can show that SiO₂ wall materials could form a continuous thermal conductivity network. Therefore, the heat transfer performance of OC/SiO₂ microcapsule PCM is improved when high heat conduction SiO₂ is used as the wall material of the microcapsule.

In order to further improve the thermal conductivity of composite PCMs, Lv et al. [115] appended EG with different contents in EP and OC to prepared OC/EP/EG composite PCMs by using vacuum impregnation method. Fig. 38 shows schematic diagram of the preparation and SEM images of OC/EP composites and OC/EP/EG composite PCMs. SEM images shown that EP has a rich pore network as well as OC is well absorbed into the pore network structure on the surface of EP and EG, which can restrict the flow of PCM in the molten state. The thermal conductivity of

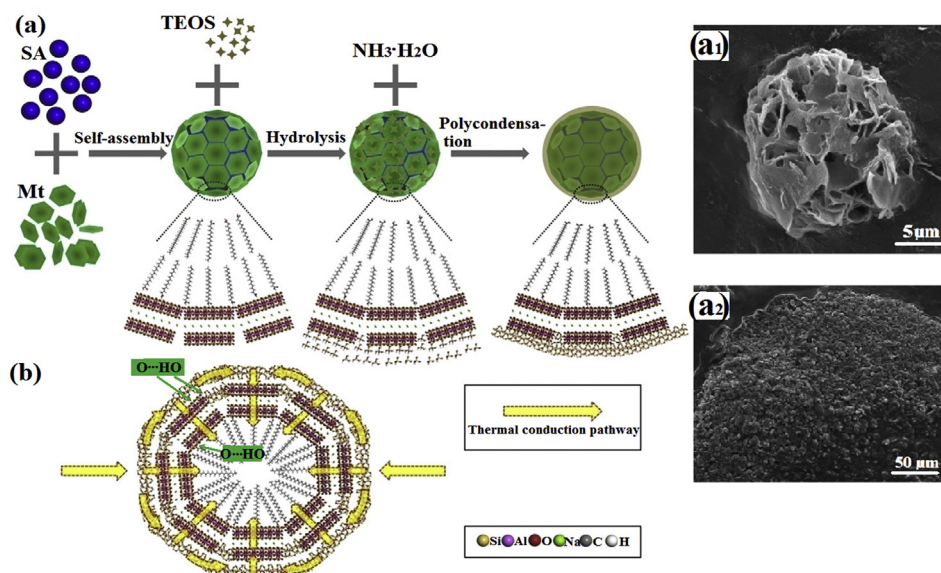


Fig. 35. (a) Schematic illustration of Mt/SA microcapsule fabrication, SEM images of (a₁) Mt/SA microcapsules and (a₂) Mt/SA composite PCMs, and (b) the interfaces and thermal conduction pathway in Mt/SA microcapsules.

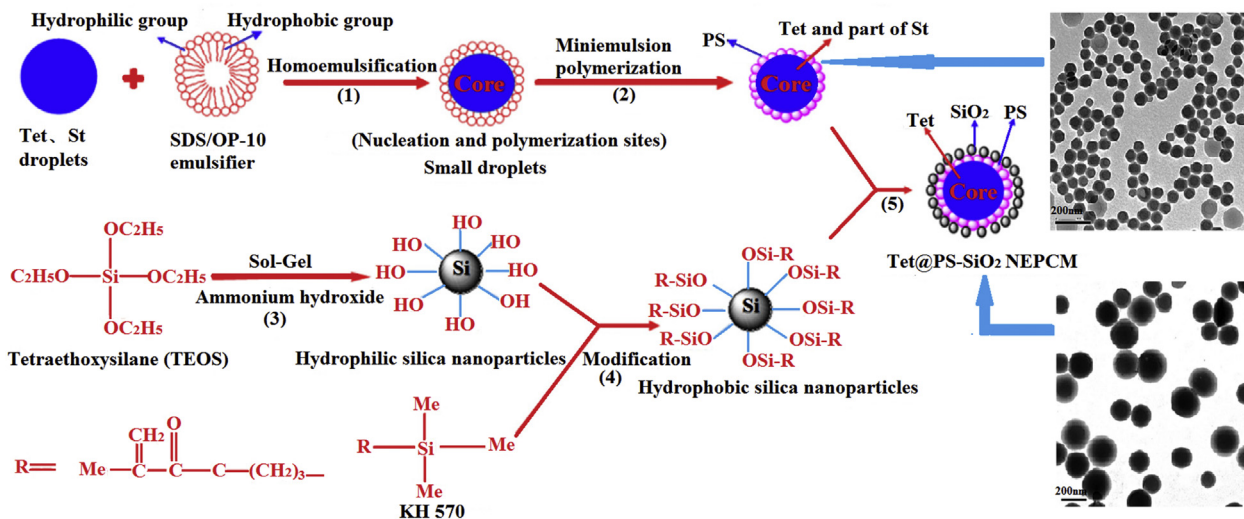


Fig. 36. Synthesis schematic of Tet@PS-SiO₂ and TEM images of Tet@PS and Tet@PS-SiO₂.

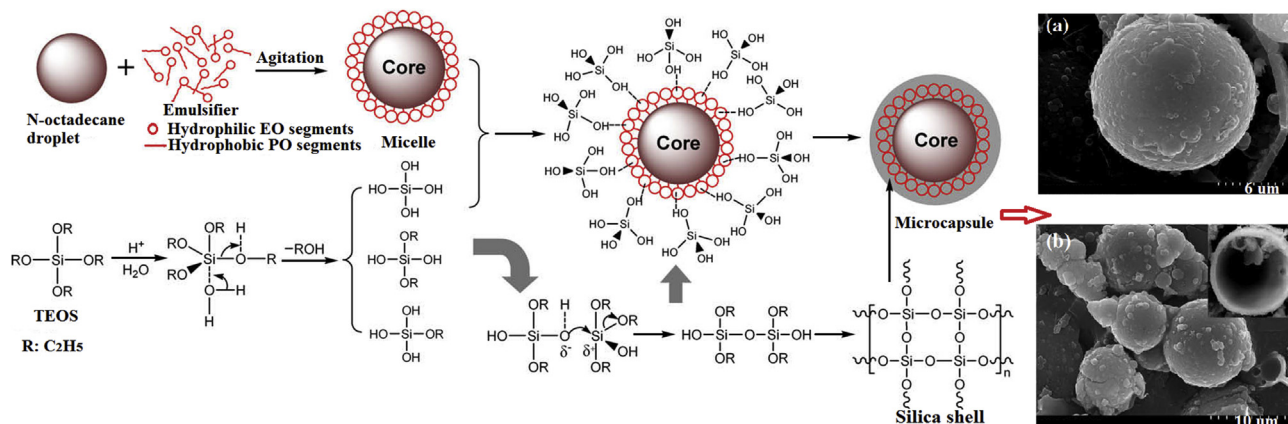


Fig. 37. Schematic formation of the silica-microencapsulated n-octadecane via a sol-gel process.

pure OC is 0.4229 W/(m·K) and decreased with the increase of temperature. However, the addition of EP in the PCM causes the thermal conductivity of the composite PCMs to slightly increase with the increase of temperature. In addition, when the addition amount of EP is 60%, the thermal conductivity of OC/EP composite PCM is 0.4336 W/(m·K). It indicated that the addition of EP improvement the thermal conductivity of OC has better effect and the thermal conductivity of OC/EP composite PCMs increased with the increase of EP mass fraction. Moreover, when the addition amount of EG is 15%, the thermal conductivity of OC/EP/EG composite PCM is 0.4336 W/(m·K). It indicated that with the increase of EG mass fraction, the thermal conductivity of OC/EP/EG composite PCMs raise and the change trend of the thermal conductivity of OC/EP composite PCMs increased. Further, it forms large number of thermal conductivity channels and promote the rapid heat transfer.

Lv et al. [116] selected kaolin with different particle sizes and mass fractions and prepared paraffin/kaolin composite PCMs by using vacuum impregnation method. Fig. 39(b) shows the microstructure of kaolin and paraffin/kaolin composite PCMs. The surface of kaolin is coated with paraffin wax or embedded in the crystal structure. Due to the irregular morphology of kaolin, the morphology of paraffin/kaolin composites is also irregular. Because of the large surface area of the paraffin/kaolin composite PCMs, the surface of the composite PCMs become compact and smooth with the addition of large particle size kaolin. Fig. 39(a) shows the thermal conductivity of paraffin/kaolin composite PCMs with different particle sizes and mass fractions. It is observed that the thermal conductivity of paraffin/kaolin composite PCMs are positively correlated with particle size, but negatively correlated with temperature. When the temperature is 20 °C, the thermal conductivity of pure paraffin is 0.287 W/(m·K) and maximum thermal conductivity of paraffin/kaolin composite PCMs is 0.547 W/(m·K). The reason is that phonon-phonon scattering is extremely easy to occur on the contact surface between paraffin and kaolin, which will generate a large amount of thermal resistance and shorten the phonon mean free path. In addition, the large number of phonons will collide with each other in the interface area, which will intensify phonon scattering and significantly reduce the thermal conductivity. Furthermore, the specific surface area of kaolin decreases with the increase of particle size, and the smaller the specific surface area is the better. Therefore, the thermal conductivity of composites with large kaolin particles is higher than that of composite PCMs with small kaolin particles. In addition, the thermal conductivity of paraffin/kaolin composite PCMs decreased with the increase of temperature, which is caused by the shortening of phonon mean free path. With

the increase of temperature, the number of phonons increases obviously, which leads to the increase of collision frequency. Therefore, phonon-phonon scattering becomes more frequent, phonon mean free path decreases, and thermal conductivity decreases.

Wei et al. [117] further studied the thermophysical properties of ternary eutectic salts with expanded vermiculite (EV) and Al_2O_3 composites. They are primarily synthesized the lauric-myristic-stearic acid (LA-MA-SA) eutectic mixture by using ultrasonic heating method, then selected EV treated with nitric acid (aEV) and further added Al_2O_3 to obtain aEV/ Al_2O_3 (aEV/AO) supporting matrix with excellent performance, and finally prepared the LA-MA-SA/aEV/AO composite PCMs by using vacuum impregnated method. As shown in Fig. 40, which shows the microscopic morphological images of EV, aEV, aEV/AO, LA-MA-SA/aEV/AO. The EV is observed to have a thick layered structure, a small amount of delamination occurred on the surface of aEV and pore structure began to further form by treating nitric acid. However, aEV/AO maintained thin laminar morphology and the number of pore matrix structure raised. Hence, In LA-MA-SA/aEV/AO composite PCMs, LA-MA-SA are fully absorbed into the pores of aEV/AO. In addition, the thermal conductivity of the composite PCMs tested results that the thermal conductivity of the aEV (0.389 W/(m·K)) is smaller than that of the EV (0.414 W/(m·K)). The reason for the decrease of thermal conductivity is that the increase of defects caused by using acid treatment would aggravate phonon scattering. Owing to increase by Al_2O_3 on the supporting matrix, the thermal conductivity of aEV/AO has reach 0.563 W/(m·K). Moreover, the thermal conductivity of LA-MA-SA/EV, LA-MA-SA/aEV and LA-MA-SA/aEV/AO composite PCMs are 0.513, 0.448 and 0.671 W/(m·K), respectively, which are 37.5%, 30.8% and 156.1% higher than that of the composite PCMs of LA-MA-SA (0.262 W/(m·K)). The reason may be that Al_2O_3 has good thermal conductivity and it is evenly dispersed in the composites, which can form a thermal conductivity chain and enhance its heat transfer performance.

Qian et al. [118] resolved silver nitrate by hydrothermal synthesis to prepare Ag nanoparticle (AgNP) additive as first step. Afterwards diatomite (Dt) was modified for filling AgNPs by alkali and sulfuric acid leaching ordinal treatment. Finally, using PEG and modified AgNPs were prepared high thermal conductivity PEG/Dt/Ag form-stable composite PCMs via melting blending method. The corresponding preparation process and morphological characteristics as well as thermal conductivity variation of composite PCMs are shown in Fig. 41. Since fewer impurities of Dt is treated via sulfuric acid leaching, which is improved obviously thermal conductivity of composite PCMs by using hydrothermal

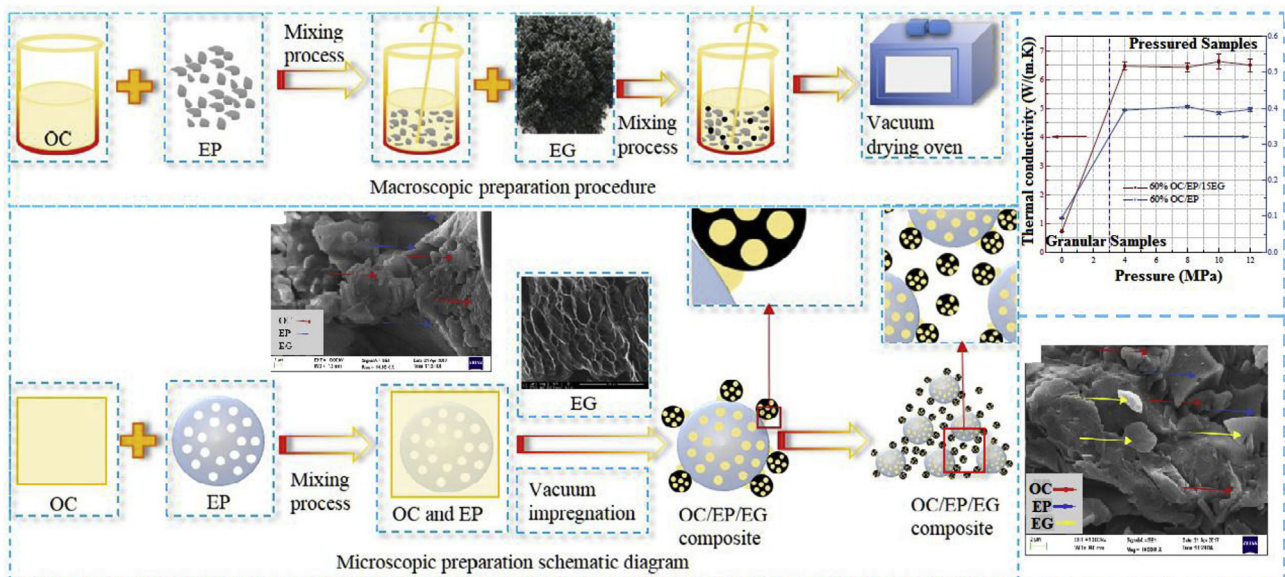


Fig. 38. Schematic diagram of the preparation and SEM images of OC/EP and OC/EP/EG composites.

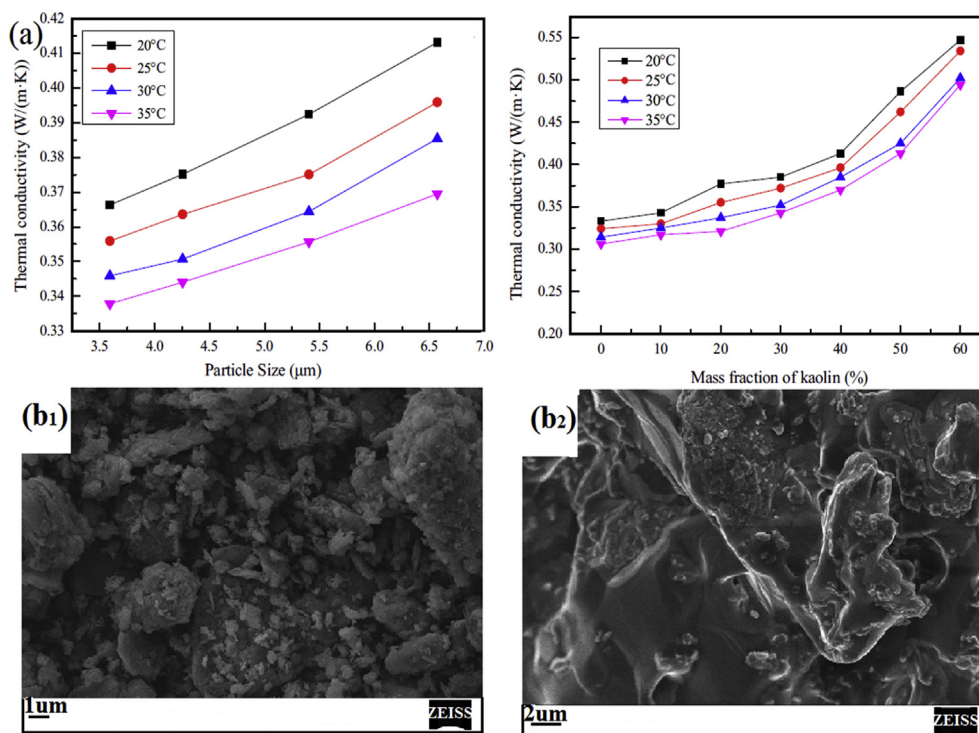


Fig. 39. Curves of the thermal conductivity of paraffin/kaolin composite PCM with diverse mass fraction and particle sizes of kaolin at different temperature and SEM images of (a) kaolin (b) paraffin/kaolin.

synthesis and vacuum impregnation. Additionally, the high thermal conductivity of metallic silver (429 W/(m·K)), the thermal conductivity of Dt itself is also high and PEG is also impregnated into the pores of Dt, its thermal conductivity is significantly improved. Interestingly, after the addition of AgNPs, the thermal conductivity of PEG/Dt/Ag composite PCMs are 127% higher than PEG/Dt composite PCMs. It indicated that the addition of AgNPs can better improve the thermal conductivity of PEG/Dt/Ag composite PCMs.

Li et al. [119] prepared three kinds of SA/FG/B, SA/FG-CNF/B and SA/FG-CNF-Bm composite PCMs by means of silane coupling agent modified bentonite (Bm) and flake graphite-carbon nanofiber (FG-CNF) through chemical bonding coupling supporting materials and melting blending with SA. The preparation process, morphological characteristics of FG-CNF-Bm and SA/FG-CNF-Bm composite PCMs are synthesized as shown in Fig. 42. It can be observed that Bm is closely attached to the surface of FG-CNF and wrapped the whole CNF, resulting in an increase in the contact area, thereby improving the thermal conductivity of FG-CNF-Bm. In addition, the porous network structure generated by CNF provides a larger filling space for SA. Significantly, the SA is uniformly fixed in the FG-CNF-Bm mesh pore structure, which improves the adsorption capacity of the composite PCMs and effectively prevents the leakage of molten SA. Consequently, the thermal conductivity of FG, FG-CNF, FG/B, FG-CNF/B, FG-CNF/Bm, FG-CNF-Bm, SA, and FG-CNF-Bm/SA composite PCMs are determined as 5.471, 5.944, 4.176, 4.499, 4.291, 4.595, 0.267 and 2.803 W/(m·K), respectively. According to the phonon thermal conductivity mechanism, excessive interfacial pores between the PCM and the added matrix will increase the surface scattering of phonons, resulting in a significant reduction in thermal conductivity [53]. Compared with FG-CNF, the thermal conductivity of CNF grown on the surface of FG (5.471 W/(m·K)) is lower than that of FG-CNF (5.944 W/(m·K)), while the thermal conductivity of FG-CNF/B (4.499 W/(m·K)) is higher than that of FG/B (4.176 W/(m·K)). It is mainly attributed that CNF is filled in the gap between FG, while the surface CNF matrix combined with microscale Bm particles can increase the contact area. Thanks to the low thermal conductivity of bentonite

(0.82 W/(m·K)) [120], the thermal conductivity of pure bentonite is lower than FG and FG-CNF samples. But the thermal conductivity of FG-CNF-Bm (4.595 W/(m·K)) is relatively higher than FG-CNF/Bm (4.291 W/(m·K)). It may be that the successful coupling of chemical bonds reduced the interface thermal resistance between FG-CNF and Bm. Hence, it can significantly improve the thermal conductivity. For this reason, CNF not only obtains a larger contact area to promote heat conduction, but also provides more heat conduction channels for heat transfer through chemical bond coupling. Compared with the thermal conductivity of pure SA (0.267 W/(m·K)), the thermal conductivity of SA/FG-CNF-Bm composite PCM (2.803 W/(m·K)) is increased by 10.50 times. Mainly, FG-CNF-Bm has larger contact area and successfully coupled chemical bonds, further forming more effective thermal conduction paths, which greatly improves the thermal conductivity of the composite PCMs as well as prominently reduces the charging and discharging time of the system.

On the whole, the EG and clay minerals of 2D layer as porous support materials to prepare high performance passive composite PCM is significant value for thermal storage. Especially, it has excellent effect in preventing the leakage of PCM in molten state. Table 3 summarizes the preparation methods of passive composite PCMs and summary of improving thermal conductivity of 2D structure additive as supporting materials.

3.2.3. Initiative composite PCMs using one-dimensional structure additives

The 1D structure additives have many morphological structures, such as carbon fiber (CF), carbon nanotubes (CNTs) and rod-shaped graphene etc. Due to its excellent 1D thermal conductivity, even just the addition of a small amount of 1D structure supporting materials can significantly improve the thermal conductivity of the composite PCMs. In addition, 1D structure additives are lighter than metal or ceramic additive materials, which has a wider range of practical applications. Besides, the 1D structure additives, which also has good dispersion in the melting state and reduce the supercooling degree, are the ideal choice for the preparation of the high performance PCM.

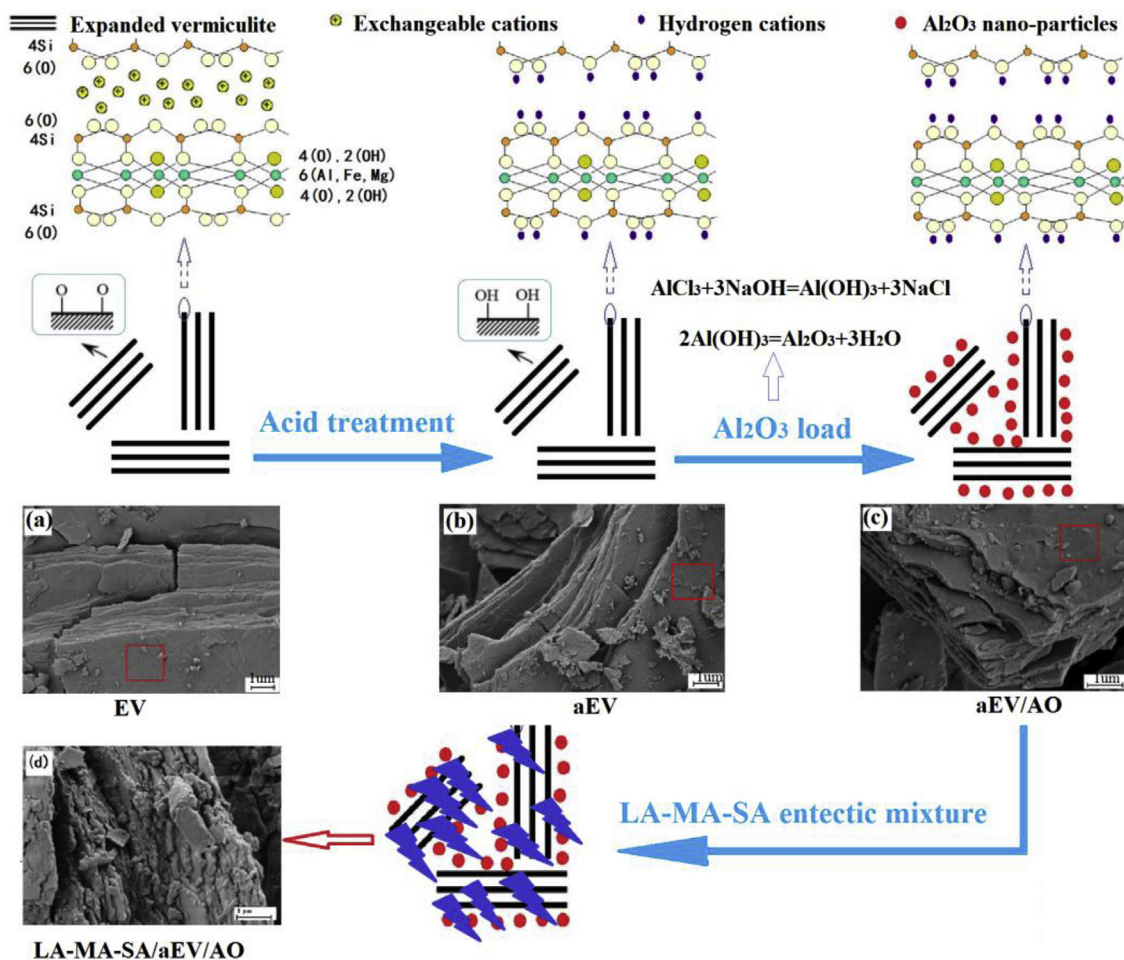


Fig. 40. Schematic diagram of the preparation process of composite PCMs and SEM images of the (a) EV, (b) aEV, (c) aEV/AO and (d) LA-MA-SA/aEV/AO.

3.2.3.1. *Addition of carbon fiber in initiative PCM.* CF and CNTs are 1D additional matrix and CF materials can be obtained by high temperature carbonization in inert gas. CF also has the advantages of low density, high

temperature corrosion resistance and high thermal conductivity. Meso-phase pitch-based CF has highly crystalline graphite structure which is parallel to the orientation of fiber axis and its longitudinal thermal

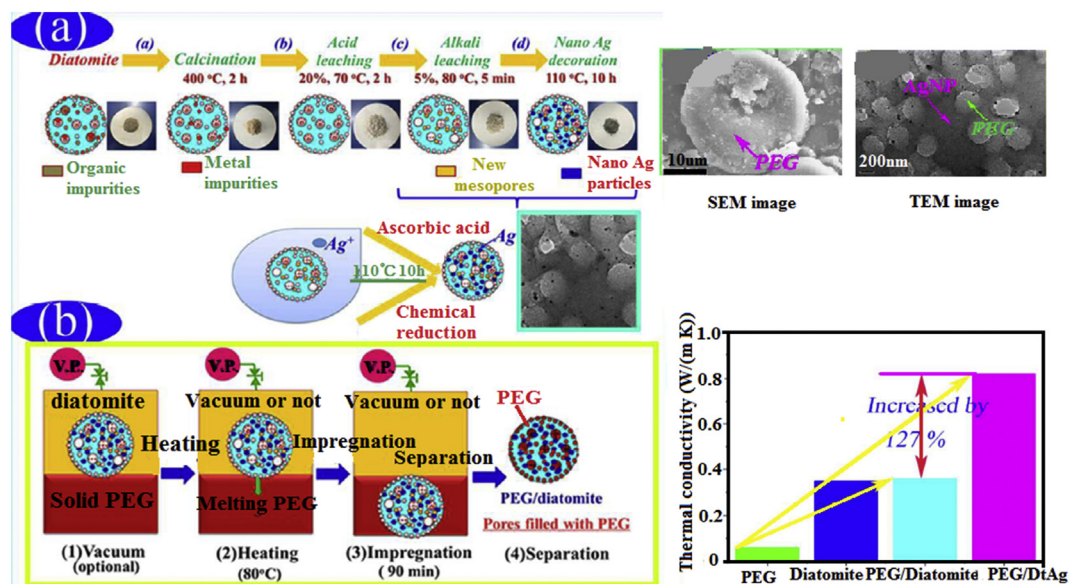


Fig. 41. (a) Principle of preparation of Dt modified by AgNP and (b) vacuum impregnation processing image of composite PCM and microstructure and thermal conductivity curve of PEG/Dt/AgNP composite PCMs.

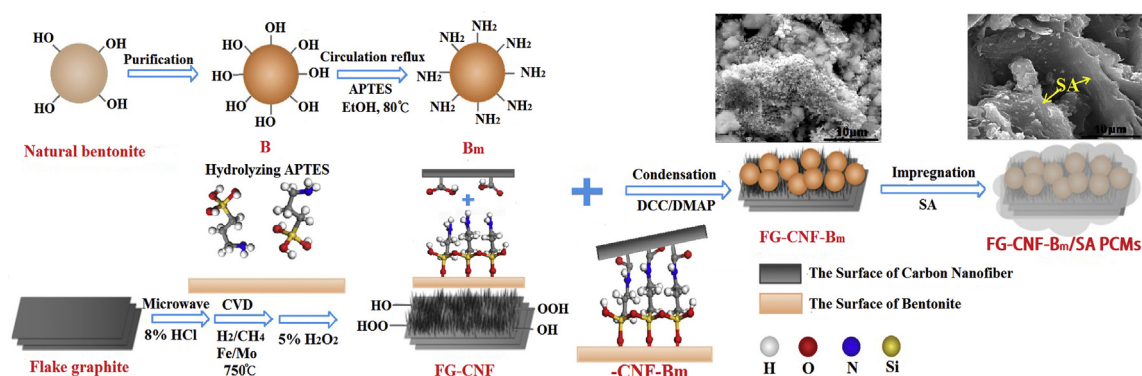


Fig. 42. Scheme of preparation route of composite as well as SEM images of FG-CNF-B_m and SA/FG-CNF-B_m composite PCM.

conductivity exceeds 1000 W/(m·K) [121]. In addition, it can absorb almost all organic PCM, so it is one of the best materials to improve the thermal conductivity of organic PCM [122]. Adding CF into PCM can significantly improve the thermal conductivity of composite PCMs, in which the length, diameter, the added mass fraction and distribution mode of CF [123] will affect the thermal conductivity of composite PCMs [121].

This has aroused the researchers' interest in studying more clearly the effect of the addition on the thermal conductivity of composite PCMs. Zhang et al. [124] prepared composite PCMs samples by filling a series of short carbon fiber (SCF) with different aspect ratios in erythritol. As shown in Fig. 43, the morphological characteristics of C₅ and C₂₅ with different mass fractions in SEM enlarged images show that C₅ and C₂₅ are easily bound to erythritol. In addition, SCFs has no aggregation and coupling, and its distribution is relatively uniform, while the addition of SCFs directly affects the size of grain. The larger the content of SCFs, the smaller the grain size. Adding more SCFs can promote the heat transfer rate of composite PCMs. As the mass fraction of SCF increases, the thermal conductivity presents a nonlinear increase. When the additive volume of C₂₅ and C₅ composite PCMs are 10%, the maximum thermal

conductivity reaches 3.92 W/(m·K) and 2.46 W/(m·K), respectively, which are 407.8% and 242.9% higher than that of pure erythritol (0.77 W/(m·K)), respectively. The larger aspect ratio of CF and the larger thermal conductivity of the composite PCMs. Because the crossbar is larger than CF, it is easier to form a thermal conduction network in composite PCMs. On the one hand, when the two kinds of aspect ratio of different additives to form composite PCMs have the same structure, the interface thermal resistance are equal, so the total thermal resistance depends on the SCFs of PCCs and erythritol total contact area of the interface, C₅ composite PCMs 8.2% larger than C₂₅ interface, interface of the large number of small diameter and C₅, lead to more of the top and bottom of the connection between the CF. On the other hand, the connection state of the additive is closely related to the composite PCMs. When low content of C₂₅ is added, CF in the matrix is connected to each other through contact nodes, forming more local networks and thus improving thermal conductivity.

Jiang et al. [125] employing CF in paraffin wax to bond with polymer binder, which formed a stable network after carbonization, prepared a kind of porous mesh carbon bonded CF composite PCMs (CBCF-CPCM) with fixed carbon bond. Fig. 44 shows the microstructure of CBCF,

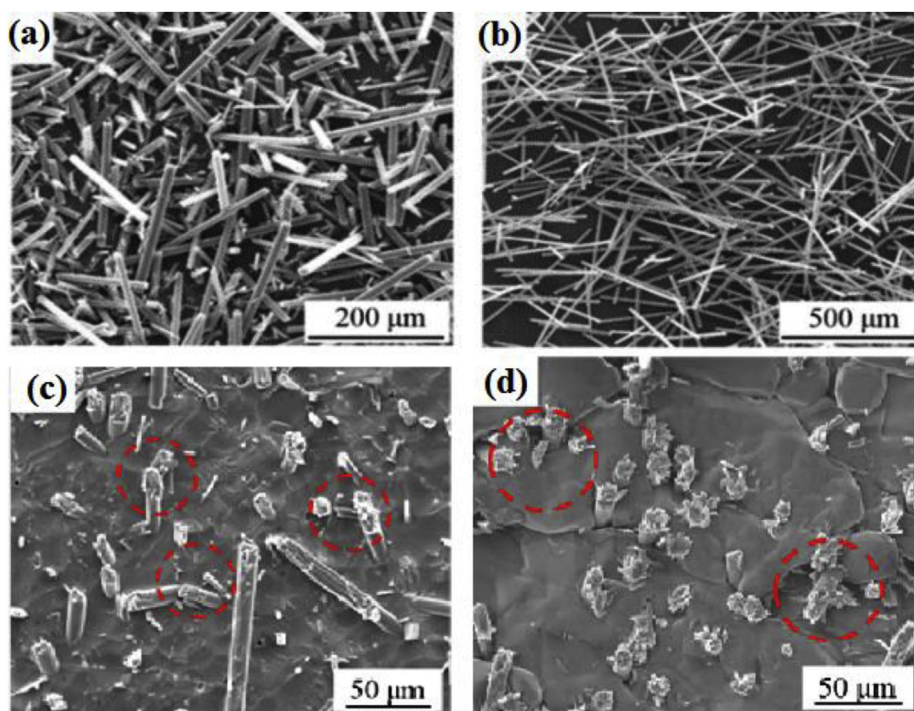


Fig. 43. SEM images of SCFs and SCFs/erythritol composite PCMs: (a, c) C₅ and C₅/erythritol as well as (b, d) C₂₅ and C₂₅/erythritol.

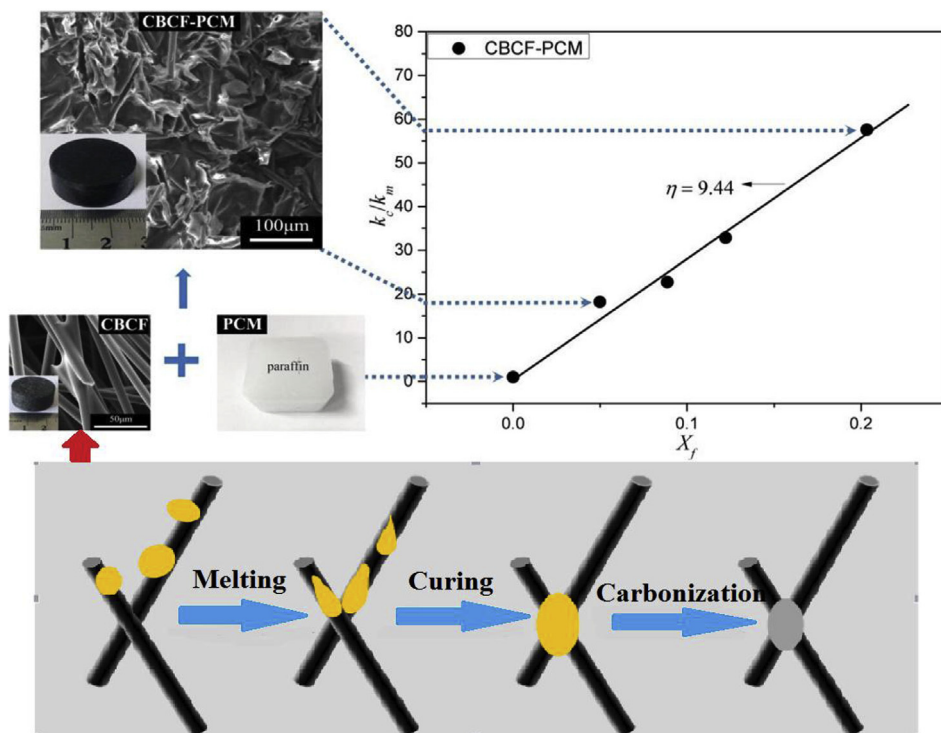


Fig. 44. Schematic diagram of CBCF binding process and microstructure and thermal conductivity curve of CBCF-PCM.

CBCF-CPCM and thermal conductivity curve. It can be seen that CF is oriented in the xy plane, as the heat transfer is more likely to be carried out along the fiber direction, resulting in the significant improvement of the thermal conductivity of CBCF-CPCM compared with pure paraffin. In the z direction, the arrangement of carbon fibers tends to overlap, so the number of nodes forming the heat conduction network is limited. In addition, the cross section of CF can directly form the heat transfer path, in which the carbon crystal plane is short and discrete according to the arrangement of the crystal layers. Therefore, the thermal conductivity outside the plane will be greatly reduced. Fig. 44 is a schematic diagram of heat conduction network formed by bonding between CFs. The powder phenolic resin is evenly distributed on the surface of CFs. When the phenolic resin melts at high temperatures, gravity causes the liquid phenolic resin to slide along the surface of the CF. Further, the liquid phenolic resin is bonded together with CF, when they reach the point of contact between the CFs. After carbonization and heat treatment, the CFs are held in place to form a stable thermal conductivity network.

Two preparation methods have been carried out by Nomura et al. [126], erythritol was employed as the PCM and CF considered as the heat conduction enhancer. Erythritol/CF composite PCMs by melt dispersion (MD) and hot press (HP) were prepared to improve the content of CF in erythritol. Fig. 45 is a schematic diagram of composite PCMs prepared by (a) MD method and (b) HP method as well as curve diagram of effective thermal conductivity of composite samples changing with the content of CFs. The results show that HP method is easy to form the percolating filler network in composite PCMs. Compared with the composite PCMs prepared by MD method, the addition of CF is less and the thermal conductivity is higher. In addition, high CF content can promote the formation of the percolating filler network. In order to control the filling ratio of CF in PCM, the diameter distribution of PCM particles can be controlled. Because in the process of HP, CF in the state of uniform mixing is distributed around the PCM particles. Because the PCM particles are stationary and solid during the HP process. When CFs are mixed with erythritol, the morphological characteristics of erythritol have great influence on the configuration of CFs. Under this condition, it is easy to form a network structure permeating CFs, because the PCM particles are

initially in contact with each other. Meanwhile, the gap around the PCM particles filled with CFs are in a seepage state. In this way, HP method can easily form the seepage structure of high thermal conductivity filler. The large filling gap between PCM particles and CF makes the composite PCMs with high filling ratio have abundant thermal conductivity network structure. Furthermore, the phonon group velocity of transfer and mean free path increased to improve the thermal conductivity of the composite PCMs.

3.2.3.2. Addition of carbon nanotubes in initiative PCM. CNTs are characterized by large specific surface area, uniform micropores, unique physical and chemical properties, low density and high thermal conductivity (2000–6000 W/(m·K)) [127], which are considered as a potential high thermal conductivity additive. MWCNTs can be used as an additive for some other chemical adsorbents to improve their heat transfer characteristics [128]. Since CNT is 1D nano-addition material, it has excellent nano-scale effect, low density and high thermal conductivity [129]. According to relevant experiments, the thermal conductivity of single CNT is higher than 3000 W/(m·K) [130]. The excellent thermal conductivity of CNTs makes it an ideal additive matrix for the preparation of composite PCMs. CNTs are divided into single wall CNT (SWCNT) and multi-wall (MWCNT), in which SWCNT is curly made of monolayer graphene. In the process of winding the graphene layer into a cylinder, the dangling bonds are rapidly combined on the boundary, which made the axis of CNTs randomly distributed. Carbon atoms are wound into a helical hexagonal grid to form CNTs. When the graphite surface is aligned longitudinally along the axis of SWCNT, a 2D geometry similarly to single-layer graphene surface is generated [131]. MWCNT has excellent thermal, electrical and mechanical properties, which have broad application prospects [132]. After calculation, at the temperature of 100 K, the thermal conductivity of single MWCNT is 37000 W/(m·K), and its macroscopic thermal conductivity can reach 6000 W/(m·K) [133]. At room temperature, the thermal conductivity of single MWCNT can be compared with the diamond, and even reach a higher value [134]. Based on the above excellent properties, mixing CNTs in PCM to improve the thermal conductivity of composite PCMs have attracted great attention of

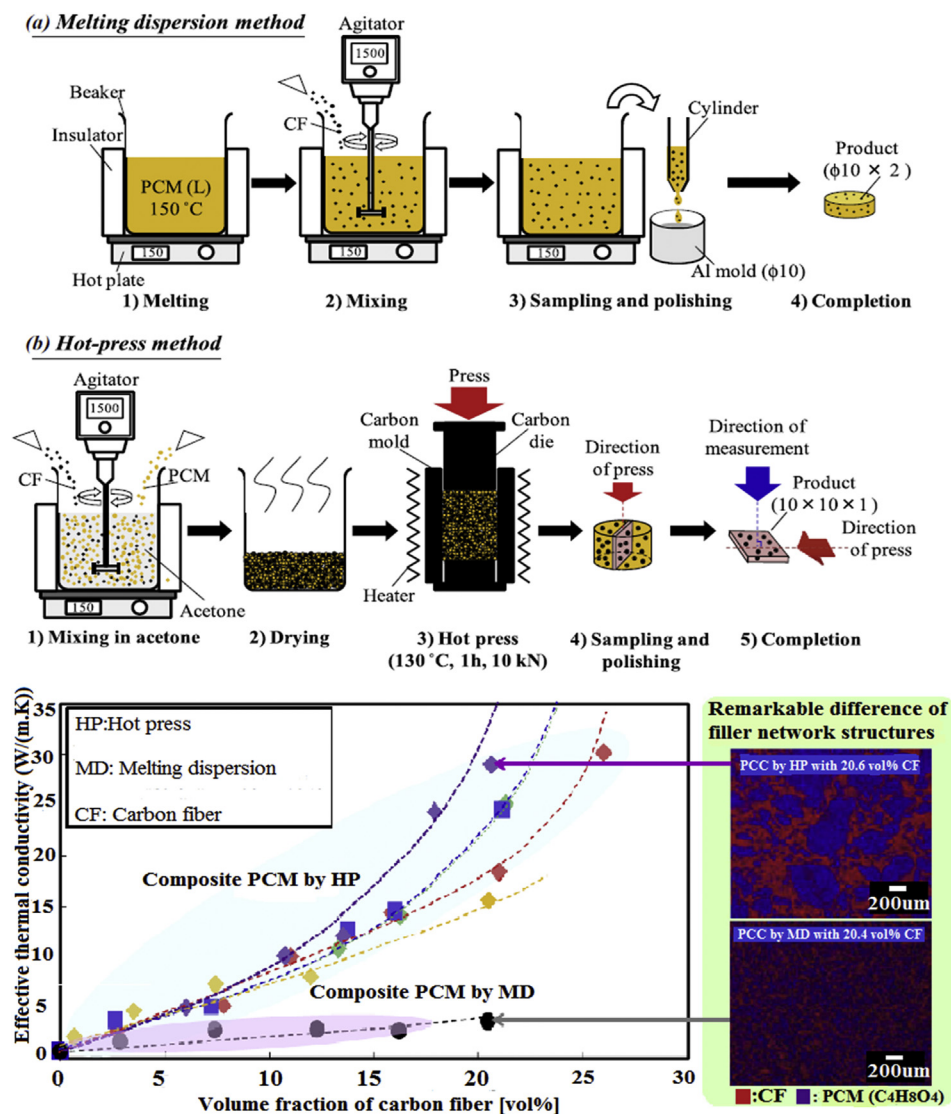


Fig. 45. Schematic diagram of composite PCM prepared by (a) melting dispersion method and (b) hot press method, and curve diagram of effective thermal conductivity of composite samples changing with CF.

researchers [135].

Zhang et al. [136] used CNTs as heat conduction enhancement agent and expanded perlite (EP) as leakage inhibitor to prepare paraffin-CNTs/EP composite PCMs by vacuum impregnation method. The preparation process of composite PCMs are shown in Fig. 46. Firstly, anhydrous ethanol is mixed with CNTs by ultrasonic treatment, and paraffin is added into the mixed solution and mixed evenly. Then, a constant temperature water bath is used for a period of times from melting state to solidification state and mixed evenly. Behind, it is transferred to a vacuum drying oven at a certain temperature and impregnated to obtain a paraffin-CNTs/EP composite PCMs. Schematic diagram of the morphological characteristics of CNTs are shown in Fig. 46(a) and (b). CNTs show a slender silky appearance, and the large number of CNTs agglomerate due to van der Waals forces. CNT is completely wrapped by paraffin and evenly dispersed in the pores and surface of EP, which can promote the formation of heat conduction channels in paraffin-CNTs/EP composite PCMs. In addition, the thermal conductivity of paraffin-CNTs/EP composite PCMs are tested. At room temperature, the thermal conductivity of pure PA and EP are 0.204 W/(m·K) and 0.058 W/(m·K), respectively. When the filling amount of CNTs is zero, the thermal conductivity of paraffin-CNTs/EP composite PCMs are only 0.107 W/(m·K), which is caused by the low

thermal conductivity of EP. With the increase of CNTs content, the thermal conductivity of paraffin-CNTs/EP composite PCMs increased. While the filling amount of CNT is 5.27%, the thermal conductivity of paraffin-CNTs/EP composite PCMs are 0.516 W/(m·K). Because the added CNTs have a high thermal conductivity and are evenly dispersed in PA, they can form a high thermal conductivity network with a high aspect ratio. With the increase of the mass fraction of CNTs, the thermal conductivity network becomes closer, which further enhances the thermal conductivity of paraffin-CNTs/EP composite PCMs.

In order to improve the thermal conductivity of polyethylene glycol, Qian et al. [137] synthesized PEG/SWCNTs nano-composite PCMs by impregnation method. Fig. 47 shows the process of preparation and the morphological characteristics of SWCNTs and PEG/SWCNTs nano-composite PCMs. It is obvious that SWCNT presented a uniform and densely distributed network composed of carbon rods and rough surfaces. In contrast, PEG/SWCNTs nano-composite PCMs, SWCNTs formed a more interconnected thermal conductivity network. On the one hand, the addition of SWCNTs to PEG/SWCNTs can improve the adsorption efficiency of PEG to a certain extent up to 98%. On the other hand, the thermal conductivity of the composites is significantly improved by increasing the content of SWCNTs. The thermal conductivity of the composites in solid state (3.43 W/(m·K)) is 1329% and in liquid state is

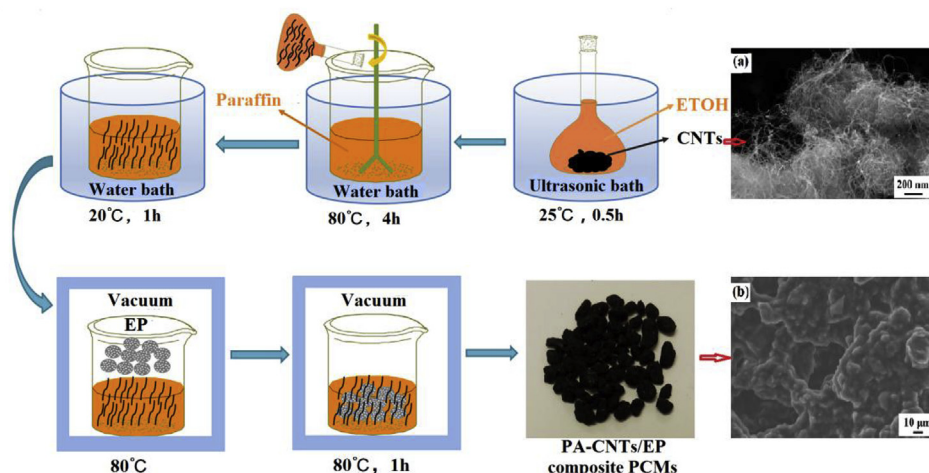


Fig. 46. Schematic diagram of composite PCMs and the morphology of (a) PA-CNTs and (b) PA-CNTs/EP composite PCMs.

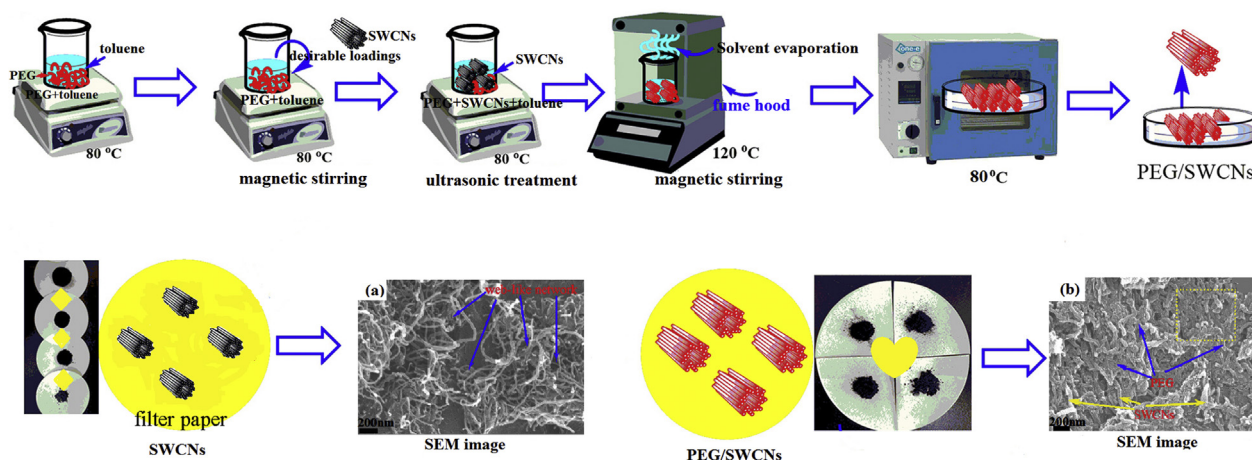


Fig. 47. Melt mixing method for preparation of PEG/SWCNs nanocomposites and microstructure of SWCNs and composite PCMs.

533% higher than pure PEG (0.24 W/(m·K)). The reason for this result may be that SWCNTs form many heat conduction channels in the composites. In fact, solid PEG is mainly heat conduction and fixed SWCNTs heat conduction channels could improve its heat transfer performance. However, the heat transfer of liquid PEG is dominated by convection and the heat conduction channels of SWCNTs is easy to agglomerate, which reduces its effect of improving heat conduction ability.

After the structural modification of MWCNT, Putra et al. [138] prepared three amorphous PCMs: untreated MWCNTs/beeswax, ball-milled MWCNTs/beeswax and acid-treated MWCNTs (A-MWCNTs)/beeswax. Fig. 48 shows the morphological characteristics of pristine MWCNT and A-MWCNT. The surface of pristine MWCNT is smooth and shaped like a rod and strong acid oxidizes MWCNT to form A-MWCNT. The surface of A-MWCNTs is shown in Fig. 48 (c): compared with pristine MWCNT, A-MWCNT rods are short and thick, which converge and become denser, and the surface is relatively rough. This is mainly due to the strong acid oxidation resulting in the reduction of MWCNT impurities. A-MWCNT can physically absorb beeswax into its pores. The higher the percentage of A-MWCNT, the more beeswax is absorbed into the pores of MWCNT, and the less beeswax in surface, thus can prevent the leakage of beeswax. On the other hand, after the addition of MWCNTs, the thermal conductivity will increase significantly. The thermal conductivity of formalized beeswax/A-MWCNT samples is higher than pure beeswax (0.25 W/(m·K)) [139] and paraffin (0.287 W/(m·K)) [140]. When the mass fraction of A-MWCNT is 20 wt% and 5 wt%, the thermal conductivity of beeswax/A-MWCNT form stable PCM samples is 0.58 and

0.46 W/(m·K), respectively, which indicated that MWCNTs could also significantly improve the thermal conductivity of beeswax. The beeswax of the acid-treated MWCNTs is tightly absorbed, which can reduce the interfacial phonon group transmission, thus improving its thermal conductivity.

Wang et al. [141] prepared CNT composites consisting of PA as the matrix and MWCNTs as the additives. The added MWCNTs are treated by four different chemical/physical methods, which synthesized P-MWCNTs (untreated) (a), B-MWCNTs (b), M-MWCNTs (c), A-MWCNTs (d), and G-MWCNTs (e). The Schematic diagram and microscopic morphology of MWCNTs processed is shown in Fig. 49. At a specified temperature under solid state, the thermal conductivity enhancement is in the order of M-MWCNT/PA > A-MWCNT/PA > B-MWCNT/PA > P-MWCNT/PA > G-MWCNT/PA. Whereas the order changes into M-MWCNT/PA > A-MWCNT/PA > G-MWCNT/PA > B-MWCNT/PA > P-MWCNT/PA at a specified temperature under liquid state. Furthermore, the thermal conductivity enhancement ratios of G-MWCNT/PA surpass the corresponding values of B-MWCNT/PA and P-MWCNT/PA. The key reason for the difference in thermal performance is the lower intrinsic CNTs conductivity after treatment, due to scattering of heat carrying phonons by interactions with the surroundings or defects and the other one is the contact thermal resistance on the surface of CNTs. In addition, in addition to the chemical/physical treatment of carbon nanotubes, the main factors affecting the thermal conductivity of the composite phase-change materials are length, thickness, mass fraction and type. Fan et al. [142] also compared different lengths of CNTs with

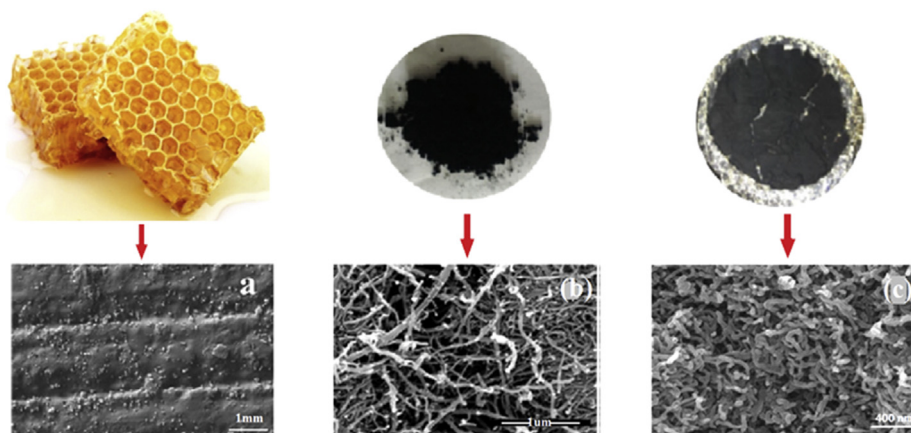


Fig. 48. Microstructure of (a) beeswax, (b) untreated CNTs and (c) acid-treated CNTs.

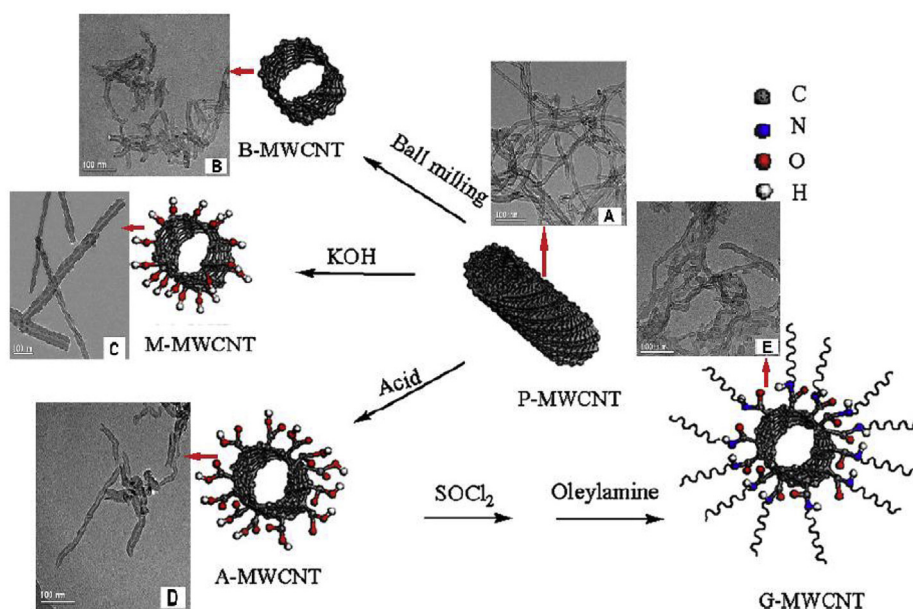


Fig. 49. Schematic diagram and microscopic morphology diagram of MWCNTs processed by physical and chemical methods.

CFs and graphene and Liu et al. [143] further studied that CNTs can significantly enhance the thermal conductivity of the microcapsule composite PCM, which has important research value for improving the heat storage performance of the microcapsule.

3.2.3.3. Addition of rod-shaped graphene in initiative PCM. Rod-shaped graphene is a new kind of monolayer 1D structure material, which is similarly to graphite nanocrystalline sheets in microstructure. It has unique physical properties such as strong mechanical stability, good thermal conductivity and large specific surface area [144]. According to recent studies, graphene has better thermal conductivity than CNTs [145]. The thermal conductivity of single-layer graphene is as high as 5300 W/(m·K), and even some studies have shown that its thermal conductivity can reach 6600 W/(m·K) [146]. Adding graphene to the PCM can improve the thermal conductivity of the composite PCMs, so graphene has great advantages in improving the thermal conductivity of the PCM. A lot of research work is being done on the preparation of high thermal conductivity graphene-based composite PCMs.

Zhong et al. [147] utilized 3D graphene aerogel (GA) and octadecanoic acid (OA) to produce GA/OA composites by the modified Hummers' method. Fig. 50 is preparation process and morphological characteristics of GA/OA composite PCMs. The inherent flexibility of graphene sheets is

a crucial property for constructing the 3D macrostructures. The 3D porous network with interconnection of unfilled GA is shown that pore sizes of the network is about several micrometers and the pore walls consist of thin layers of stacked graphene sheets. Approximately 95% of the pores in GA are interconnected, which allows it to be easily saturated with the molten OA. Calculated results indicated that the OA filled about 98% of the available pore volume in the GA. The fracture surface of GA/OA composite PCMs are indicated that basketwork of the GA remains very well and no evidence is seen of interfacial separation or cracking in the foam structures. After being filled with OA, the GA is clothed in the OA as well as interfaces of the OA and the graphene combine compactly due to the good wetting property between the two phases. Compared with the CNT, the GA is a homogeneous holder material for the PCM, which can avoid the agglomeration, structural deformation and thermal contact resistance.

Yang et al. [148] used graphene oxide (GO) and GNP as additives, polyethylene glycol (PEG) as PCM to prepare PEG/HGA composite PCMs via vacuum impregnation method. GO nanosheets formed a 3D supporting network to keep the form-stable PEG during phase change and GNP dispersed uniformly along the network structure of GO and thus a thermal conductive path way is constructed. The aromatic regions offered active sites to make it possible to interact with other aromatic

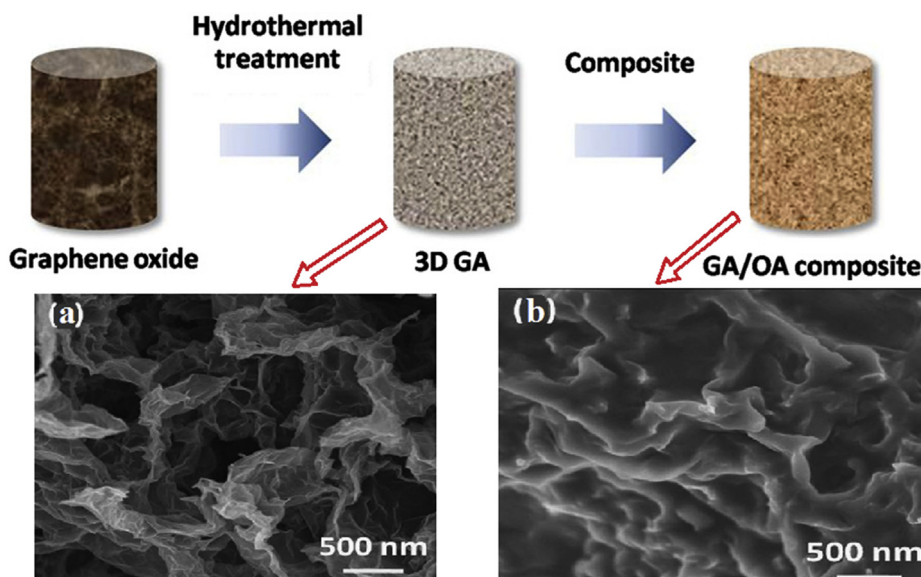


Fig. 50. Preparation process of GA/OA composite PCMs and microstructure of: (a) porous structure of 3D GA, (b) composite PCMs.

molecules through π - π supramolecular interactions. Preparation and morphological images of the PEG/HGA composite PCMs are shown in Fig. 51. The thermal conductivity of PCM can be improved remarkably main reason that GNP, as a thermal conductive filler, established a well-developed 3D heat conduction pathway with the support of GO and the structure of HGA can be kept after freeze drying and vacuum impregnation. The 3D thermal conduction pathway formed is more perfect and the thermal resistance is reduced. Furthermore, Choi et al. [149] compared the effects of three carbon-based additives: MWCNTs, EG and graphene, on the thermal conductivity improvement of PCM. They found that graphene has the best thermal conductivity for PCM, which is attributed to its smaller interfacial thermal resistance and faster thermal conduction. In addition to the above researches on 1D graphene, there are also researches on porous graphene foams, such as Chen et al. [150] and Li et al. [151]. We can predict that graphene foam is also a novel type of high thermal conductivity additive with great potential, which will play an important role in the preparation of high performance composite PCMs.

To sum up, 1D structure additives have the advantages of high thermal conductivity, low density and stable chemical property, etc. Obviously, the high thermal conductivity of matrices is profitable for the improvement of thermal conductivity in PCM. Further, many factors affect the high thermal conductivity of the added substrate, such as porosity, specific surface area and length-width ratio, etc. CFs, CNTs and rod-shaped graphene possess large length-width ratio, where CFs and CNTs have chemical attack resistance and corrosion resistance, and rod-shaped graphene with a special structure of a single layer 1D structure, shows distinctive chemical and physical properties. Thermal conductivity enhancement of abundant 1D structure additives is listed in Table 4, which manifested that 1D structure additives are wonderful matrices in

the aspect of improving the thermal conductivity of initiative PCM.

3.2.4. Initiative composite PCMs using zero-dimensional structure additives

In addition to the above additives, the 0D structure additive, such as ceramic-based and nanoscale-based nano particles, have received considerable attention in the preparation of composite PCMs due to large specific surface area, strong interface interaction and homogeneity as well as dispersion. It also includes a mixture of bio-based additives concentrated carbon nano-materials [152]. Owing to possess high thermal conductivity, the addition of a small amount of 0D structure additives supporting materials can significantly improve the thermal conductivity of the initiative composite PCMs [153]. As a result, it can be used as an additive for form stable PCM and as a heat conduction of microcapsule composites. Further, the heat transfer performance of composite PCMs can be significantly improved by adding nanoparticles to PCM in a certain way and proportion. For the lack of free electrons in the process of heat conduction, heat flow is predominantly transmitted by phonons. In addition to BeO, most metal oxide additives such as aluminum oxide (Al_2O_3), titanium dioxide (TiO_2) and copper oxide (CuO) have relatively high thermal conductivity [154–157]. Non-oxide additives such as boron nitride (BN) [158–161], silicon nitride (Si_3N_4) [162] silicon carbide (SiC) [163,164] and MOFs [44,45] have strong interatomic bond energy and crystal structure, which can significantly reduce phonon scattering and show high thermal conductivity. In particularly, AlN and BN additives are widely used to prepare high performance composite PCMs.

Tang et al. [165] prepared composite PCMs by in-situ doping Al_2O_3 and PEG with ultrasonic assisted Sol-gel method, which significantly enhanced the thermal conductivity of PEG/ SiO_2 - Al_2O_3 hybrid stereotyped composite PCMs. The reason may be that one end of spherical

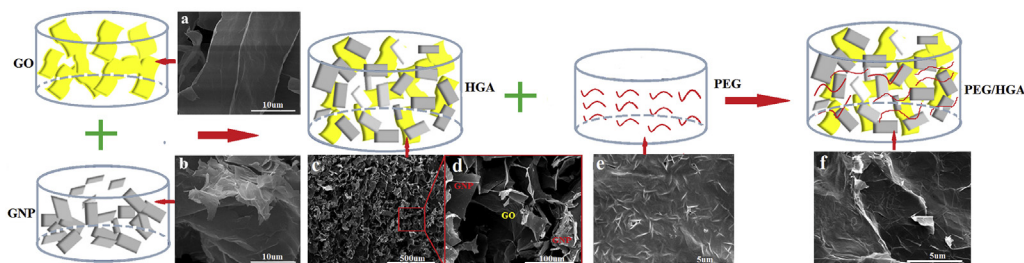


Fig. 51. Preparation of PEG/HGA composite PCMs and corresponding SEM images of (a) GO, (b) GNP, (c, d) HGA, (e) pure PEG and (f) PEG/HGA composite PCMs.

Table 4

Preparation methods of initiative composite PCMs and summary of improving thermal conductivity of one-dimensional structure additives.

PCM type	Thermal conductivity of PCM k_p W/(m·K)	One-dimensional structure additives	Preparation methods	Thermal conductivity of composite PCM k_c W/(m·K)	Fraction of matrices	Improvement (%)	Ref.
Erythritol	0.77	SCF (C5, C25)	Melting blend - solidification form stable	3.92	C25 = 10 wt%	509.1%	[124]
Paraffin wax	0.24	Graphite fiber	Slurry molding method	$k_{//} = 13.82$, $k_{\perp} = 1.31$	Graphite fiber = 38.6 wt%	5758.3%, 545.8%	[125]
Erythritol	0.733	CF	Melt-dispersion method, Hot-press method	$k_{MD} \approx 3.5$, $k_{HP} \approx 28$	CF _{HP} = 20.6 vol%, CF _{MD} = 20.4 vol%	3820%, 478%	[126]
Eicosane	0.22	CNTs	Ultrasonic vacuum impregnation method	0.32	CNTs = 1 wt%	145.5%	[127]
Li ₂ CO ₃ – K ₂ CO ₃ eutectic salts	0.618	SWCNT	Solution evaporation method	0.97	SWCNT = 1.5 wt%	156.98%	[129]
Paraffin wax	0.25	MWCNT	Two-step technique	0.92	MWCNT = 5 vol%	367%	[133]
Paraffin	0.204	CNTs	Vacuum impregnation method	0.516	CNTs = 5.27 wt%	253%	[136]
PEG	0.24	SWCNTs	melt-mixing method	3.43	SWCNTs = 10 wt%	1429.17%	[137]
Beeswax	0.25	MWCNT	Impregnation method	0.58	MWCNT = 20 wt%	232%	[138]
Beeswax	0.25	Graphene	Ultrasonic method	2.89	Graphene = 0.3 wt%	1156%	[139]
Paraffin wax	Liquid: 0.305 Solid: 0.287	GO sheets	Vacuum impregnation method	0.985	GO sheets = 51.7 wt%	Liquid: 323%, Solid: 343%	[140]
Palmitic acid (PA)	0.223 (25 °C)	MWCNTs	Acid oxidation method, Mechanochemical reaction method, Ball milling method, grafting method	0.339	MWCNTs = 1.0 wt%	151.6%	[141]
Dodecanol	0.1296	Graphene oxide (GO) and carbon nanotube (CNT)	In-situ polymerization method	0.3821	GO and CNT hybrid filler = 0.6 wt%	295%	[143]
Octadecanoic acid (OA)	0.184	3D graphene aerogel (GA)	Modified Hummers' method	2.635	3D GA = 20 vol%	1432.1%	[147]
Polyethylene glycol (PEG)	0.31	Hybrid graphene aerogels (HGA)	Vacuum impregnation method	1.43	GO = 0.45 wt% and GNP = 1.8 wt%	461%	[148]
Paraffin	0.21	Porous Al ₂ O ₃ @graphite foams (PAGFs)	Vacuum impregnation method.	0.76	Porosity: 84%	362%	[151]

micron particles is anchored on the surface of Al₂O₃ and the other end diverges outward. Evidently, this composite structure greatly improves the dispersion of PEG/SiO₂. Furthermore, compared with PEG/SiO₂, the thermal conductivity of Al₂O₃ with 3.3 wt% in the original PCM increases by 12.8%. Since Al₂O₃ skeleton greatly reduced the contact thermal resistance and interface thermal resistance, it can form more stable thermal conduction channels in the matrix, which is beneficial to phonon transmission. Additionally, PEG/SiO₂–Al₂O₃ hybrid structure not only improves PEG/SiO₂ dispersion and eliminates agglomeration, but also it constructs a continuous PEG/SiO₂ heat conduction chain with Al₂O₃ micron particles as anchor points. The AlN shell layer is formed on the surface of Al₂O₃ and SiO₂ by carbon thermal reduction and nitriding, which can produce the nucleated shell particles of Al₂O₃@AlN and SiO₂@Si₃N₄, which can enhance the heat transfer between AlN and Al₂O₃ as well as enhance its thermal conductivity.

Nourani et al. [166] also prepared a novel nano-composite PCMs (NCPs) using paraffin as PCM, Al₂O₃ nano-particles as fillers and sodium stearyl lactate (SSL) as surfactant. When the content of Al₂O₃ is 2.5, 5, 7.5, 10 wt% and the mass fraction of SSL/Al₂O₃ is 1:3.5, the dispersion of nano-particles in paraffin wax is significantly improved. The experimental results show that the effective thermal conductivity enhancement ratio of samples containing 10.0 wt% nano-Al₂O₃ in solid and liquid state is 31% and 13%. What is more, the melting rate of samples increased by 27%. In addition, the thermal conductivity tested in the 45–55 °C, its effective thermal conductivity increased under the temperature of 55 °C. Since the influence of temperature, it can be changed the orderly structure of solid composite PCMs and the number of molecules of vibrations probably increased. When NCPs changes to liquid state, the ordered solid phase structure changes into disordered liquid phase structure and it may be diminished the effective thermal conductivity of the composites. Compared with the disordered structure in liquid state, the ordered solid structure can enhance the heat transfer between nano-Al₂O₃

particles and paraffin wax. Therefore, the effective thermal conductivity increases more in solid state than in liquid state.

Mohamed et al. [167] synthesized composite PCMs by fully adsorption of paraffin wax and microcrystalline wax loaded with α -Al₂O₃ (0.5, 1 and 2 wt%) with different mass fraction further studied their thermal conductivity and thermal storage performance. The experimental results show that the higher the loading mass fraction of α -Al₂O₃ in paraffin wax composite PCMs, the better its thermal conductivity and latent heat storage performance. At the temperature of 293 K–323 K, the nano-Al₂O₃ particles with different mass fraction are loaded into the PCMs, which resulted in significant enhancement of thermal conductivity. Therefore, nano-alumina particles could obviously improve the thermal conductivity of the composite PCMs. Furthermore, the thermal conductivity of the composites increases with the increase of phase change temperature. In addition, the thermal conductivity of six kinds of composite PCMs are higher than that of pure paraffin wax and microcrystalline wax. It indicates that the addition of nano-Al₂O₃ and paraffin can regulate the range of phase change temperature in the composite PCMs and enhance thermal load response rate of heat storage.

Jiang et al. [168] employed paraffin wax core and poly (methyl methacrylate comethyl acrylate) shell modifying with nano alumina (nano-Al₂O₃) inlay to produce microcapsule PCMs via emulsion polymerization. The SEM images of paraffin morphological characteristics modified with different percentages of nano-Al₂O₃ are shown in Fig. 52. It can be seen that the average particle size apparently became small for the sample synthesized at the nano-Al₂O₃ mass of more than 16%. Since nano-Al₂O₃ is equivalent to nucleating agent, when the number of nucleating agents increase and the particle size will decrease. Furthermore, thermal conductivity rapidly improves by the mass fraction of nano-Al₂O₃ increasing, which is nearly linear correlation between the thermal conductivity and the addition level of nano-Al₂O₃. The high thermal conductivity channel formed by nano-Al₂O₃ evenly distributed

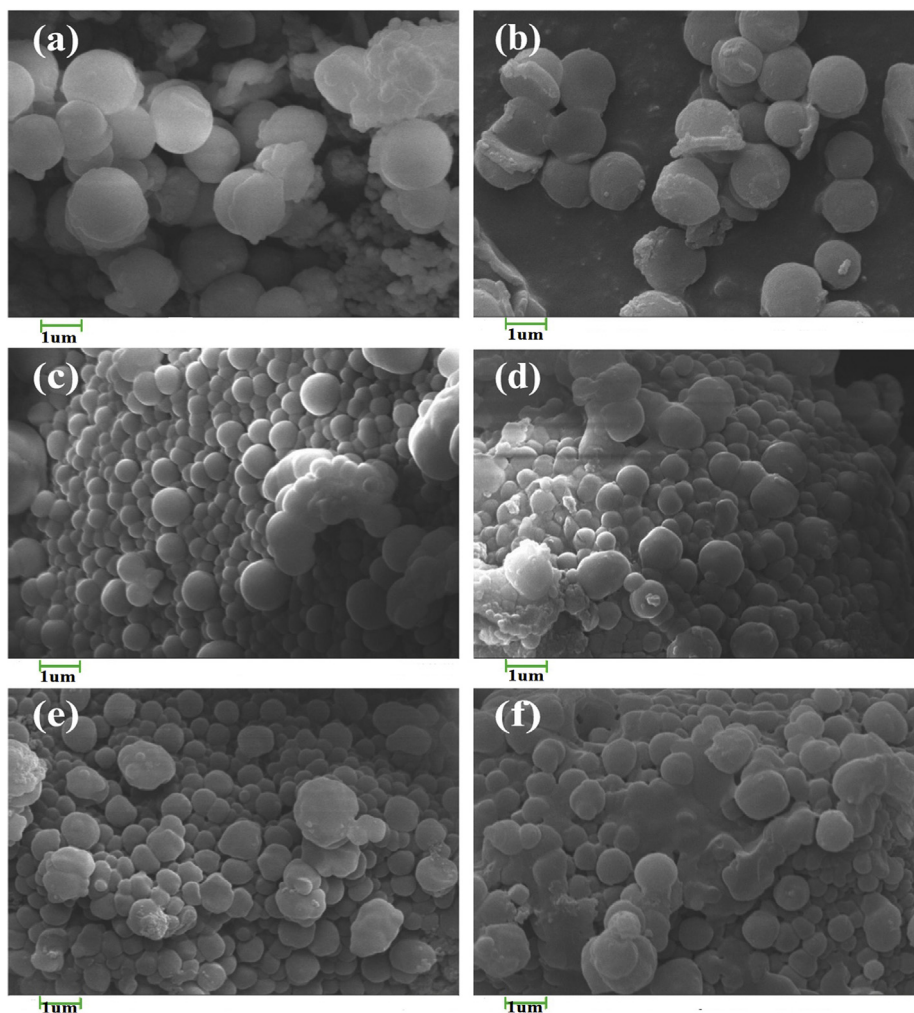


Fig. 52. SEM images of paraffin microcapsules modified with different percentages of nano- Al_2O_3 : (a) 0%, (b) 5%, (c) 16%, (d) 27%, (e) 33% and (f) 38%.

in PCM is the fundamental reason for promoting the improvement of thermal conductivity.

Sharma et al. [169] prepared nano-enhanced organic PCMs of PA/ TiO_2 by melting blending method using PA and TiO_2 nano particles with different mass fractions. The preparation process and morphological characteristics of the sample of the composite PCMs are shown in Fig. 53. Since the repulsion of the surfactant, TiO_2 nanoparticles are evenly dispersed in PA. In some regions, the melting of the PCM near the PA block, which is mainly caused by the concentration of electron beams at a specific location. Moreover, thermal conductivity of PCM significantly improve by the mass fraction of nanoparticles increasing. When 0.5, 1, 3, and 5 wt% of TiO_2 nano particles are supplemented in all else being equal, the thermal conductivity increased by 12.7%, 20.6%, 46.6% and 80%, respectively. Significantly, the thermal conductivity of composite PCMs are approximately proportional to the mass fraction of TiO_2 nano particles. Since the agglomeration of abundant TiO_2 nano particles occurs in mixture, which results in the nonlinear increase of thermal conductivity. Further, the thermal conductivity of the composite PCMs are larger than that of pure PA and it can promote to melt and solidification of the composite PCMs. At the same time, it may be that TiO_2 nano particles are evenly dispersed in PA, which is equivalent to the addition of nucleating agent, thus reducing the degree of supercooling and further shortening the melting and solidification time.

Su et al. [170] prepared composite PCMs containing hexagon boron nitride (HBN) eutectic crystals using OC and SA, and introduced HBN to significantly improve the thermal conductivity of the composite PCMs.

The chemical adhesion, crystalline phase and morphology of eutectic mixture/HBN composite PCMs are further tested. Fig. 54 shows the morphological characteristics of the HBN, SA-ODE eutectics, PCM1, PCM2 and PCM3 composites. From the diagram shows that there is no chemical reaction between SA-ODE eutectic and HBN, and the composite PCMs had good thermal stability. Moreover, the thermal conductivity of melting/solidifying composite PCMs increased significantly by addition of HBN medium, while the latent heat of phase change decreased slightly. Furthermore, under liquid phase conditions, the mass fraction 5, 10, 15 wt% of HBNs added into SA-ODE to form composite PCMs and the thermal conductivity of eutectic mixture is 0.1586, 0.1764, and 0.1863 W/(m·K). After the introduction of HBN, the thermal conductivity increased by 0.0074, 0.0252 and 0.0351 W/(m·K), which means that the thermal conductivity increased by 4.9%, 16.7% and 23.2%, respectively. Undoubtedly, the new composite PCMs have higher thermal conductivity than other composite PCMs. Obviously, the introduction of HBN will improve the thermal conductivity of SA-ODE eutectic mixtures in both solid and liquid phase environments. The reason is that HBN packing with high aspect ratio is easy to form network structure and improve thermal conductivity. Interestingly, the quasi-percolation behavior can be observed when the HBN particle network is constructed, but the thermal conductivity does not follow the percolation rule. Moreover, hybrid micron HBN and SA-ODE are filled with eutectic mixtures meanwhile the hybrid particles uniformly distributed in the matrix are easy to establish a heat conduction network when the appropriate specific sequence is suitable. Eventually, the HBN particles

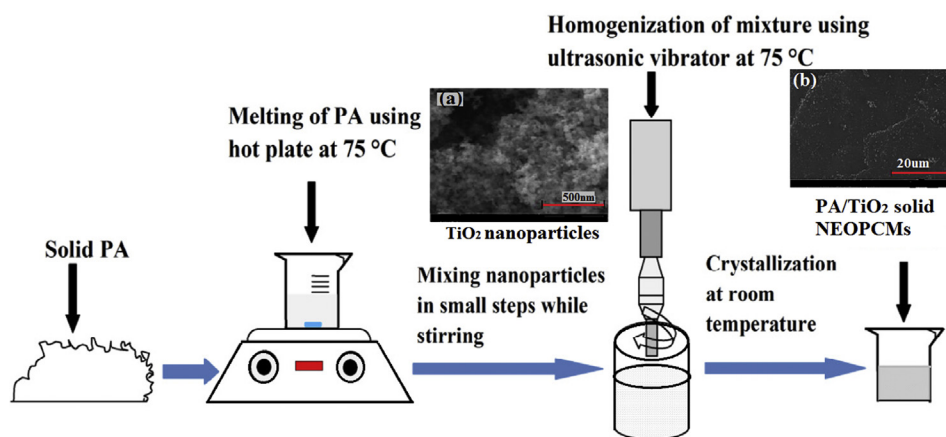


Fig. 53. Preparation steps of PA/TiO₂ composite PCMs and FESEM images of (a) TiO₂, (b) PA/TiO₂ nano-enhanced organic PCMs.

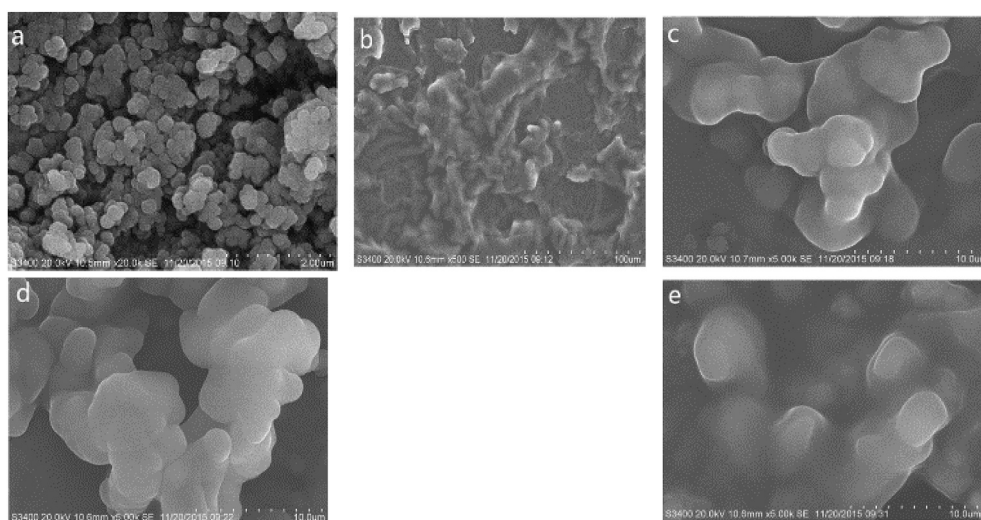


Fig. 54. SEM images of the (a) HBN, (b) SA – n-octadecane eutectics, (c) CPCM1, (d) CPCM2 and (e) CPCM3.

form an effective connection point between SA-ODE, which promotes the formation of heat conduction network and the effect of interfacial phonon transmission.

Fang et al. [171] filled HBN nano-sheets with paraffin wax to produce a series of novel paraffin-based composite PCM samples by supplement HBN nano-sheets (mass fraction: 0, 1, 2, 5 and 10 wt%). The presence of HBN nano-tubes (up to 10 wt%) resulted that the thermal conductivity increases reach 60% and the latent heat of melting decreases less than 12%. Since the thermal conductivity of the HBN nano-sheets has been improved and the melting/frozen rate also has been increased by 25%, which can prove the superior applicability of composite PCMs. Fig. 55 shows the microscopic morphology characteristics of HBN nano-sheet. Because the thermal conductivity increase largely depends on the shape, size and thickness of HBN nano-sheets, which may become a research hotspot in the future for the influence of these geometric parameters.

Yang et al. [172] used OC as the core material and polymethyl methacrylate shell as wall material to synthesize a novel microcapsule PCMs by supplemented with modified Si₃N₄ powder. When modified microcapsules fill a certain amount of Si₃N₄, the heating rate is positively correlated with the addition level of Si₃N₄, which indicated that Si₃N₄ can significantly improve the thermal conductivity of the microcapsules. Fig. 56 shows an improved Si₃N₄ microcapsule synthesis method and morphological characteristics of microcapsule composite PCMs. Frankly, adding Si₃N₄ particles will increase the thermal conductivity. When the

amount of Si₃N₄ increased to 10g, the thermal conductivity increased by 56.8% to 0.3630 W/(m·K). The most reason is that the microcapsule isolation structure reduces the interfacial thermal resistance between the thermal conducting particles and the matrix, which is wrapped around the matrix particles rather than uniformly entering and distributing inside the matrix particles, and it can significantly improve its thermal conductivity at a lower addition level.

Yuan et al. [173] prepared a novel composite PCMs by introducing SiC/EG heat conducting skeleton and adding it to the mixture of PA and epoxy resin (ER). Since the thermal conductivity of SiC is similarly to EG, but the volume resistivity is much higher than that of EG, the high thermal conductivity and volume resistivity of SiC to prepare the composite PCMs have high thermal conductivity and low electrical conductivity. Fig. 57 shows the morphological characteristics and preparation of SiC and composite PCMs. It can be clearly seen that the size of SiC particles is evenly distributed. After adding granular SiC, the morphology of the samples becomes sufficiently dense and SiC particles can be effectively filled in the network pores. The thermal conductivity of the composite PCMs increases with the increase of SiC added content reach 15 wt %, the thermal conductivity of 2.0 W/(m·K) continues to grow and the volume resistivity reach the maximum of $12.5 \times 10^{11} \Omega \text{ cm}$, and the composite PCMs have the best comprehensive performance. Moreover, when the content of SiC from 15 wt% increase to 20 wt%, the thermal conductivity increased from 2.0 W/(m·K) to 2.8 W/(m·K) and volume resistivity decreased to $7.7 \times 10^{11} \Omega \text{ cm}$ from $12.5 \times 10^{11} \Omega \text{ cm}$. When

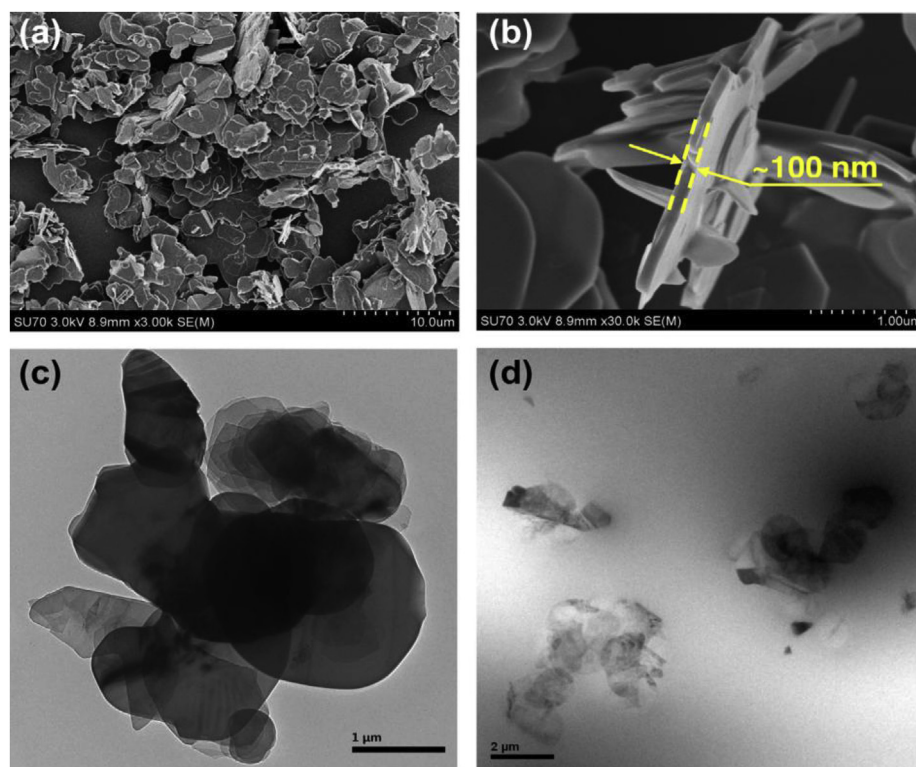


Fig. 55. Microscopic characterization of the h-BN nano-sheets. (a) SEM image on the h-BN nano-sheets and (b) isolated h-BN nano-sheet as well as (c) TEM image of h-BN nano-sheets and (d) dispersion of the h-BN nano-sheets in paraffin.

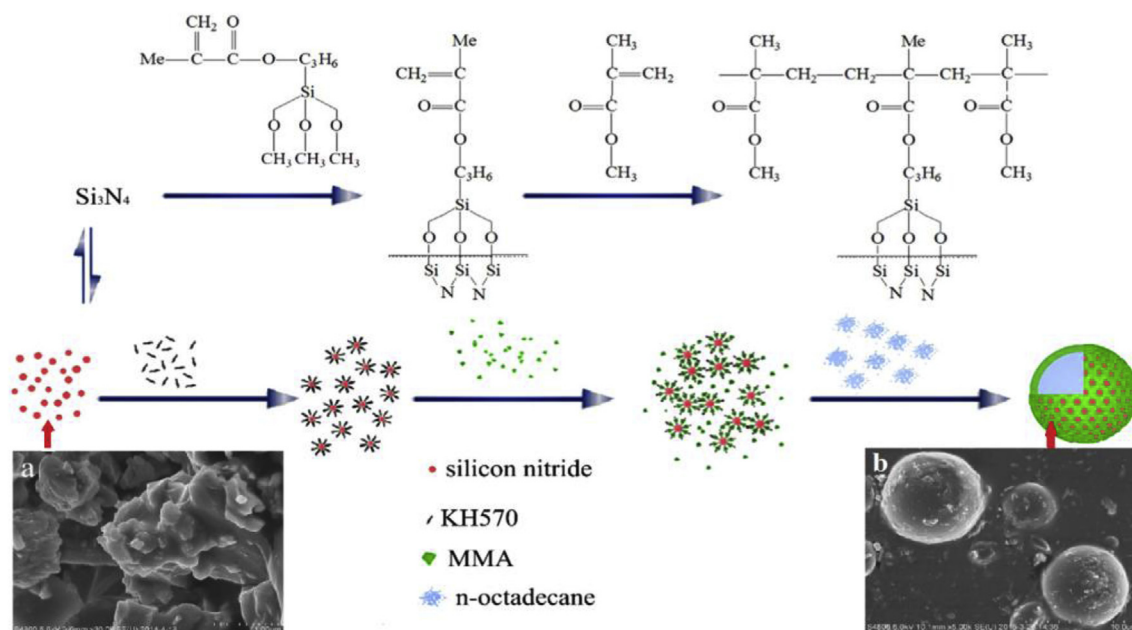


Fig. 56. Schematic of synthetic strategy for microcapsules with modified silicon nitride as well as SEM micrographs of (a) silicon nitride and (b) microcapsule samples. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

adding appropriate amount of SiC in the composite PCMs, SiC particles in EG pores may block the electron transmission path, thus increasing the volume resistivity. In the case of high silicon carbide content, EG and SiC form the large number of continuous skeletons in the composite PCMs, which may lead to a sharp increase in thermal conductivity and a significant decrease in volume resistivity.

Chen et al. [174] used Cr-MIL-101-NH₂ (MOFs) as porous support material, carbon quantum dot (CQD) as fluorescence active material and

SA as heat storage material to design and prepare MOFs/CQD/SA photoluminescence-functionalized composite PCMs by melting impregnation method. Fig. 58 shows the preparation (a) and excitation fluorescence (b) schematic diagram, and phase change enthalpy (c) of photoluminescence functionalized composite PCMs. From the figure, the unique structure of the material can effectively prevent the conventional polymerization induced fluorescence quenching process to obtain excellent heat storage and fluorescence capture properties. After adding

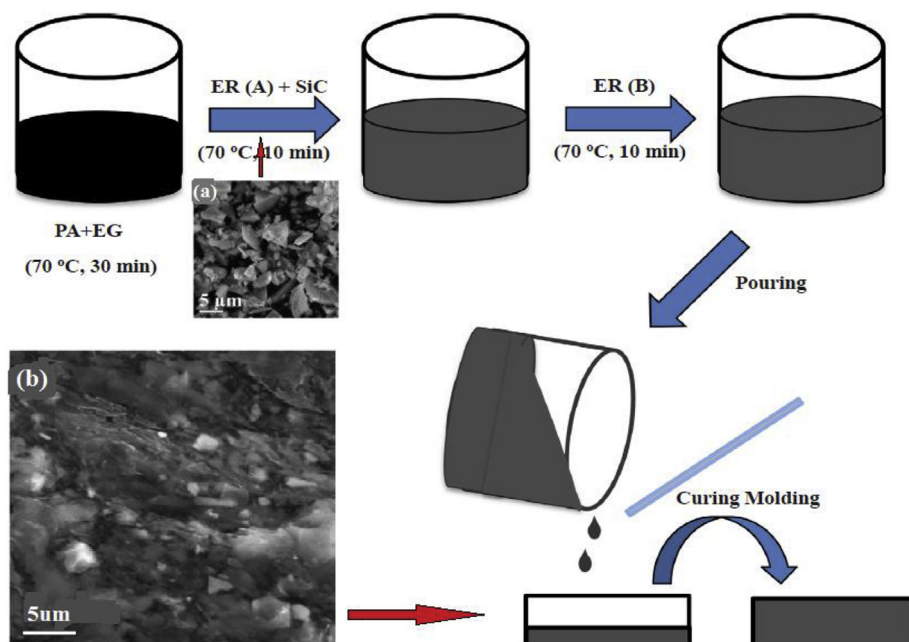


Fig. 57. Schematic diagram for the preparation of the composite PCMs and SEM images of (a) SiC, (b) 50 wt% PA, 20 wt% SiC, 3 wt% EG, 27 wt% ER composite PCMs.

CQDs to MOFs, as CQDs is inserted into some micro/meso pores of MOF scaffolds, CQDs occupied the pores of MOF and its specific surface area as well as pore volume are relatively reduced. Because the molecular length of stearic acid is small, the average pore diameter of the MOFs/CQD

composite scaffold is large enough that the SA molecule can rotate freely. Therefore, the high heat storage density of SA can be ensured by adjusting the width of the channel. Because of the large capillary force of porous MOFs/CQD skeleton, a large amount of SA can be fixed in the

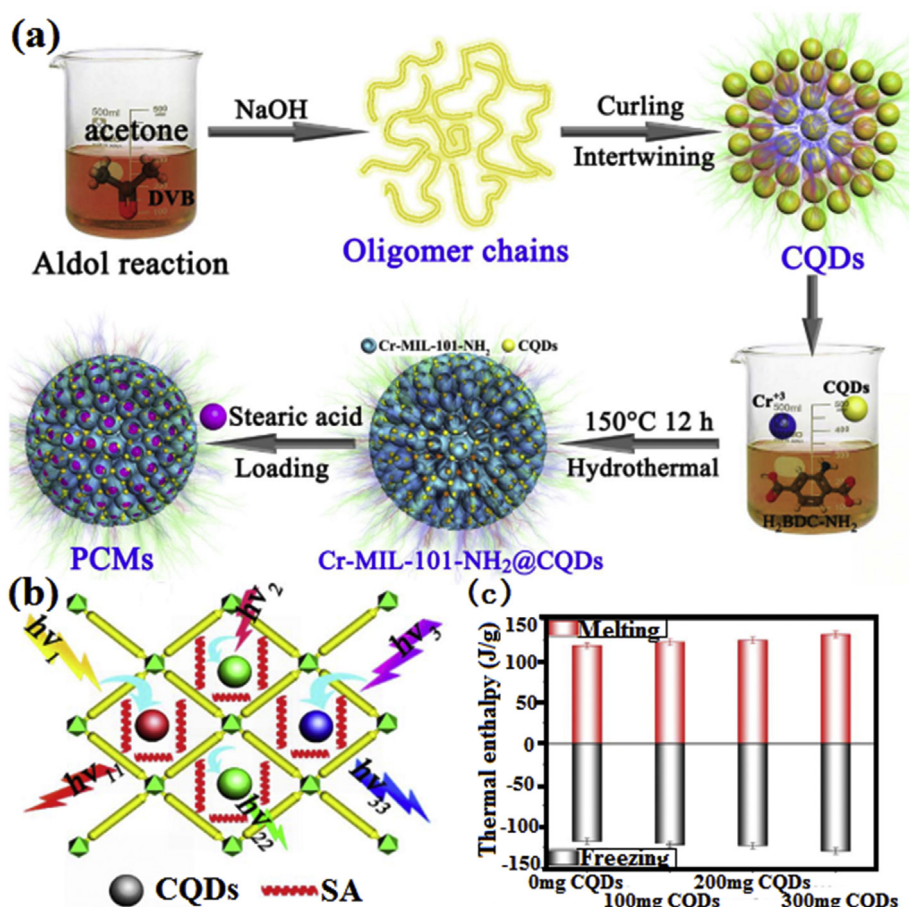


Fig. 58. Preparation (a) and excitation fluorescence (b) schematic diagram, and phase change enthalpy (c) of photoluminescence functionalized composite PCMs.

pores of MOFs/CQD, which can not only prevent the leakage of liquid SA, but also improve the transmission capacity of photons and heat. Moreover, the hydrogen bond between SA and MOFs helps to adsorb more SA and increase the load of SA. However, the hydrogen bond is the stronger and SA chain is restricted thought the free stretching and crystallization to some extent. Therefore, the latent heat is lower than the theoretical value. Further, the insertion of CQDs will weaken the hydrogen bond between SA and MOFs, making SA more capable of free stretching and crystallization, thus it can be increased the latent heat of SA/MOFs/CQD composite PCMs. With the increase of CQD content, the hydrogen bond interaction between SA and MOFs gradually weakens. Therefore, increasing more CQDs are helpful to increase the latent heat value of composite PCMs, which is gradually close to the theoretical latent heat.

Tang et al. [175] synthesized high porosity carbons (HPCs) by controlling carbonized MOFs method and added into PEG to prepare PEG/HPCs composite PCMs. The migration and aggregation of ZnO fine particles by evaporating in the carbon matrix produced many micropores, mesoporous and nano-channels, thus obtaining HPCs with high specific surface area and large pore volume. The material has high adsorption capacity and storage capacity of PEG, and the microporous and mesoporous pore channels can induce large surface tension and capillary force, thus ensuring the high thermal stability of the composites. In addition, the graphite of disordered porous carbon has a small structure, large specific surface area and relatively high thermal conductivity. The thermal conductivity of the PEG/HPCs composite PCMs (0.42 W/(m·K)) are about 56% higher than that of pure PEG (0.27 W/(m·K)). It is suggested that HPCs can effectively improve the thermal conductivity of PEG.

Li et al. [176] firstly employed in situ deposition method and calcining method to load ZIF-67 on EG surface to generate $\text{Co}_3\text{O}_4/\text{EG}$ with 3D interconnected network structure. Further, it can be used melting impregnation method and dispersed it in SA to prepare SA/ $\text{Co}_3\text{O}_4/\text{EG}$ amorphous composite PCMs. Due to the abundant pores and large specific surface area of $\text{Co}_3\text{O}_4/\text{EG}$, the thermal performance of SA is effectively improved to achieve good form stable effect and prevented the leakage of molten SA. The preparation process, SEM images and thermal conductivities are shown in Fig. 59. From the figure, the thermal conductivity of SA/ $\text{Co}_3\text{O}_4/\text{EG}$ composite PCM containing 10% $\text{Co}_3\text{O}_4/\text{EG}$ of 2.53 W/(m·K) is higher than 7.67 times with pure SA. The high surface area and abundant fractional pore volume of $\text{Co}_3\text{O}_4/\text{EG}$ additives provide an effective fractional heat transfer pathway for SA, thus can improve its thermal performance.

The 0D structure particles with high thermal conductivity also play an important role in improving the thermal conductivity of the PCM. They mainly increase the heat flow by forming a thermal conductivity chain evenly dispersed in the PCM and increase the phonon group velocity and mean free path to a certain extent. In addition, during the heterogeneous

nucleation process of PCM, nucleation work reduces with the increased the nucleation area and the solidification process is accelerated, thus significantly reducing the supercooling degree of the nano particles. The results of 0D structure additives/PCM composites are listed in Table 5. Comparing with pure PCM, there is a remarkable improvement in thermal conductivity.

4. Conclusions and perspectives

Phase change energy storage technology, with potentially high energy density and stable output temperature and energy, is regarded as one of cost-effective and large-scale heat storage methods. The key to develop low cost and compact heat storage system is to develop high performance PCM for more efficient storage of industrial waste heat and solar energy heat. In the way to obtain high performance PCM, some challenges still remain:

1. Thermal conductivity is one of the critical parameters of PCM and thermal energy storage system.

Unlike sensible and thermochemical materials, thermal storage efficiency of composite PCMs greatly depends on thermal conductivity; however, in previous reviews of PCM, little attention has been devoted to summarizing high thermal conductivity additives. Another issue concerning interface between PCM and heat conducting matrix is random, making it hard to accurate predict for the thermal conductivity of composite PCMs. Therefore, it is necessary to further study the combination forms of two or multi-phase composite PCMs.

2. Due to the particularity of the PCM in the structure as well as physical and chemical properties, it is complicated to explain the thermal conductivity of the PCM by using the phonon thermal conductivity mechanism.

Though the electrons of most inorganic non-metallic materials are bound, which cannot be used as heat conduction carriers and phonon heat conduction mechanism is used to explain the phenomenon of heat conduction. However, the heat conducting matrix and the PCM may not belong to the same heat conducting mechanism, so their experimental results are far from satisfactory. Based on these experiences, unified research is recommended for phonon, electron and photon thermal conductivity mechanism.

3. Since the heat flow follows the principle of least resistance, the addition of high thermal conductivity matrix in the composite PCM is bound to cause the heat to flow along the thermal conductivity skeleton. The heat flow intensity and heat conduction rate of common

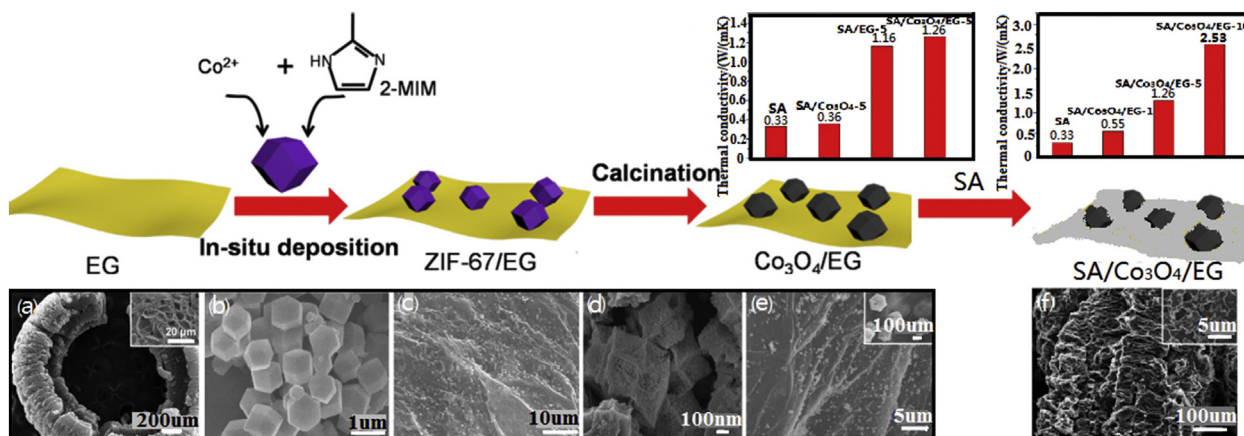


Fig. 59. Preparation process, SEM images and thermal conductivities of (a) EG, (b) ZIF-67, (c) ZIF-67/EG, (d) Co_3O_4 , (e) $\text{Co}_3\text{O}_4/\text{EG}$, (f) SA/ $\text{Co}_3\text{O}_4/\text{EG}$.

Table 5

Main results of zero-dimensional structure additives/PCM initiative composite PCMs.

PCM type	Thermal conductivity of PCM k_p W/(m·K)	High thermal conductivity additive	Preparation methods	Fraction of matrices	Thermal conductivity of composite PCM k_c W/(m·K)	Improvement (%)	Ref.
Paraffin	Solid: 0.25 Liquid: 0.16	Graphite-based nanocomposites	Molten embed	Solid: 2.75 Liquid: 0.16	Graphite = 10 wt%	1100%	[152]
Erythritol	0.70	EG	Impregnation, compression and sintering	12.51	EG = 10 wt%	1738%	[153]
Paraffin	0.147	TiO ₂ , sodium stearoyl lactylate (SSL)	Ultrasonic dispersion method	3 wt%	Paraffin/TiO ₂ -SSL:0.195	132.7%	[154]
Eutectic hydrate salt (EHS)	0.131	Boron nitride foams (BNFs)	Physical blending and impregnation method	4 wt%	EHS/BNF: 1.359	1037%	[158]
Paraffin	0.12	h-BN porous scaffolds	Vacuum impregnation method	18 wt%	Paraffin/h-BN: 0.85	708.3%	[159]
Polyethylene glycol (PEG)	0.273	Boron nitride (BN) and graphene oxide (GO)	Facile solution blending method	4 wt% GO and 30 wt% BN	PEG/BN-GO: 3.00	1100%	[160]
n-eicosane	0.33	Boron nitride (BN)	Molecular dynamics simulation	10 wt%	n-eicosane/BN: 5.18	1569.7%	[161]
Paraffin	0.2293	Nano-Si ₃ N ₄	Ultrasonic dispersion method	10 wt%	Paraffin/nano-Si ₃ N ₄ : 0.309	135%	[162]
Sodium acetate trihydrate (SAT)	0.79	Silicon carbide (SiC), bentonite	Vacuum impregnation method	15 wt%	SAT/bentonite-SiC: 0.9614	121.7%	[163]
Polyethylene glycol (PEG)	0.06	Silicon carbide nanowires network (SiC NWs) and (EVM)	Vacuum impregnation method	3.29 wt%, 23.59 wt%	PEG/EVM-SiC: 0.53	880%	[164]
PEG	0.297	SiO ₂ -Al ₂ O ₃	Ultrasound-assisted sol-gel method	12.6 wt %	PEG/SiO ₂ -Al ₂ O ₃ : 0.435	146.5%	[165]
Paraffin	Solid:0.197, Liquid:0.148	Nano-Al ₂ O ₃	Blending and vacuum impregnation method	10.0 wt%	Solid:0.259, Liquid:0.167	131.5% 112.8%	[166]
Paraffin wax, Microcrystalline wax	0.33; 0.18	α nano-alumina	Blending and vacuum impregnation method	2.0 wt %	Paraffin/ α nano-alumina:0.53	160.6% 761.1%	[167]
Paraffin wax	0.2442	Nano-Al ₂ O ₃	Emulsion polymerization method	38 wt%	Microcrystalline/ α nano-alumina:1.37	156.3%	[168]
Palmitic acid (PA)	0.194	TiO ₂ nanoparticles	Two-step method	5 wt%	Paraffin/nano-Al ₂ O ₃ : 0.3816	180%	[169]
n-octadecane/stearic acid eutectics (ODE-SA)	0.2982	Hexagonal boron nitride (HBN)	Melt mixing-stirring adsorption method	15 wt%	PA/TiO ₂ nanoparticles = 0.35	109.7%	[170]
Paraffin	0.265	HBN nanosheets	Two-step method	10 wt%	ODE-SA/HBN: 0.3272	200%	[171]
n-octadecane (ODE)	0.232	Silicon nitride (Si ₃ N ₄)	Ultrasound-modified method	30 wt%	Paraffin/HBN nanosheets: 0.53	156.8%	[172]
Paraffin	0.239	SiC-EG	Melt mixing-stirring adsorption method	28 wt%	ODE/Si ₃ N ₄ : 0.3630	1338.9%	[173]
SA	0.26	MOFs/CQD	Physical mixing and impregnation method	30%	Paraffin/SiC-EG: 3.2	158%	[174]
PEG	0.27	High porosity carbons (HPCs)	Physical blending and impregnation method	7.5%	SA/MOFs/CQD: 0.41	156%	[175]
SA	0.33	Co ₃ O ₄ /EG	Physical blending And vacuum impregnation methods.	10%	PEG/HPCs: 0.42	767%	[176]

additives are extremely small for the high-performance heat conduction matrix with a large number of heat conduction chains. Therefore, some high thermal conductivity additives, such as highly graphitized network carbon, graphene foam and HBN nanoparticles, are proposed to improve the thermal conductivity of the composite PCMs. Nevertheless, because the difficulty in preparing these high thermal conductivity additives, the porous skeleton is prone to collapse and the reaction conditions need to be controlled with great discretion.

4. Broadening the heat flow channel is one of the methods to reduce phonon scattering, which is extremely key to improve the effective thermal conductivity of composite PCMs.

Heat flow channel provides an important carrier for phonon transfer. Widening heat flow channel can be considered an effective way to improve the thermal conductivity of composite PCMs. New additives,

such as 3D network carbon, TiO₂ foam and MOFs with different pore sizes and porosity have been proposed, whose effectiveness was verified through experimental testing. It is expected that heat storage system will yield improved performance by reducing the time which requires to be heated during both charging and discharging. Composite PCMs have been adopted in some heat storage systems to realize simple and compact design and low-cost serial production.

The challenges mentioned here indicate not only that industrial waste heat and solar energy storage by PCM is nothing near reaching the optimal thermal storage performance, but they also point the directions toward future investigation of high-value composite PCM and cost-effective thermal storage system. Right now research activities devoted to thermal conductivity of composite PCMs are extensively increasing. As a positive outcome, the preparation method, microstructure and phonon thermal conduction mechanism of the composite PCMs and various heat storage systems with high performance have been studied and tested.

These projects demonstrate that a heat storage system designed using PCM is a high-value and large-scale heat storage option. To bring the phase change heat storage solution into a broader market, more intensive studies in fields of phonon thermal conductivity mechanism, development of high performance composite PCMs and efficient and compact phase change heat storage system are still required.

Declaration of competing interest

The authors declare no competing financial interest.

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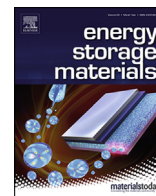
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Corrigendum to: “Thermal conductivity enhancement on phase change materials for thermal energy storage: A review”



Shaofei Wu^{a,b}, Ting Yan^{a,b,*}, Zihan Kuai^{a,b}, Weiguo Pan^{a,b,**}

^a College of Energy and Mechanical Engineering, Shanghai University of Electric Power, Shanghai, 200090, China

^b Institute of Distributed Energy System and Energy Storage, Shanghai, 200090, China

The authors sincerely regret that the title of this paper bears strong similarities to the previously published paper entitled “Thermal conductivity enhancement of phase change materials for thermal energy storage: A review” by Liwu Fan and J.M. Khodadadi

Renewable and Sustainable Energy Reviews, 15: 24–46, January 2011, <https://doi.org/10.1016/j.rser.2010.08.007>. [RSER 2011].

The authors regret the similarities of the title in EnSM 2020 which was done without any intent to copy or plagiarize. All other aspects of EnSM 2020 are original.

To avoid confusion, the title of EnSM 2020 is hereby updated to:

“Review on thermal conductivity enhancement in phase change materials: Mechanism, approaches and perspectives”.

The authors would like to apologize to the previous authors of RSER 2011 and the readers of this journal for any inconvenience this may cause.1.

Declaration of competing interest

All authors of the manuscript have no real or potential conflicts of interest related to their work, including any financial, personal or relationship with others or other organizations, that have no adverse effect on their work during the three years prior to the submission of the manuscript.

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* Corresponding author. College of Energy and Mechanical Engineering, Shanghai University of Electric Power, Shanghai, 200090, China.

** Corresponding author. College of Energy and Mechanical Engineering, Shanghai University of Electric Power, Shanghai, 200090, China.

E-mail addresses: yt81725@126.com (T. Yan), pweiguo@163.com (W. Pan).

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